Functional Materials

Electrical materials

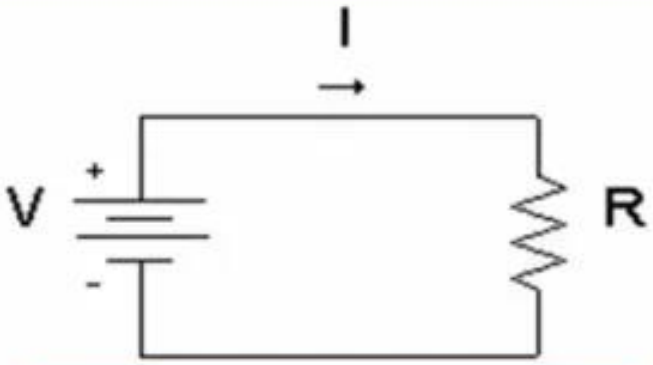
Contents

- Ohm’s law, resistivity,
- classification –
- metals
- semiconductors (intrinsic & extrinsic-Si)
- insulators
- superconductors



Georg Simon Ohm

Electronics and You



Basic Electrical Circuit

$$I = \frac{V}{R}$$

Ohm's Law

Great German Physicist and Mathematician 1789 - 1854.

Ohm's Law and resistivity

One of the most important electrical characteristics of a solid material is the ease with which it transmits an electric current. Ohm's law relates the potential difference, V (in volts), with the electrical resistance, R (in ohms and the electrical current, I (in amp). as follows:

$$V = IR \qquad 1$$

This law is applicable to most but not all materials. The resistance R is the resistance of the material through which the current is passing. The value of R is influenced by specimen configuration (size, shape); materials properties; and for many materials is independent of current.

$$R = \rho \frac{l}{A} \qquad 2$$

where l is the length (cm) of the resistor, A is the cross-sectional area (cm^2) of the resistor. The ρ is the resistivity or Electrical resistivity (Ohm.cm or Ω.cm). From the expression for Ohm's law and Eq. (2)

$$\rho = \frac{RA}{l} = \frac{VA}{Il}$$

Factors affecting resistivity:

a)Temperature

b)Alloying

c)Mechanical stress

d)Age Hardening

e)Cold Working

Classification Energy Band Theory:-

In a single isolated atom, the electrons in each orbit have definite energy associated with it. But in case of solids all the atoms are close to each other, so the energy levels of outermost orbit electrons are affected by the neighboring atoms. When two single or isolated atoms are brought close to each other then the outermost orbit electrons of two atoms interact or shared with each other. i.e., the electrons in the outermost orbit of one atom experience a attractive force from the nearest or neighboring atomic nucleus. Due to this the energies of the electrons will not be in same level, the energy levels of electrons are changed to a value which is higher or lower than that of the original energy level of the electron.

The electrons in same orbit exhibits different energy levels. The grouping of this different energy levels is called energy band.

However, the energy levels of inner orbit electrons are not much affected by the presence of neighboring atoms.

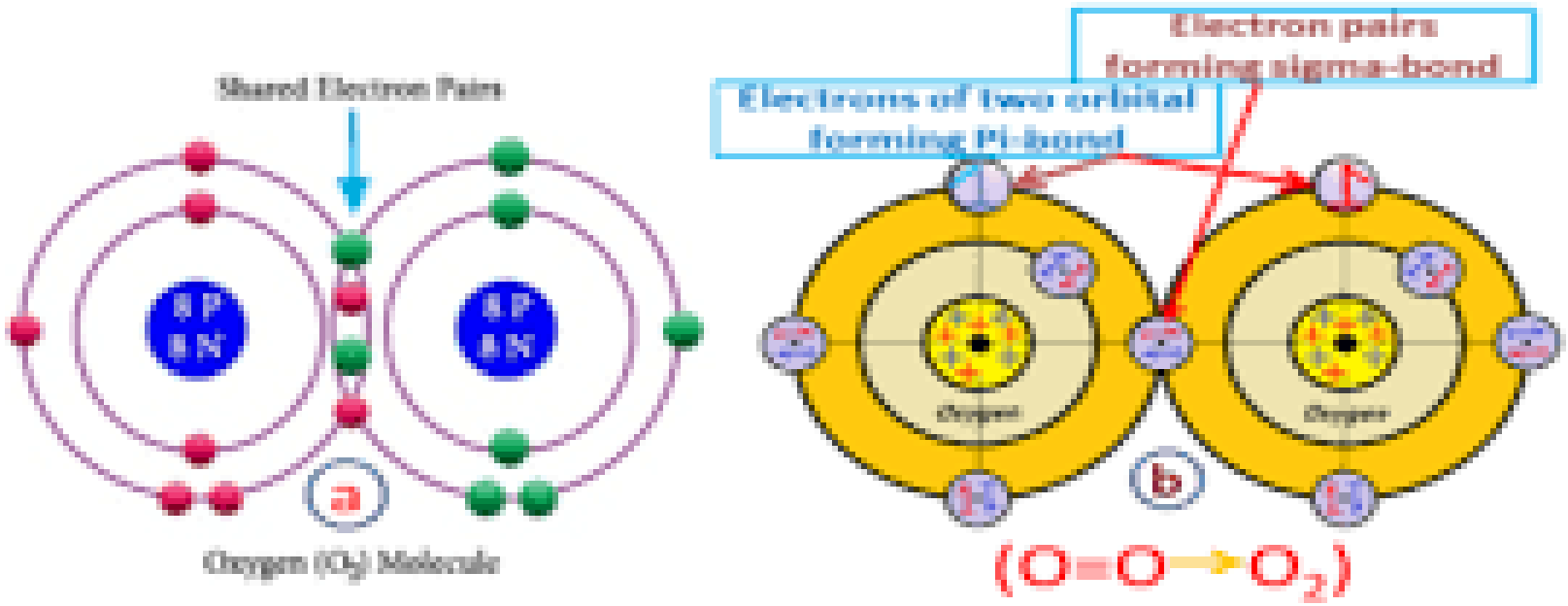
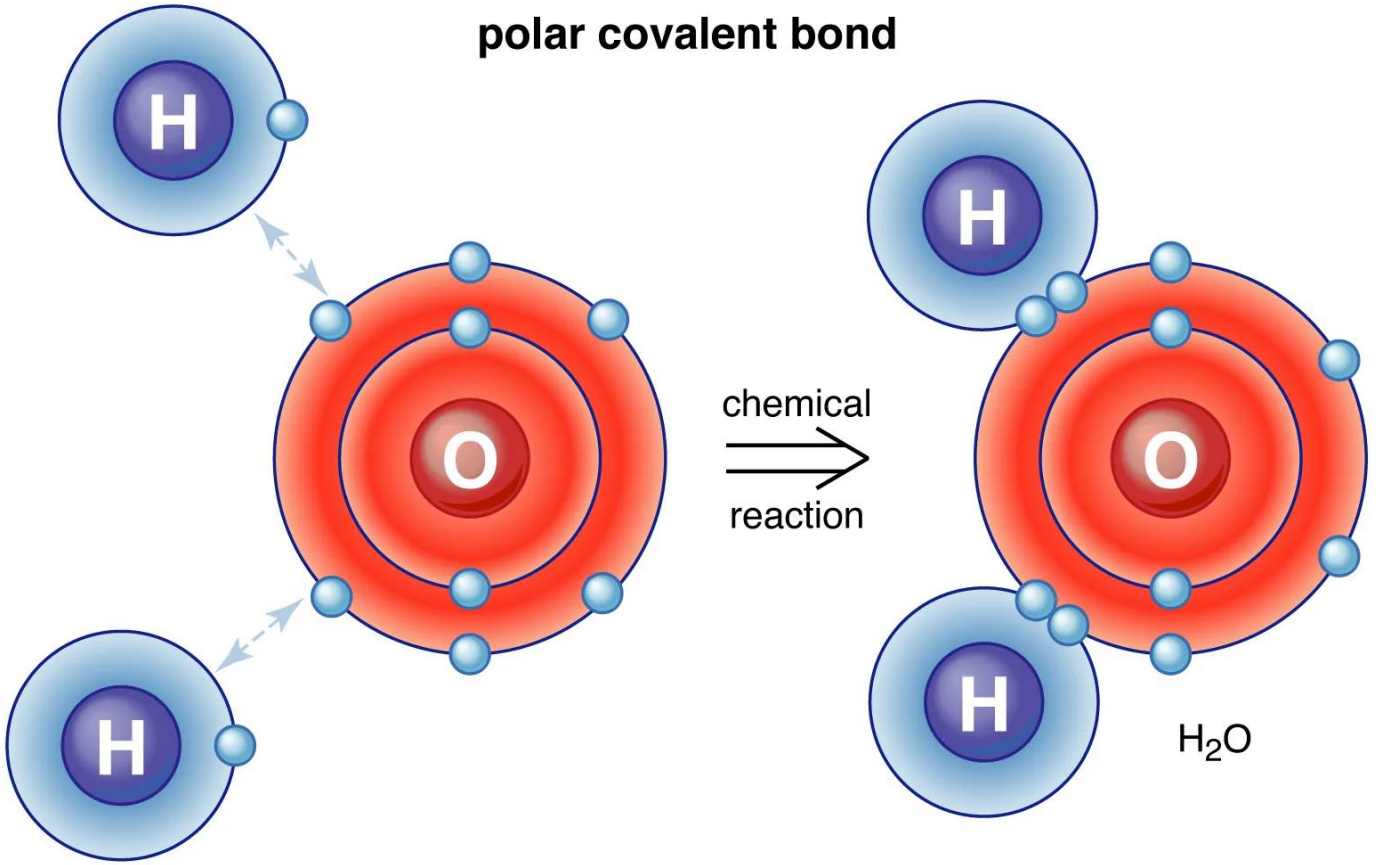


Figure 8: (a) Present molecular orbital structural showing two shared paired electrons in the same overlapped chamber with head to head overlapping results two sigma bonds; (b) New molecular orbital structure clearly showing two different Sigma and pi-bond proves the existing of double bond in Oxygen molecule also showing clear IMF line between two bonded Oxygen nucleus.

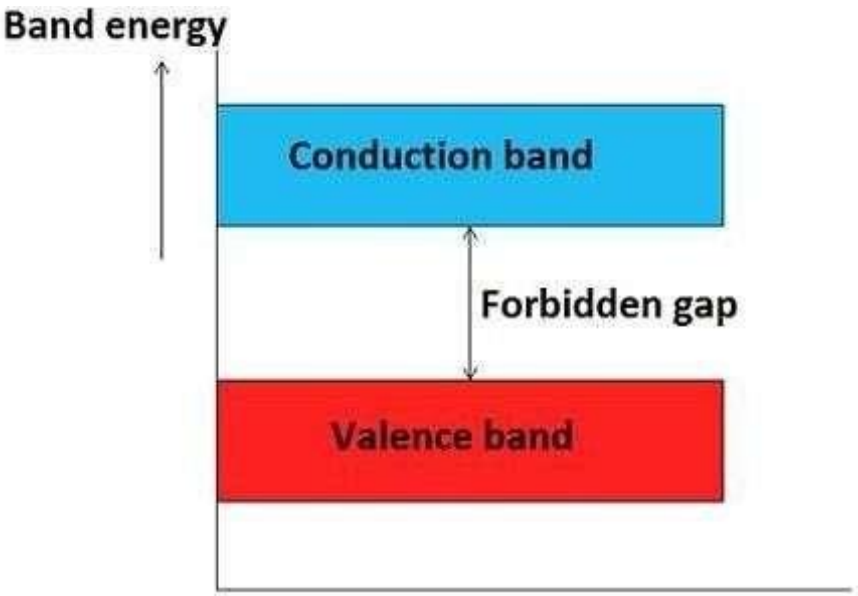
Interactions

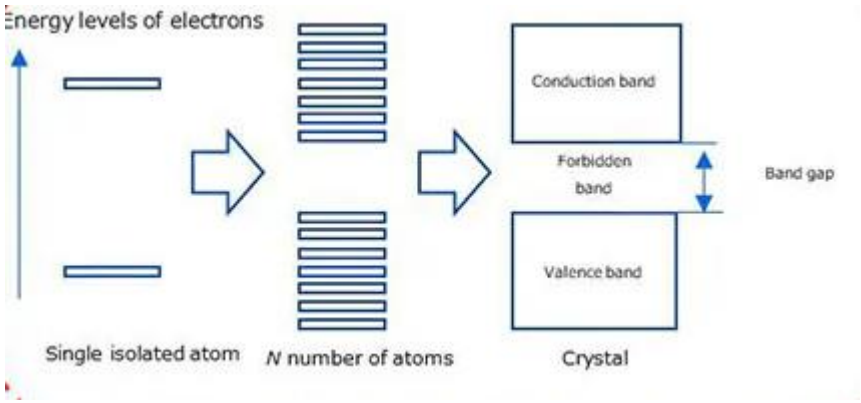


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Important Energy Bands in Solids:

- There are number of energy bands in solids but three of them are very important. These three energy bands are important to understand the behavior of solids. These energy bands are
- Valence band
- Conduction band
- Forbidden band or forbidden gap





• Valence band

The energy band which is formed by grouping the range of energy levels of the valence electrons or outermost orbit electrons is called as valence band. Valence band is present below the conduction band as shown in figure below. Electrons in the valence band have lower energy than the electrons in conduction band. The electrons present in the valence band are loosely bound to the nucleus of atom.

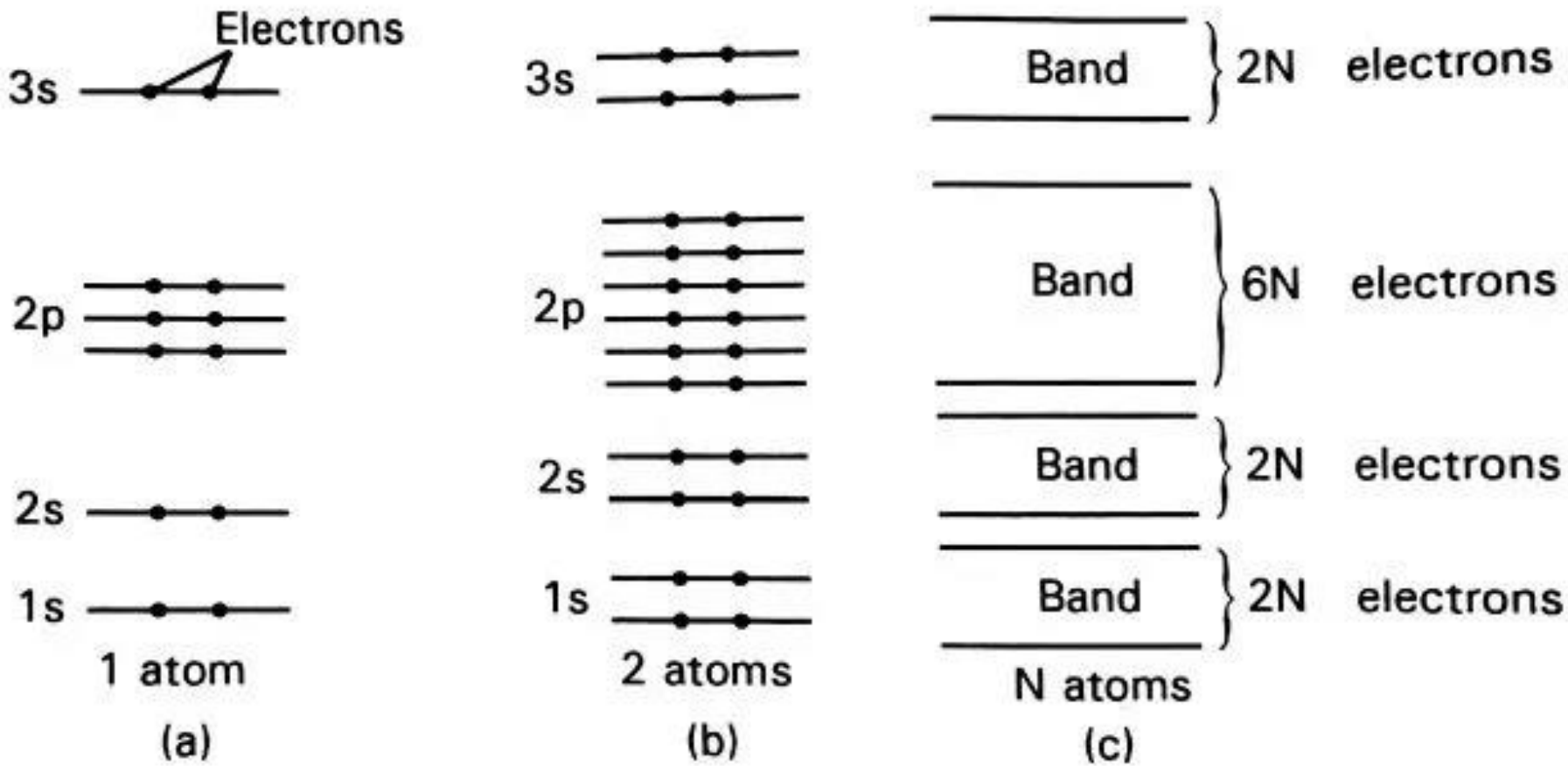
• Conduction band

The energy band which is formed by grouping the range of energy levels of the free electrons is called as conduction band. Generally, the conduction band is empty but when external energy is applied the electrons in the valence band jumps in to the conduction band and becomes free electrons. Electrons in the conduction band have higher energy than the electrons in valence band. The conduction band electrons are not bound to the nucleus of atom.

• Forbidden gap

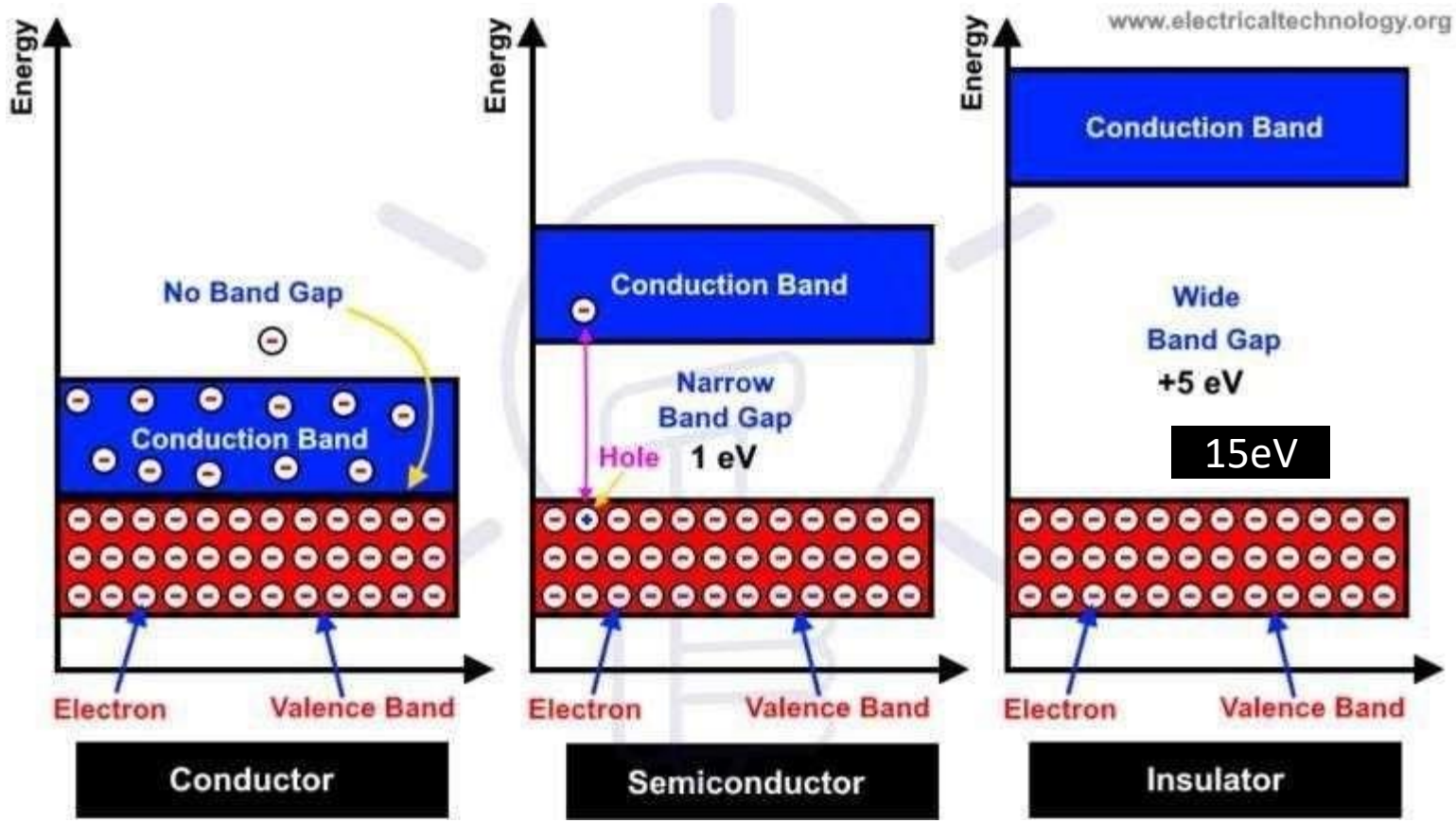
The energy gap which is present between the valence band and conduction band by separating these two energy bands is called as forbidden band or forbidden gap.

Energy Bands



The energy levels broaden into energy bands

Classification of Conductors, Semiconductors and Insulators on the basis of Energy Band Diagram:-



1.Conductors:-

In conductors , conduction band is overlapped with valence band. So , large number of electrons can jump from valence band to conduction band with the application of small potential difference. The energy band diagram is shown in the figure above.

2. Semiconductors:-

In these materials , valence band is full of electrons while conduction band is empty. In this case, energy gap is very small (1 eV).With the application of small potential difference, electrons can jump from valence band to conduction band. The semiconductor behaves as an insulator at room temperature or low temperature. As the temperature increases, more electrons pass through valence band to conduction band resulting conductivity increases. Energy band diagram for semiconductor shown above.

Insulators:-

In insulators , conduction band remain is empty, while valence band has many electrons. The forbidden energy gap between valence band and conduction band is quite large (15 eV). So it is not easy to conduct electric current through them. The electrons of the valance band require very high electric field in order to cross it and move in the conduction band. The electrons of the valence band do not have enough energy to cross and move towards conduction band therefore the conductivity of the insulator is almost zero.

comparison between conductors, insulators, and semi-conductors

S.No	Conductor	Insulator	Semi-conductor
1	The materials which allow electrical current through them very easily are known as Conductors.	The materials which does not allow electrical current through them are known as Insulators.	The materials which allow electrical current partially through them are known as Semi-conductors.
2	There are more free electrons to jump from one atom to another.	There are no free electrons to jump from one atom to another.	There are few free electrons to jump from one atom to another.
3	Good conductors of electricity.	Bad conductors of electricity.	Partial conductors of electricity.
4	The resistivity lies between 10^{-8} to 10^{-6} ohm-m.	The resistivity lies between 10^{12} to 10^{18} ohm-m.	The resistivity lies between 10^{0} to 10^{2} ohm-m.
5	Electrical conductivity is very high.	Electrical conductivity is negligible.	Electrical conductivity is very less.
6	Number of valence electrons are less than four.	Number of valence electrons are more than four.	Number of valence electrons are equal to four.
7	No forbidden gap between valence band and conduction band (i.e., $E_G = 0$ ev)	Wider forbidden gap between valence band and conduction band (i.e., $E_G > 5$ ev)	Narrow forbidden gap between valence band and conduction band (i.e., $E_G > 5$ ev)
8	Very less energy is required for electrons to pass from valence band to conduction band.	Very high energy is required for electrons to pass from valence band to conduction band.	High energy is required for electrons to pass from valence band to conduction band.
9	They have positive temperature co-efficient of resistance	They have negative temperature co-efficient of resistance	They have negative temperature co-efficient of resistance
10	Examples are Copper, Aluminium, Tungsten, Silver, etc.	Examples are Paper, Wood, Glass, Plastics, etc.	Examples are Germanium, Silicon, Selenium, Carbon, etc.

Conducting materials – metals – low resistivity materials :

Material having low resistivity or high conductivity are very useful for manufacturing electrical engineering machines or equipment's. These material used as conductors for all kind of winding required in electrical machines, apparatus and devices. These material are also used as conductor in transmission and distribution of electrical energy.

Required Properties in Low Resistivity or High Conductivity Conducting Material

Highest possible conductivity (ideally zero).
Least possible temperature coefficient of resistance (ideally zero).
High melting point.
High mechanical strength.
High ductility, so that can be drawn in the form of wire easily.
High corrosion resistance (free from oxidation).
Solder ability, so that can be soldered easily to join the conductors.
Low cost.
Long life or durable.
High flexibility.

The above required properties varies with the purpose for which material is being used. Any impurity whether metallic or non metallic increase the resistivity of metals. Even an impurity of low resistivity will increase the resistivity of metal. The reason behind this is that the addition of slight impurity creates imperfections in the crystal lattice which disturb the flow of electrons through metals.

Some of low resistivity or high conductivity materials and their resistivity with temperature coefficient of resistance are given in table below:-

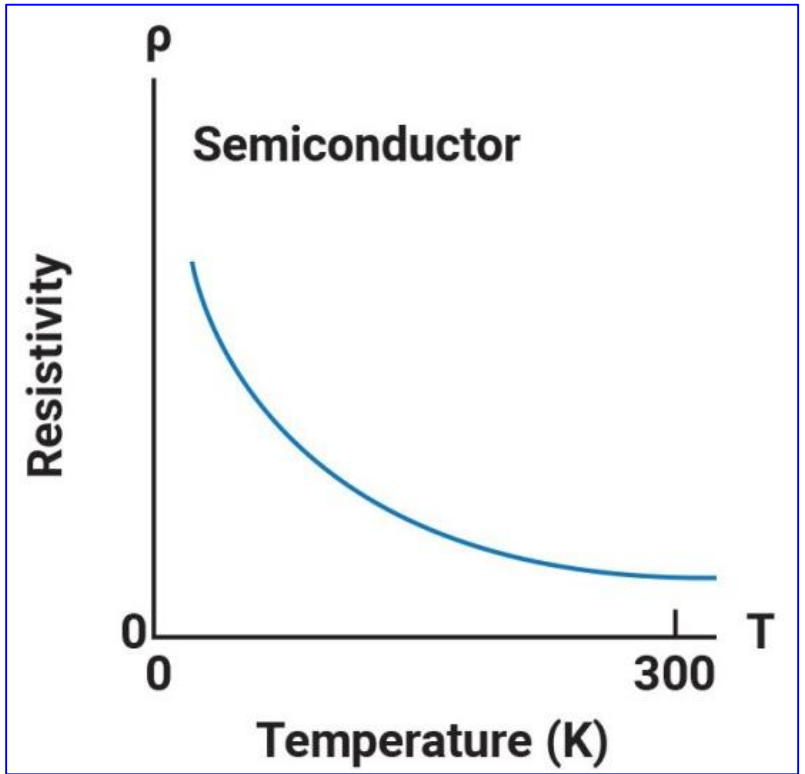
Sl No.	Metals	Resistivity (μΩ -cm)	Temperature Coefficients of Resistance (/ ^{0}C)
1	Silver	1.58	0.0038
2	Copper	1.68	0.00386
3	Gold	2.21	0.0034
4	Aluminium	2.65	0.00429

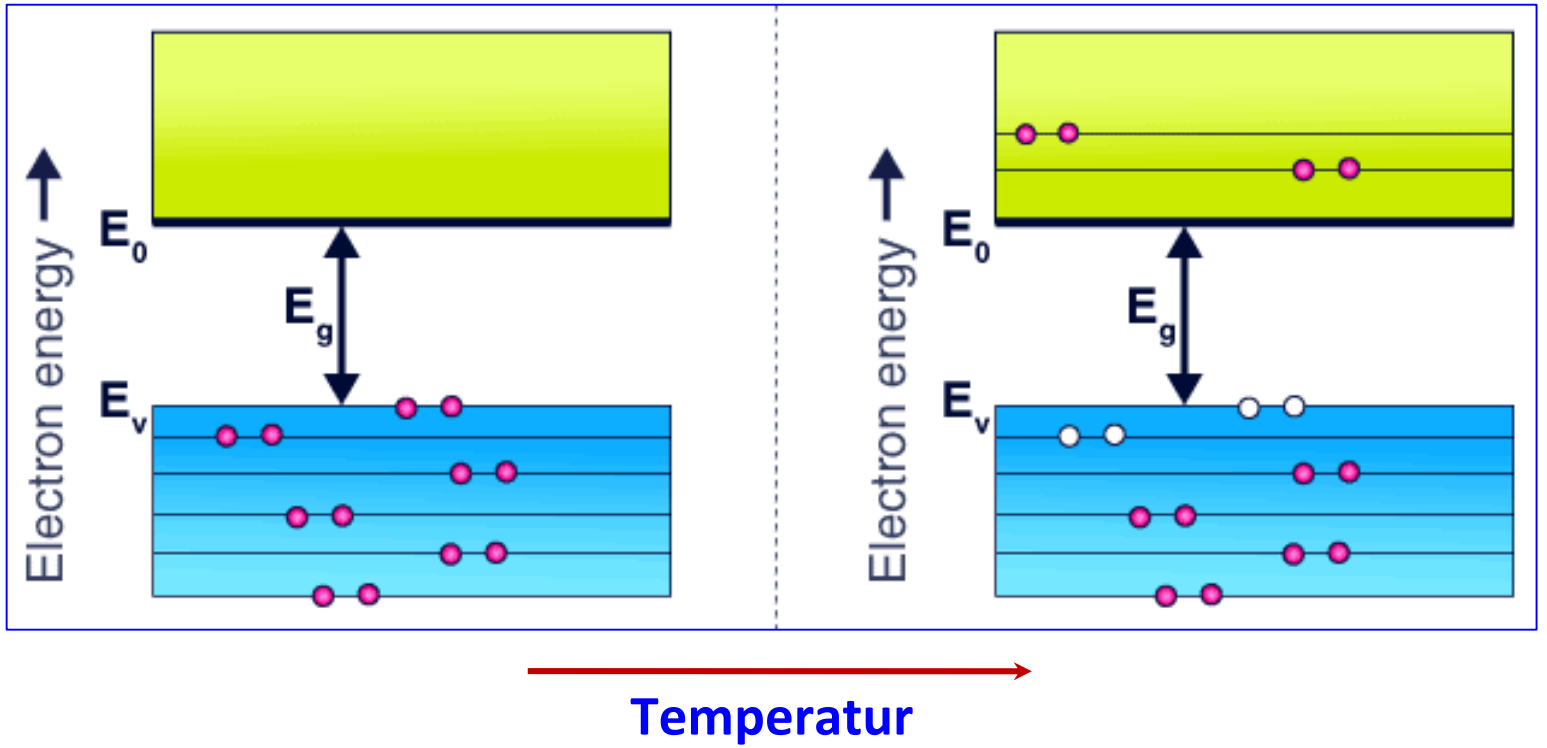
Semiconductors

- A semiconductor is a substance, usually a solid chemical element or compound that can conduct electricity under some conditions but not others, making it a good medium for the control of electrical current.
- It has almost filled valence band, empty conduction band and very narrow energy gap i.e., of the order of 1 eV. Energy gap of Silicon (Si) and Germanium (Ge) are 1.0 and 0.7 eV respectively. Consequently Si and Ge are semiconductors.

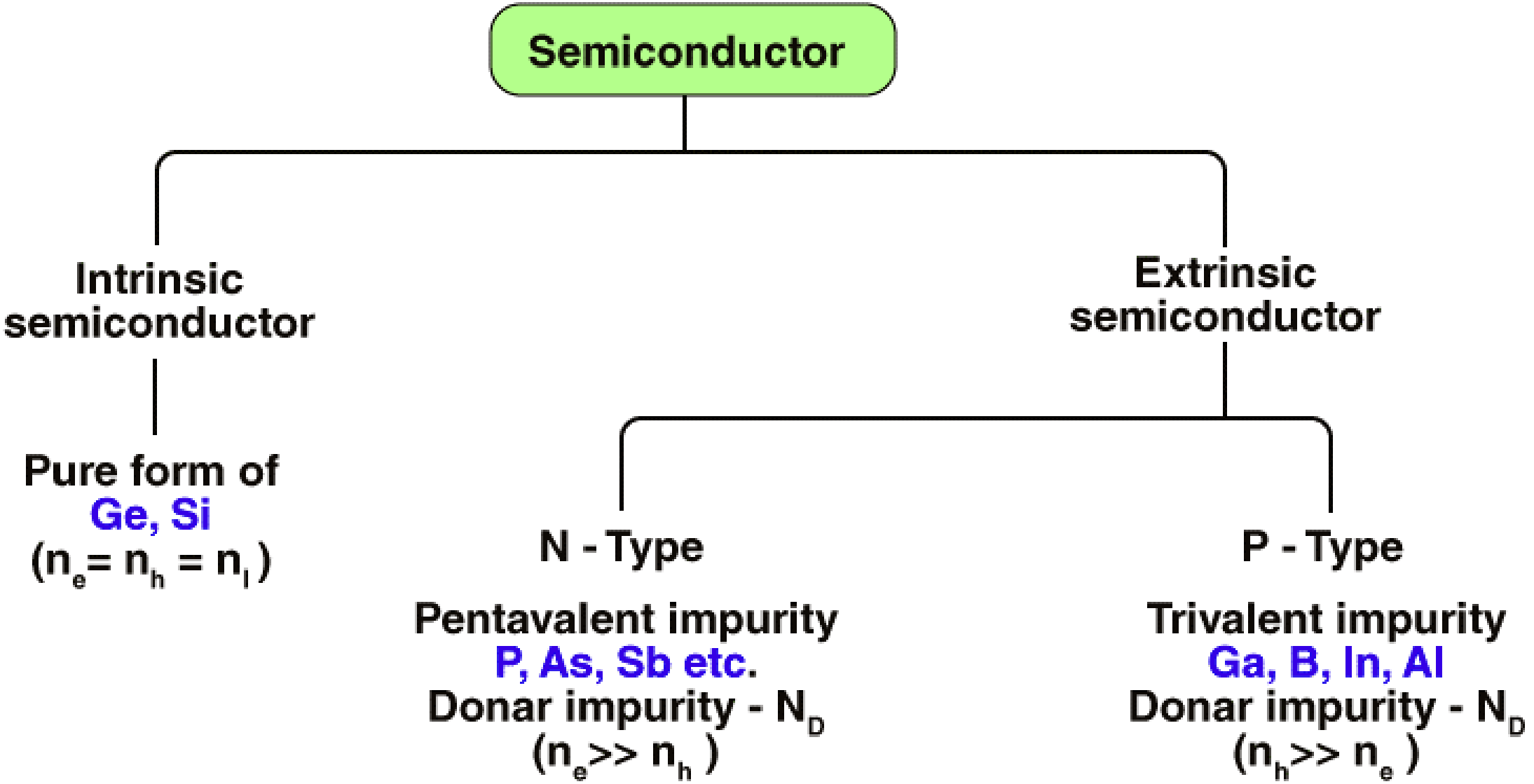
Effect of temperature on conductivity of semiconductors:

- At 0 K electrons freeze at valence band and hence all semiconductors are insulators.
- Electrical conductivity of a semiconductor material increases with increasing temperature as resistivity decreases.
- At higher temperature transition from the valence band to the conduction band gets facilitated ⇒ higher conductivity or lower resistivity.



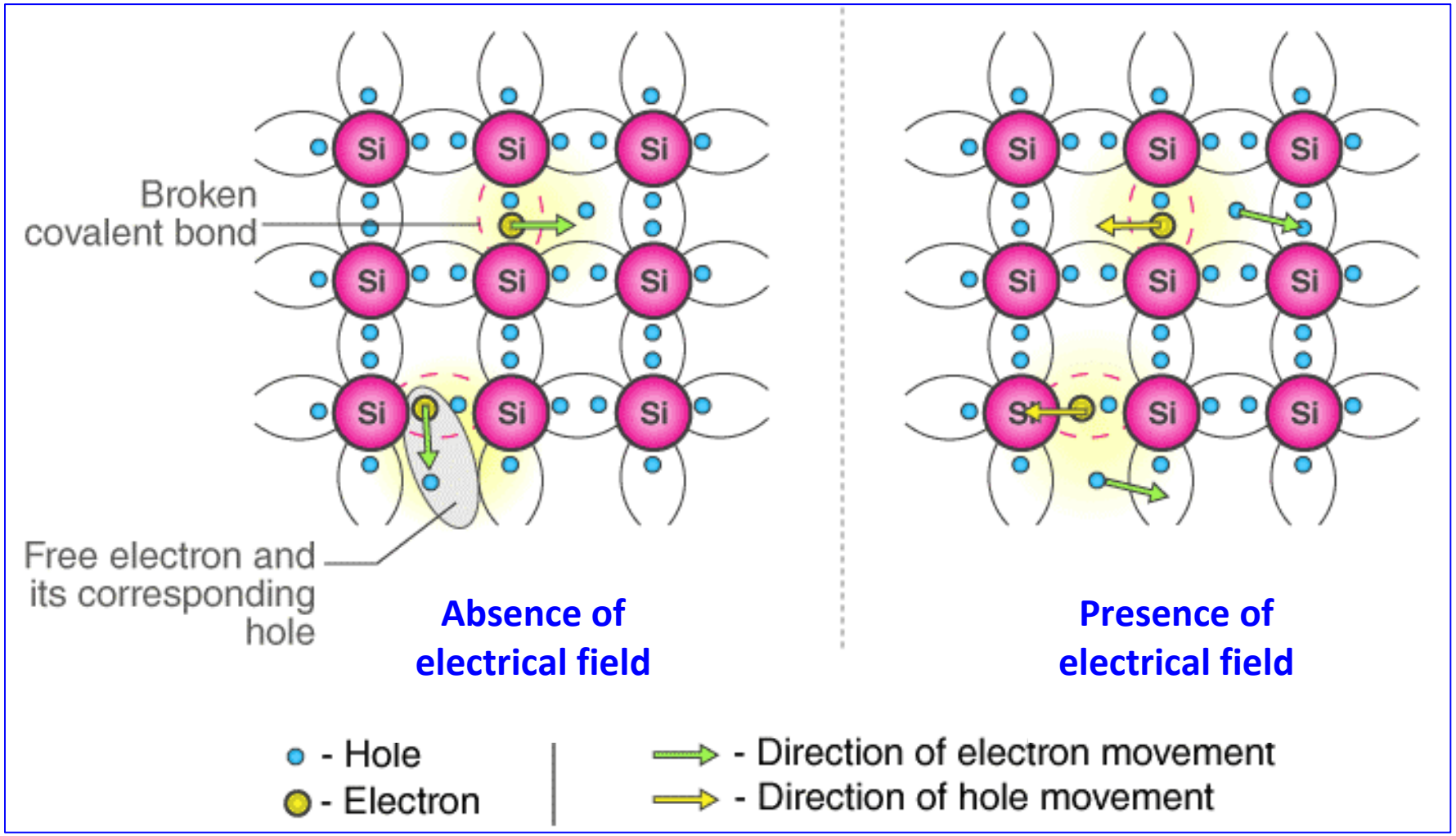


Types of


Semiconductor
Intrinsic semiconductor
Pure form of
Ge, Si
($n_e = n_h = n_i$)
Extrinsic semiconductor
N - Type
Pentavalent impurity
P, As, Sb etc.
Donar impurity - N_D
($n_e >> n_h$)
P - Type
Trivalent impurity
Ga, B, In, Al
Donar impurity - N_D
($n_h >> n_e$)

Intrinsic Semiconductor

- Intrinsic semiconductor material is chemically very pure and possesses poor conductivity.
- It has equal numbers of negative carriers (electrons) and positive carriers (holes)

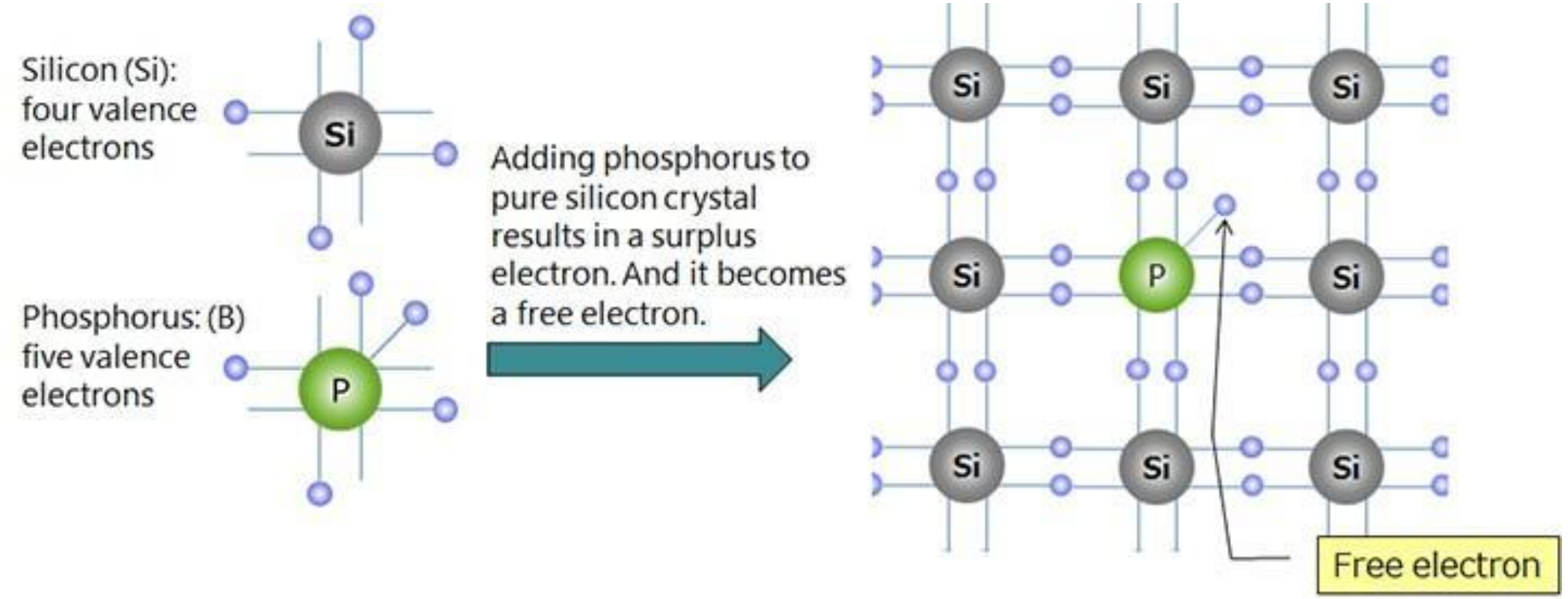


Extrinsic Semiconductor.

- **Extrinsic semiconductor is an improved intrinsic semiconductor with a small amount of impurities added by a process, known as doping, which alters the electrical properties of the semiconductor and improves its conductivity.**
- **Introducing impurities into the semiconductor materials (doping process) can control their conductivity.**
- **Doping process produces two groups of semiconductors:**
  - **The negative charge conductor (*n*-type)**
  - **the positive charge conductor (*p*-type)**
- **Semiconductors are available as either elements or compounds.**
- **Silicon and Germanium are the most common elemental semiconductors.**
- **Compound Semiconductors include InSb, InAs, GaP, GaSb, GaAs, SiC, and GaN.**

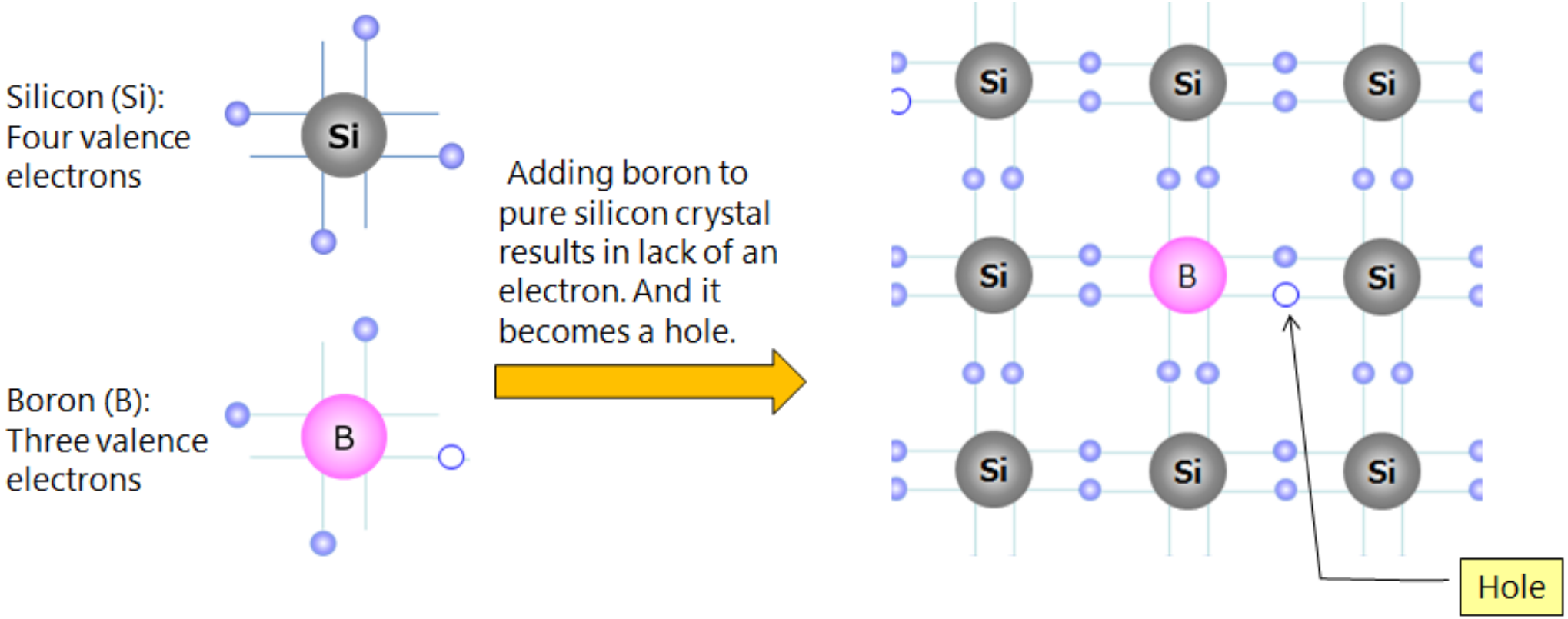
n-type Semiconductor

- **An n-type semiconductor is an intrinsic semiconductor doped with pentavalent impurity, such as, P, As, Sb, etc.**
- **If a small amount of phosphorus is added to a pure silicon crystal, one of the valence electrons of phosphorus becomes free to move around (free electron) as a surplus electron.**



p-type Semiconductor

- An p-type semiconductor is an intrinsic semiconductor doped with trivalent impurity, such as, B, Al, In, etc.
- If a small amount of boron is doped to a single crystal of silicon, valence electrons will be insufficient at one position to bond silicon and boron, resulting in holes that lack electrons.



comparisons between N and P type semiconductors

S.No	N-Type Semiconductor	P-Type Semiconductor
1	Pentavalent impurity is added to pure semiconductor to form N-Type semiconductor.	Trivalent impurity is added to pure semiconductor to form P-Type semiconductor.
2	The impurity atom has one extra electron i.e., five.	The impurity atom has one short fall electron i.e., three.
3	The doping agent is called donor.	The doping agent is called acceptor.
4	Large number electrons are produced by doping.	Large number holes are produced by doping.
5	In this electrons are the majority carriers and holes are minority carriers.	In this holes are the majority carriers and electrons are minority carriers.
6	Current conduction is predominantly by elctrons.	Current conduction is predominantly by holes.
7	The impurity atom becomes positively ionized by donating an electron.	The impurity atom becomes negatively ionized by accepting an electron.
8	Examples of donors: Arsenic, Antimony, Bismuth, Phosphorous, etc.	Examples of acceptors; Boron, Gallium, Indium, Aluminium, etc.

- **Applications**

- Temperature sensors are made with semiconductor devices.
- They are used in 3D printing machines
- Used in microchips and self-driving cars
- Used in calculators, solar plates, computers and other electronic devices.

- Transistor and MOSFET(metal-oxide-semiconductor field-effect transistor) used as a switch in Electrical Circuits are manufactured using the semiconductors.

Insulators

In insulators , conduction band remain is empty, while valence band has many electrons. The forbidden energy gap between valence band and conduction band is quite large (15 eV). So it is not easy to conduct electric current through them. The electrons of the valance band require very high electric field in order to cross it and move in the conduction band. The electrons of the valance band do not have enough energy to cross and move towards conduction band therefore the conductivity of the insulator is almost zero.

These materials which have very high resistivity i.e. offers a very high resistance to the flow of electric current. Insulating materials play an important part in various electrical and electronic circuits. In domestic wiring insulating material protect us from shock and prevent leakage current.

Examples

Asbestos and asbestos products

Glass

Insulating resins and their products

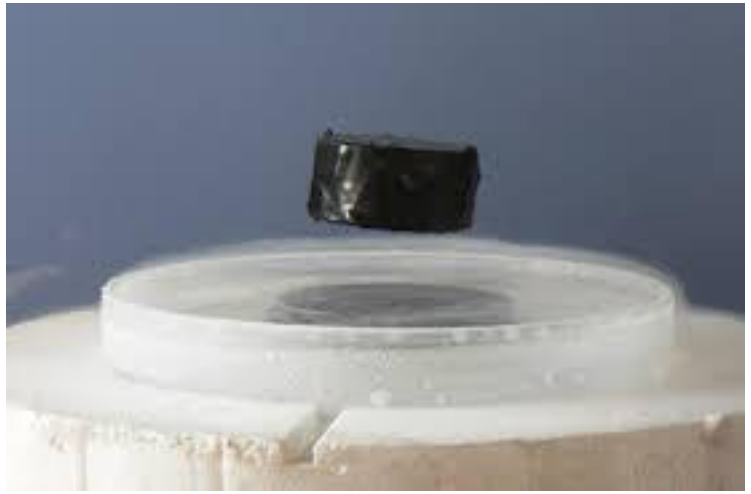
Laminates, adhesives, enamels

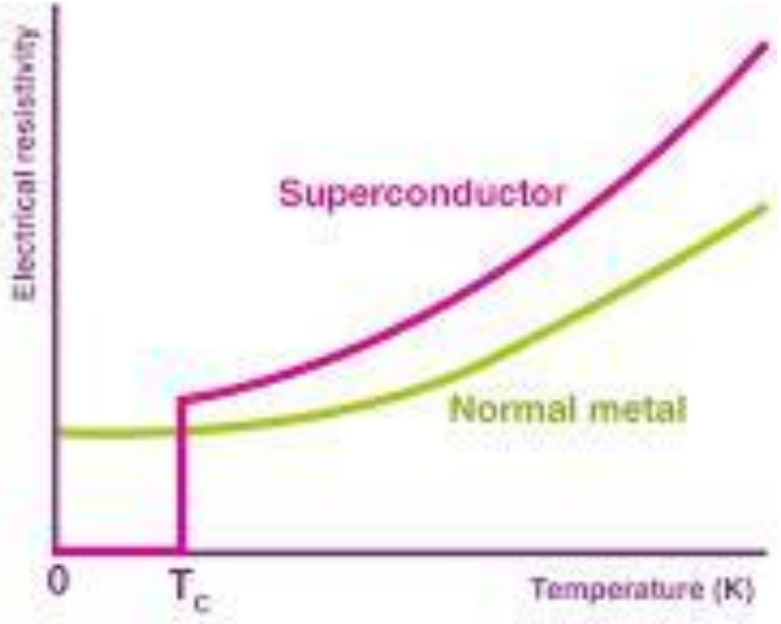
superconductors



Superconductivity was discovered by Dutch Physicist Heike Kamerlingh Onnes in 1911 in Leiden. He was awarded the Nobel Prize in Physics in 1913 for his low-temperature research. Some materials when they are cooled, below certain temperature their resistivity get abolished means they exhibit the infinite conductivity.

The temperature at which the metals change from normal conducting state to superconducting state, is called critical temperature/transition temperature. An example of superconductors, is Mercury. It becomes superconductor at 4k. In superconducting state the materials expel the magnetic field.



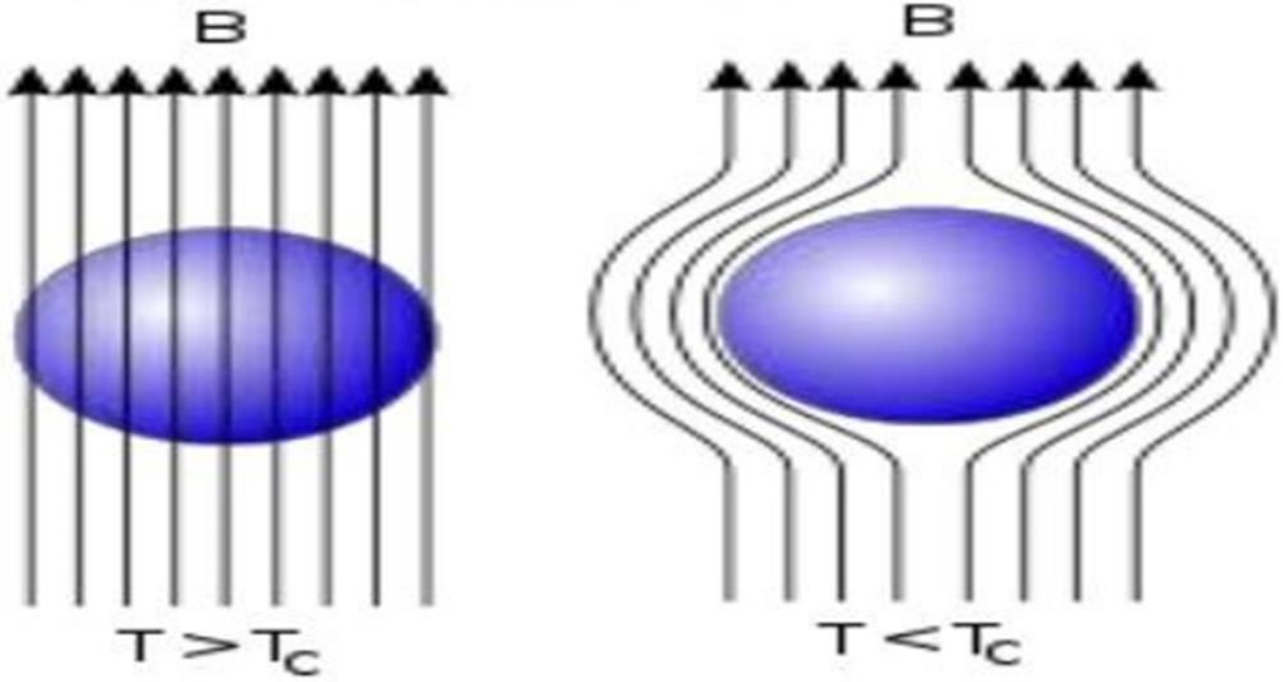


Meissner effect

Walther Meissner and Robert Ochsenfeld in 1933 by measuring the magnetic field distribution outside superconducting tin and lead samples.

A superconductor with little or no magnetic field within it is said to be in the Meissner state.

- The phenomenon of exclusion of magnetic flux or ejection of lines of magnetic induction from the interior of bulk superconductors, when they are cooled below the transition temperature is called Meissner's effect.



The Meissner Effect

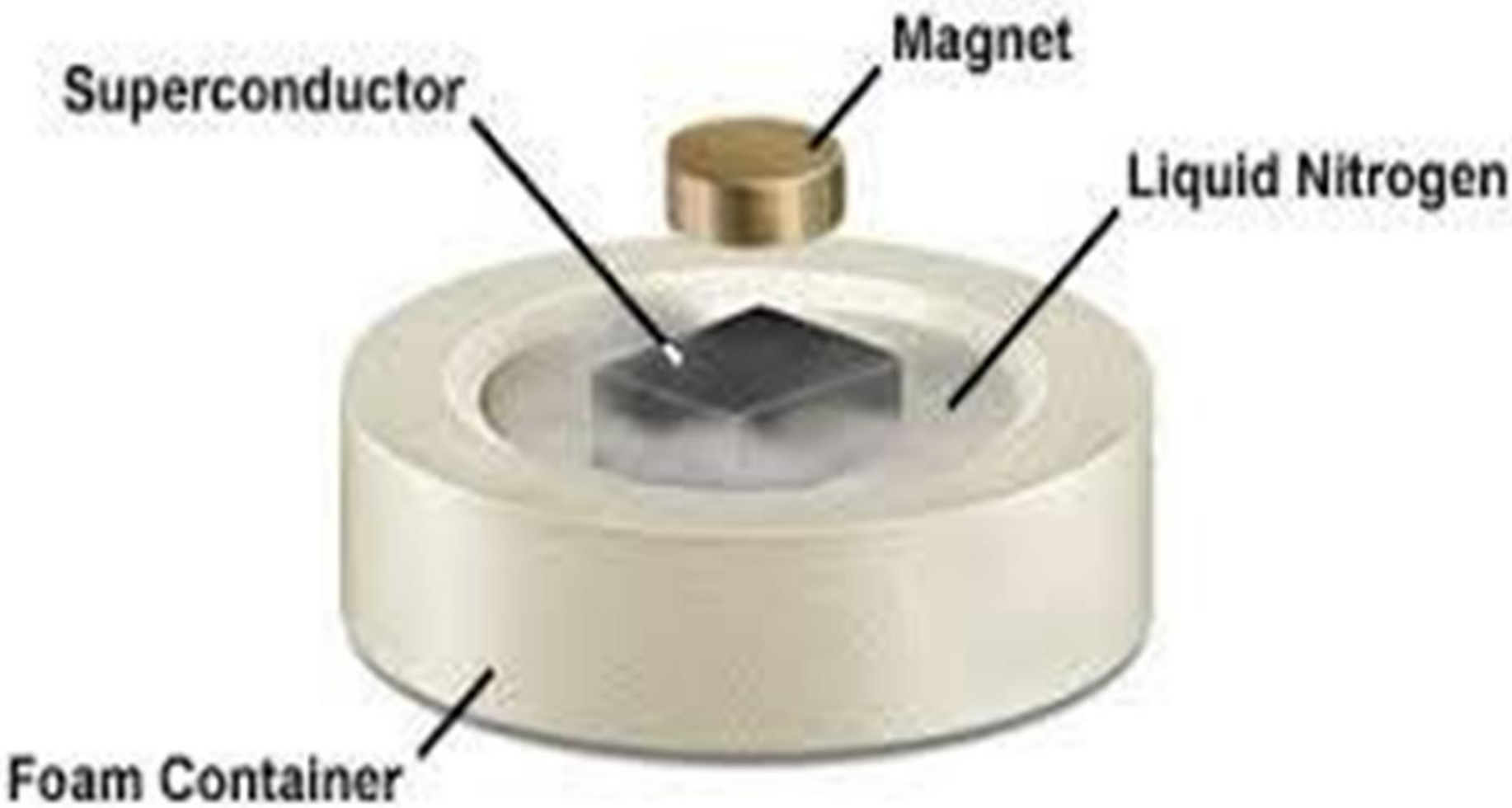
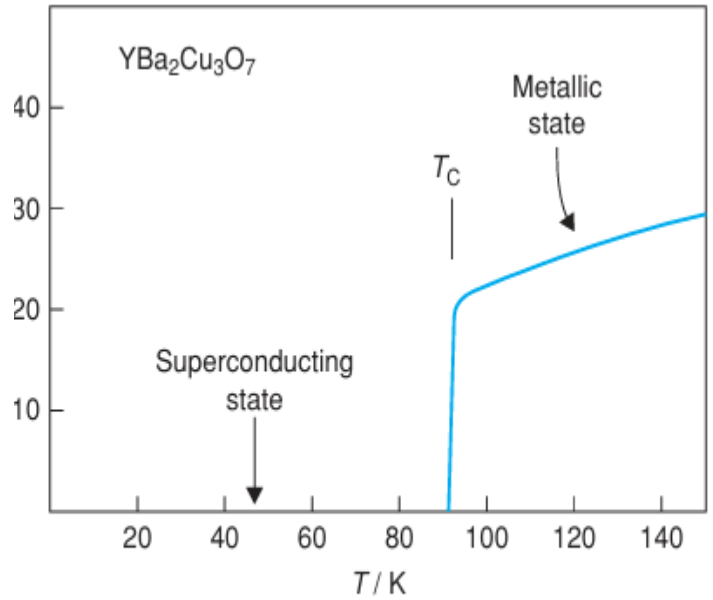


Table 8.3 ***Some superconducting materials***

Material	Substance	T_c/K	Substance	T_c/K
Metals	Hg	4	Nb	9.5
	V	5.4	Tc	7.8
	Pb	7.2		
Alloys	Nb_3Ge	23.2	V_3Si	17
	Nb_3Sn	18	V_3Ga	16.5
	Nb_3Al	17.5	NbN	16
Non-cuprate oxides	$LiTi_2O_4$	12	$BaPb_{0.75}Bi_{0.25}O_3$	13
			$Ba_{0.6}K_{0.4}BiO_3$	30
Cuprates	$La_{1.85}Sr_{0.15}CuO_4$	40	$Tl_2Ba_2CuO_6$	80
	$YBa_2Cu_3O_7$	93	$Tl_2Ba_2CaCu_2O_8$	105
	$Bi_2Sr_2Ca_2Cu_3O_{10}$ (BiSCCO)	110	$Tl_2Ba_2Ca_2Cu_3O_{10}$	125
	$HgBa_2Ca_2Cu_3O_{10}$	138[a]	$Tl_2Ba_2Ca_3Cu_4O_{12}$	115
	Sr_2CuO_4	1.5		
Organics	$(ET)_2Cu(NCS_2)$	11.4		
Others	YNi_2B_2C	15.5	MgB_2	39
	Fullerenes, e.g. Cs_2RbC_{60}	33	$LaO_{1-x}F_xFeAs$	28
	Chevrel phases, e.g. $HoMo_6S_8$	0.7–1.8	$SmO_{1-x}F_xFeAs$	55
	Sulfur: at 93 GPa	10	$ErRh_4B_4$	1.0–8.7
	at 160 GPa	17[b]		



ectrical resistance of $YBa_2Cu_3O_7$ as a function of temperatu

Application

Transportation:

- Maglev Trains: Superconducting magnets enable magnetic levitation, allowing trains to float above the tracks and travel at high speeds with minimal friction and energy loss.

Medical Imaging:

- Superconducting magnets in MRI (Magnetic Resonance Imaging) machines create strong, stable magnetic fields for high-resolution medical imaging.

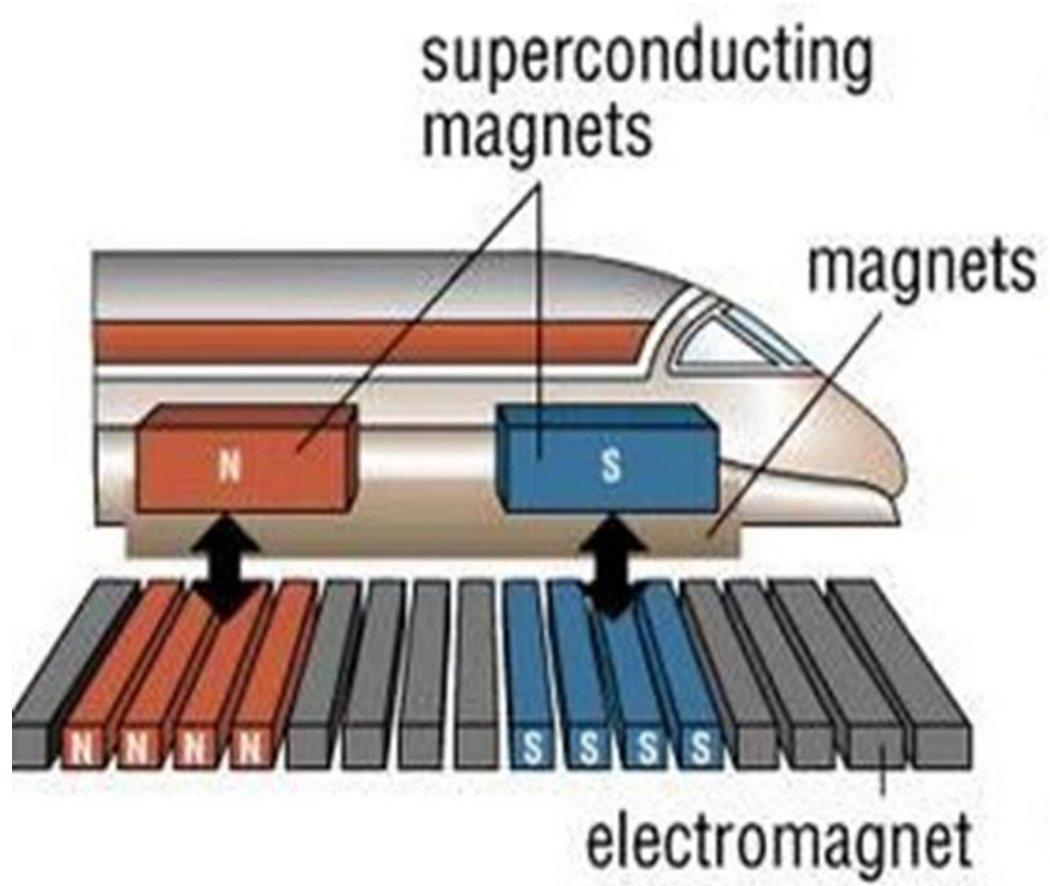
Power Transmission and Energy Storage:

- Superconducting cables can transmit electricity over long distances with minimal energy loss, offering significant efficiency gains.

Computing:

- Superconducting circuits are used in the development of quantum computers, offering potential for faster and more powerful computation



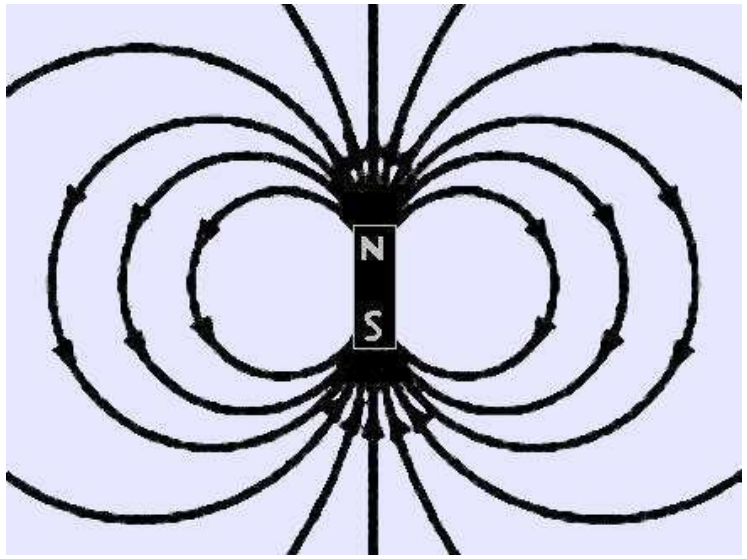

superconducting magnets
magnets
N
S
N N N N
S S S S
electromagnet

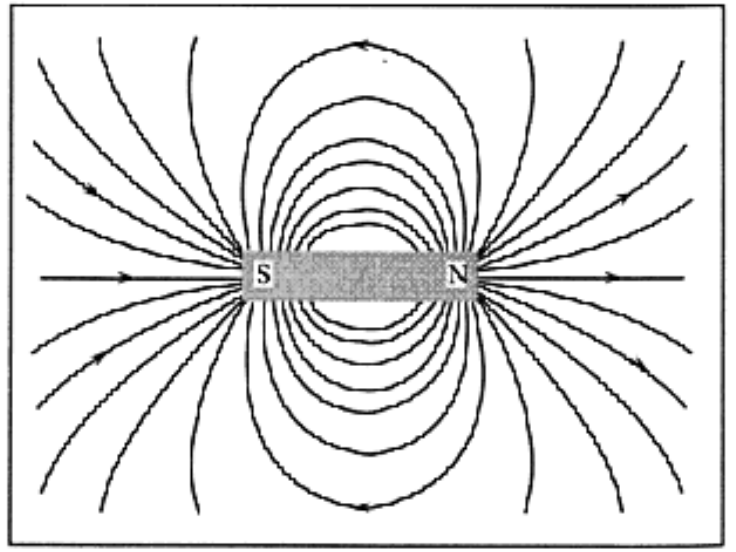
Magnetic Materials

, Magnetic: para, dia, ferro & anti ferro magnetism, applications in storage devices

MAGNETIC DIPOLE:

The ends of a magnet are where the magnetic effect is the strongest. These are called "poles." Each magnet has 2 poles – 1 north, 1 south.





Any two opposite poles separated by a finite distance constitute a magnetic dipole.

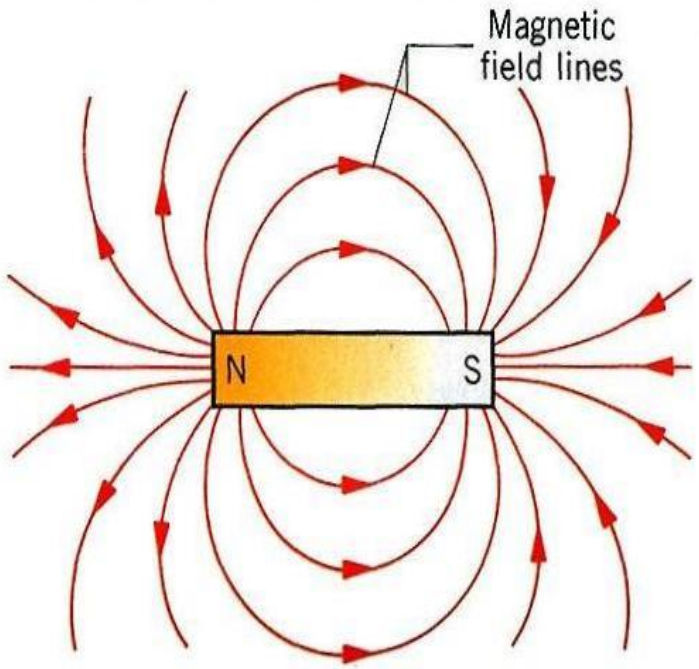


Michael Faraday

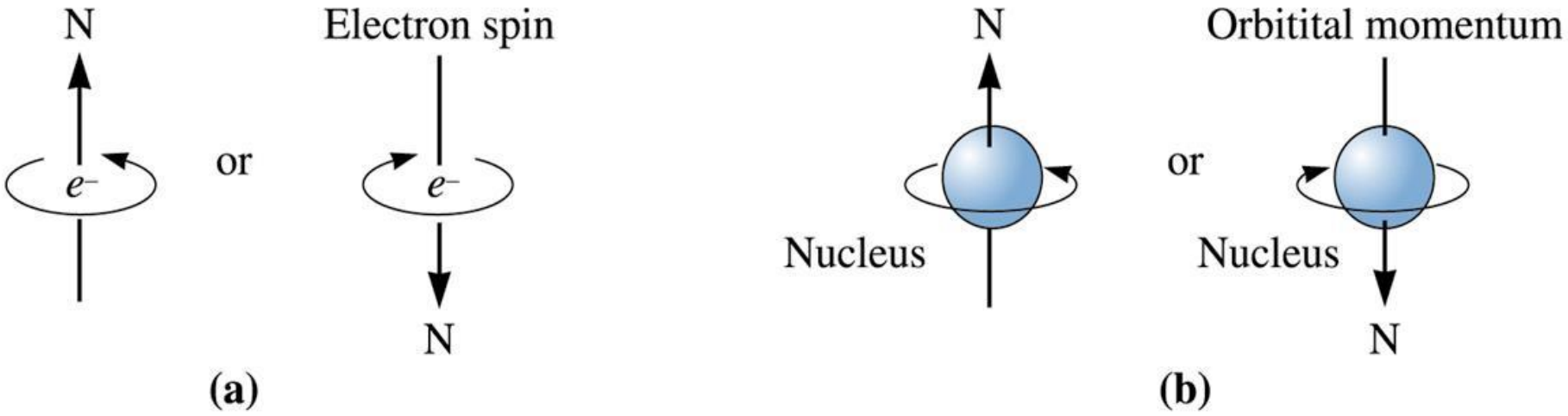
Response of materails to a magnetic field can be different and studied in magnetism.

Magnetic field:

The space surrounding the magnet upto which its influence felt is known as magnetic field.



- The magnetic behavior of materials can be traced to the structure of atoms.
- Bohr magneton - The strength of a magnetic moment of an electron (μ_B) due to electron spin.



Origin of magnetic dipoles: (a) The spin of the electron produces a magnetic field with a direction dependent on the quantum number m_s. (b) Electrons orbiting around the nucleus create a magnetic field around the atom.

Origin of magnetic moment

1. Orbital Motion - Orbital magnetic moment
2. Spin motion – Spin magnetic moment

- Magnetic moment (μ_{BM})
- $\mu_{eff} = \mu_{s.o} = 2\sqrt{S(S+1)} = \sqrt{n(n+2)}$ BM
- n number of un paired electrons
- s.o- Spin only value

- If there is a possibility for contribution from the orbital angular momentum,then
- **$\mu = \sqrt{L(L+1) + 4S(S+1)}$**

TABLE 19-1 ■ ***The electron spins in the 3d energy level in transition metals, with arrows indicating the direction of spin***

Metal	3*d*					4*s*
Sc	↑					↑↓
Ti	↑	↑				↑↓
V	↑	↑	↑			↑↓
Cr	↑	↑	↑	↑	↑	↑
Mn	↑	↑	↑	↑	↑	↑↓
Fe	↑↓	↑	↑	↑	↑	↑↓
Co	↑↓	↑↓	↑	↑	↑	↑↓
Ni	↑↓	↑↓	↑↓	↑	↑	↑↓
Cu	↑↓	↑↓	↑↓	↑↓	↑↓	↑

Magnetization, Permeability, and the Magnetic Field

- Magnetic permeability - The ratio between inductance and magnetic field. It is a measure of the ease with which magnetic flux lines can "flow" through a material.
- Magnetization (M) - The total magnetic moment per unit volume.
- Magnetic susceptibility (χ_m,)- The ratio between magnetization and the applied field. $\chi_m = M / H$

- H is magnetic field

Permeability: (μ)

The Magnetic induction B is proportional to the applied Magnetic field intensity H.

$$B \alpha H$$

$$B = \mu H$$

$$\mu = \frac{B}{H}$$

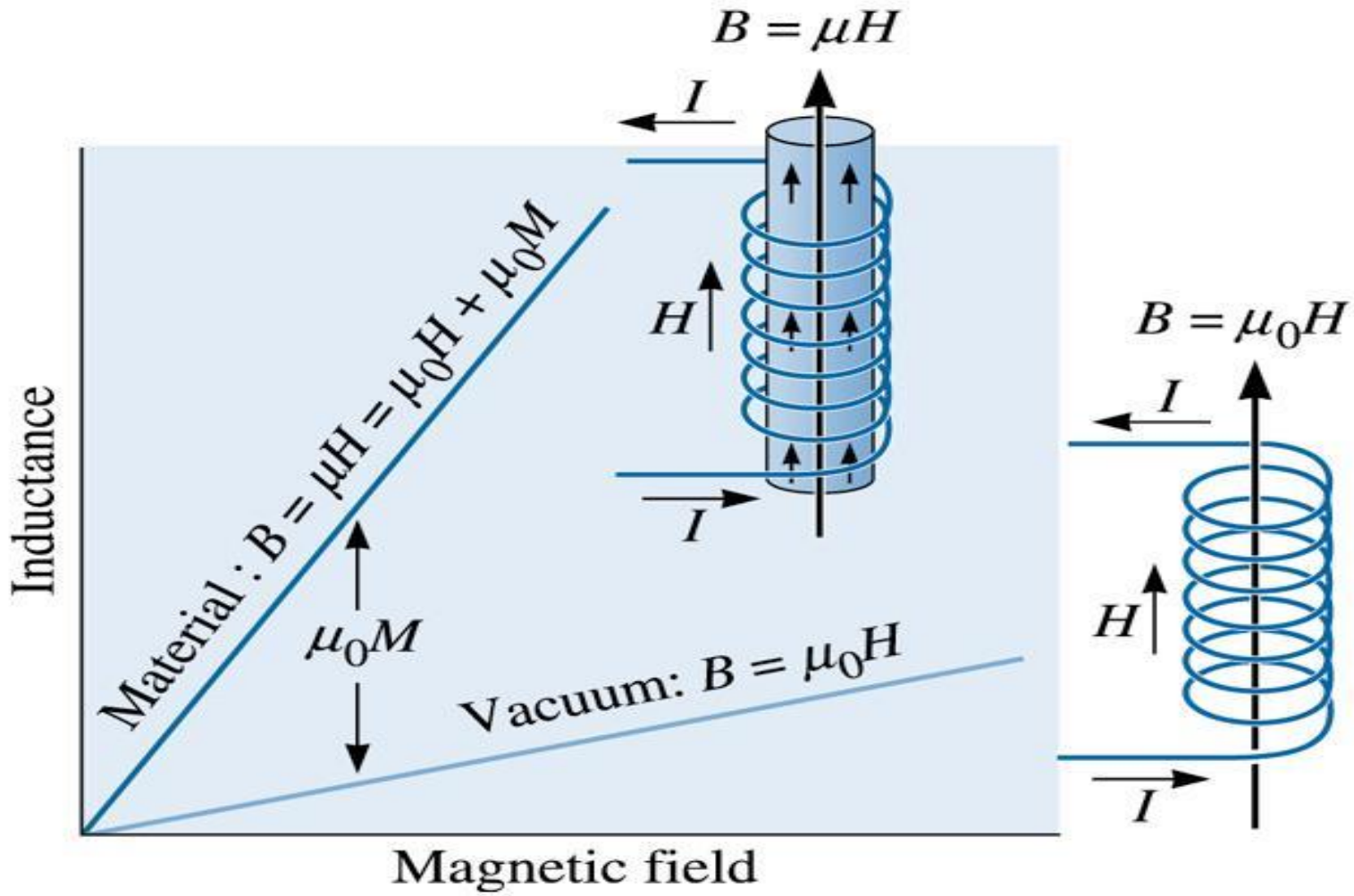
Where μ permeability of a medium

Relative permeability μ_r

The ratio of permeability of medium to the permeability of free space is called relative permeability μ_r of the solid.

$$\mu_r = \frac{\mu}{\mu_0}$$

$$\mu_r = \frac{\frac{B}{H}}{\frac{B_0}{H}} = \frac{B}{B_0}$$



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A current passing through a coil sets up a magnetic field *H* with a flux density *B*. The flux density is higher when a magnetic core is placed within the coil.

Magnetization

- Magnetization refers to the process of converting a non-magnetic material into a Magnetic material.
- The intensity of Magnetization is directly related to the applied field H.

$$M \alpha H$$

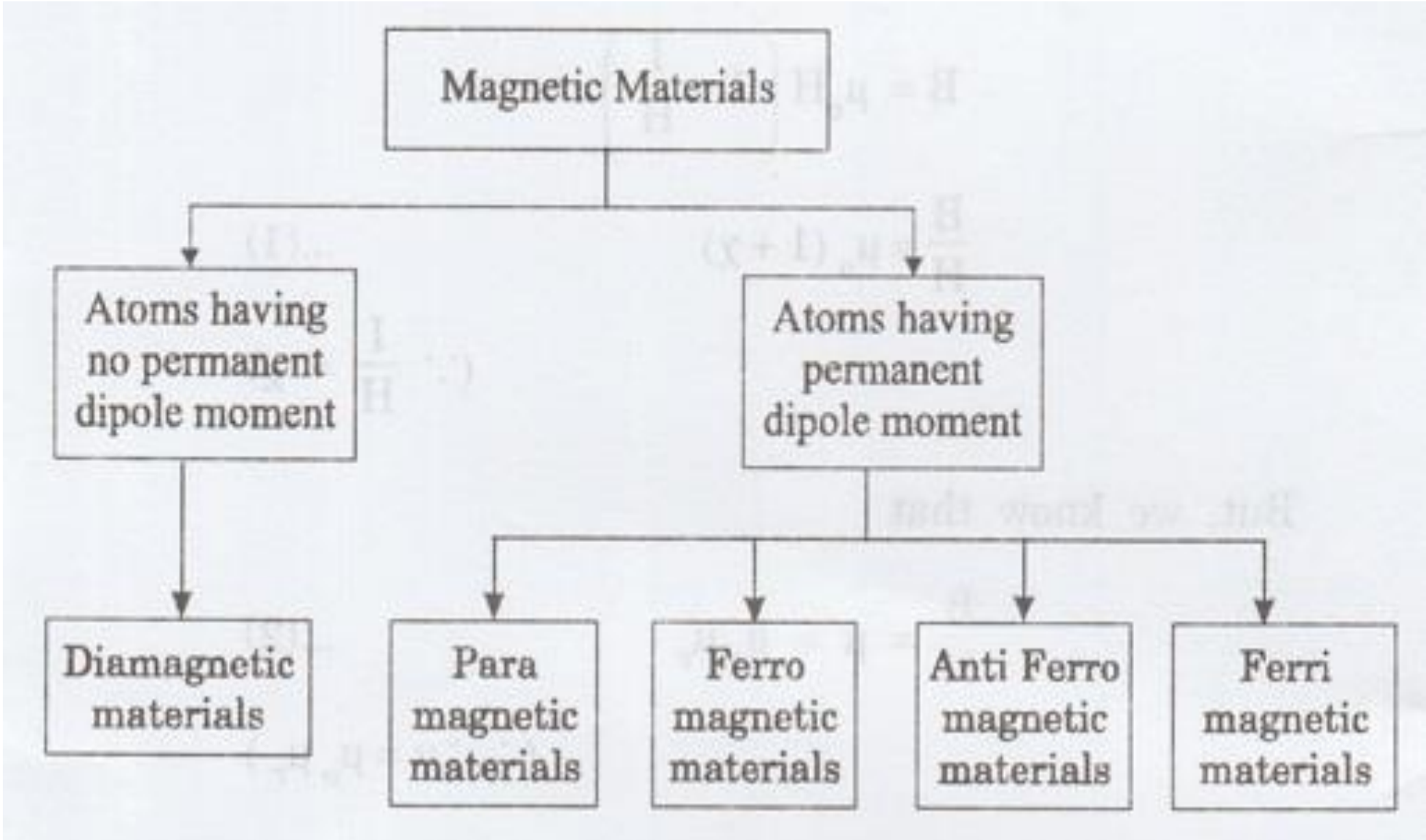
$$M = \chi_m H$$

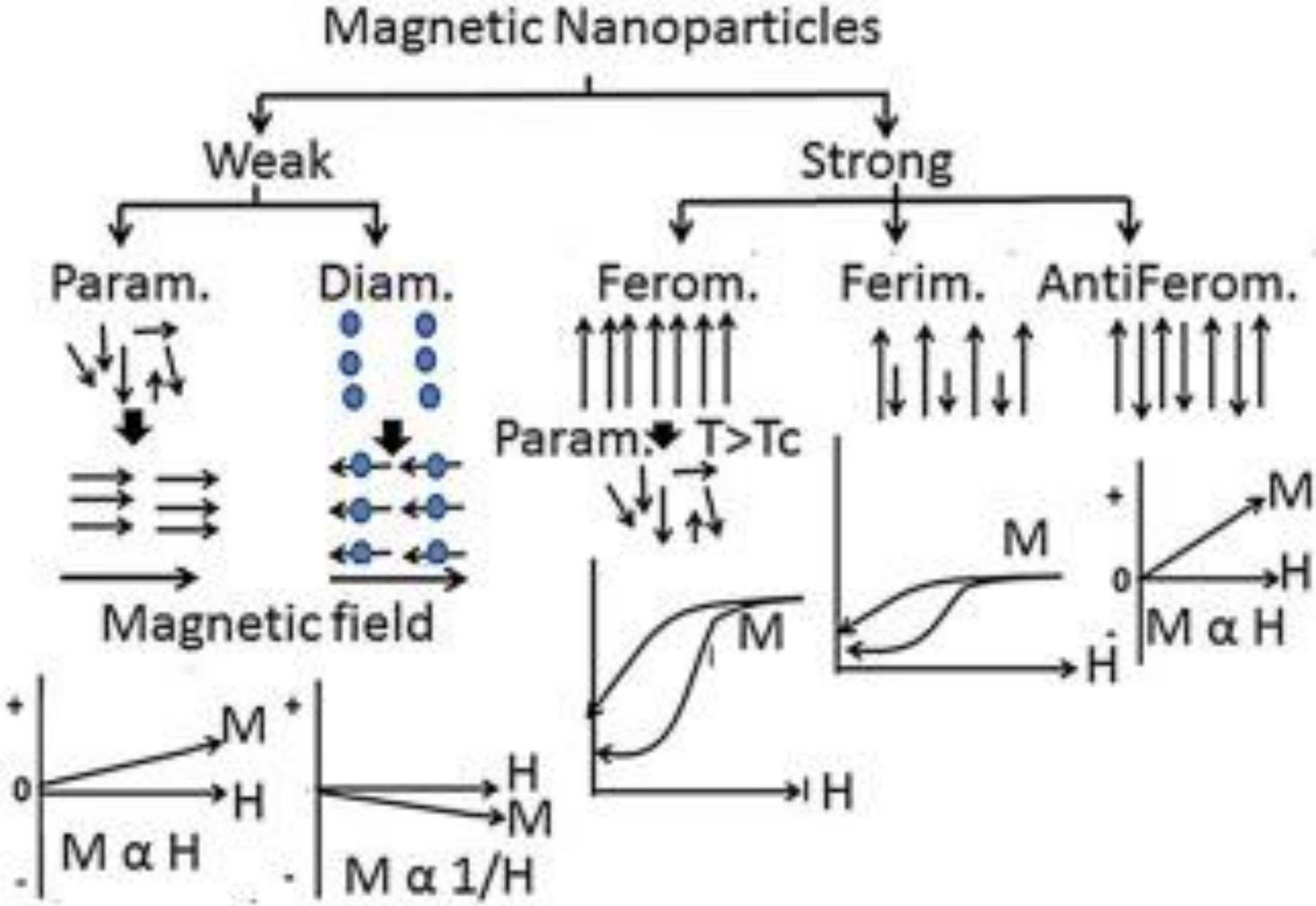
$$\text{magnetic susceptibility} \quad \chi_m = \frac{M}{H}$$

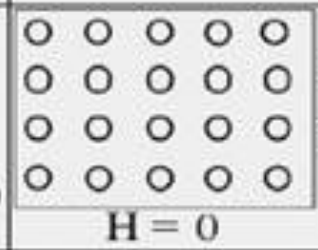
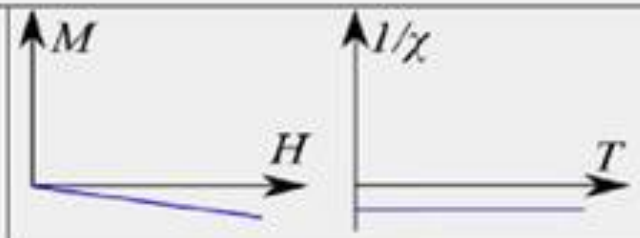
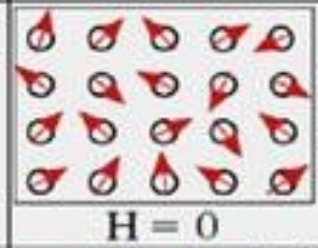
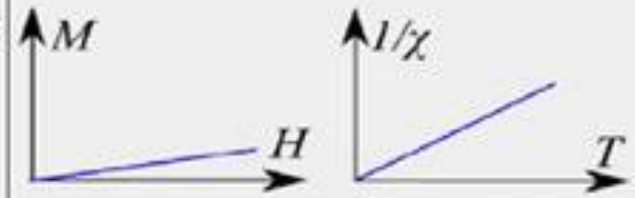
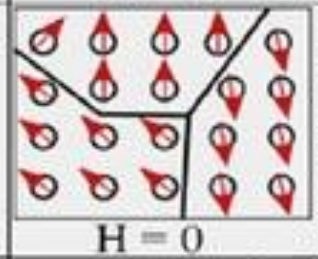
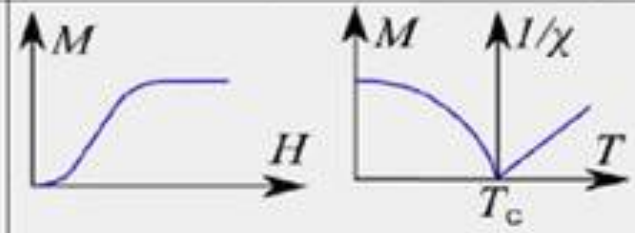
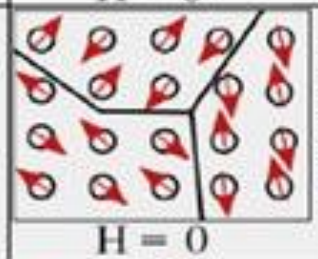
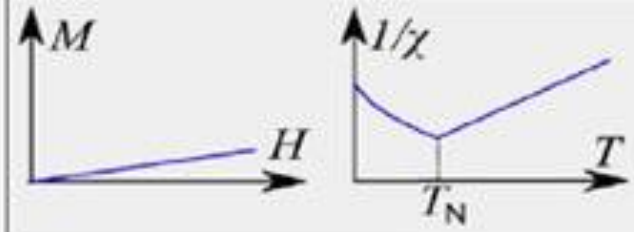
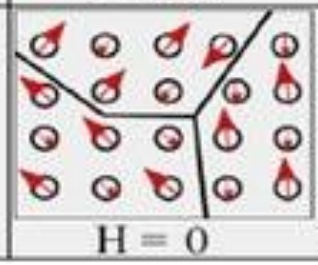
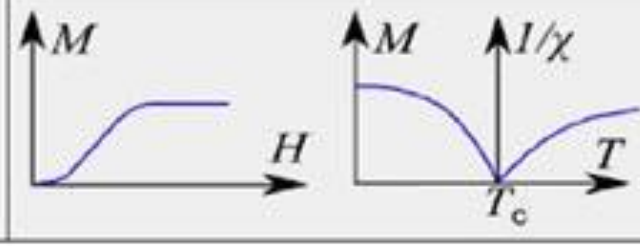
✓ Every substance is formed from an assembly of atoms which can be either non-magnetic or magnetic.

The magnetic materials are classified into five groups depending on their response to the magnetic field.

1. Diamagnetic Materials
2. Paramagnetic Materials
3. Ferromagnetic Materials
4. Antiferromagnetic Materials
5. Ferrimagnetic Materials


Magnetic Materials
Atoms having no permanent dipole moment
Atoms having permanent dipole moment
Diamagnetic materials
Para magnetic materials
Ferro magnetic materials
Anti Ferro magnetic materials
Ferri magnetic materials


Magnetic Nanoparticles
Weak
Strong
Param.
Diam.
Ferom.
Ferim.
AntiFerom.
Param.
T>Tc
Magnetic field
M
H
M α H
M α 1/H

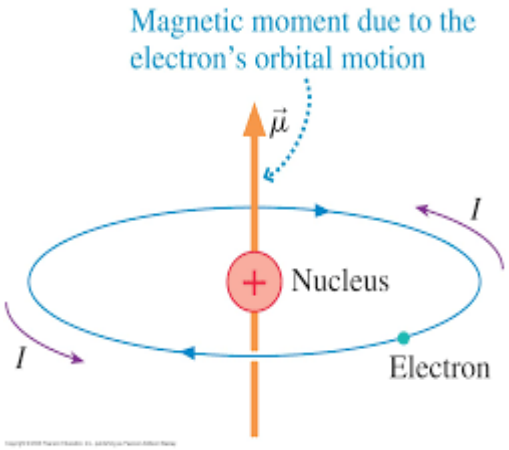
Magnetism	Examples	Magnetic behaviour		
Diamagnetism	Bi, Si, Cu, inert gases Susceptibility small and negative (-10^{-6} to -10^{-5})		Atoms have no magnetic moments.	
Paramagnetism	Al, O_2, MnBi Susceptibilty small and positive (10^{-5} to 10^{-3})		Atoms have randomly oriented magnetic moments.	
Ferromagnetism	Fe, Ni, Co, Gd Susceptibility large (generally > 100)		Atoms are organized in domains which have parallel aligned magnetic moments.	
Antiferromagnetism	Cr, MnO, FeO Susceptibilty small and positive (10^{-5} to 10^{-3})		Atoms are organized in domains which have antiparallel aligned moments.	
Ferrimagnetism	Fe_3O_4, $MnFe_2O_4$, $NiFe_2O_4$ Susceptibility large (generally > 100)		Atoms are organized in domains which have a mixture of unequal antiparallel aligned moments.	

DIAMAGNETISM

✓ Diamagnetism characterizes the substances that have only non-magnetic atoms. These don't have unpaired electrons.

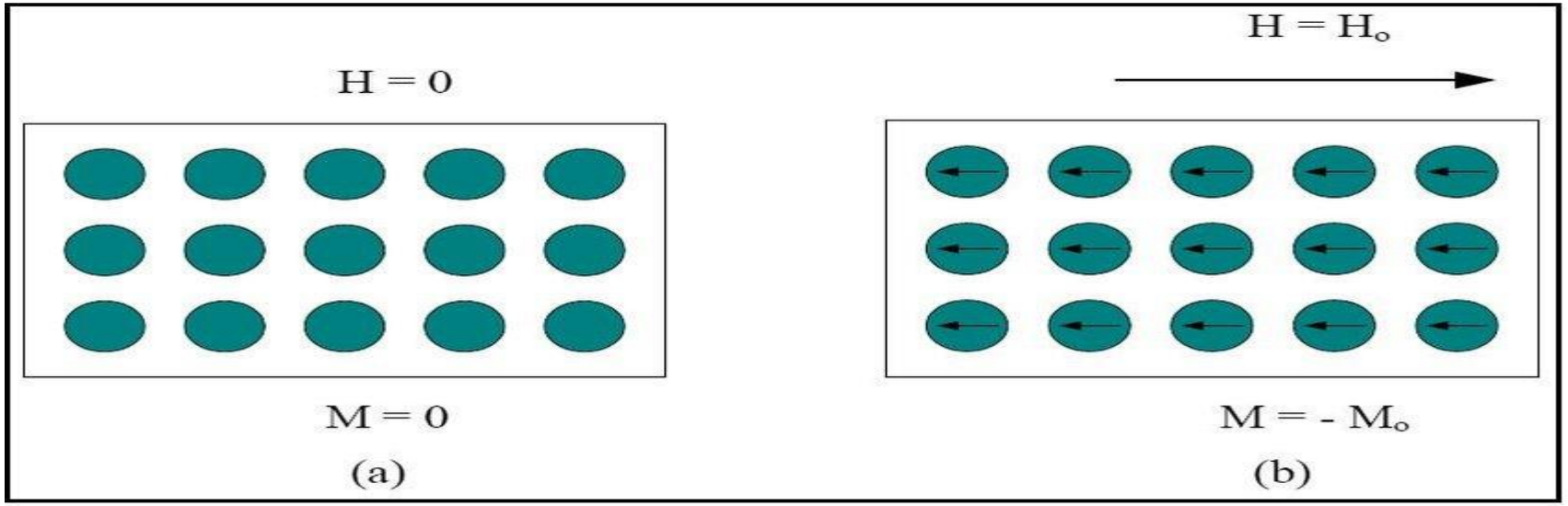
✓ Origin:

- Diamagnetism is associated with orbital motion of filled (paired) electrons .
- Due to different orientations of various orbits of an atom, the net magnetic moment is zero in diamagnetic materials.
- When an external field is applied the motion of electrons in their orbits changes resulting in induced magnetic moment in a direction opposite to the direction of applied field.



DIAMAGNETISM

- ✓ The magnetization induced by the applied magnetic field is very weak and the magnetic lines of force are repelled.
- ✓ This magnetism also exists in substances with magnetic atoms, but very weak and completely masked by the contribution of magnetic atoms.
- ✓ Relative permeability is slightly less than unity.
- ✓ The magnetic susceptibility is independent of applied magnetic field strength.



DIAMAGNETISM

Magnitude of susceptibility	**Temperature dependence**	**Examples**
Small & negative	Independent	Organic materials, light elements
Intermediate & negative	Below 20K varies with field and temperature	Alkali earhs, Bismuth
Large & Negative	Exists only below critical temperature (Meissner effect)	Superconducting materials

PARAMAGNETISM

✓ Paramagnetic materials consists of magnetic atoms that posses unpaired electrons oriented at random on different atoms.

✓ Origin

These unpaired spins may interact with each other leading to cooperartive phenomenon like ferromagnetism

- When the shells are unfilled there is net magnetic moment.
- In the absence of the external field the net moments of the atoms are arranged in random directions because of thermal fluctuations. Hence there is no magnetization.
- When external magnetic field is applied, there is tendency for the dipoles to align with the field giving rise to an induced positive dipole moment.
- The induced magnetism is the source for paramagnetic behaviour.

Paramagnetism

Magnetic Field Absent

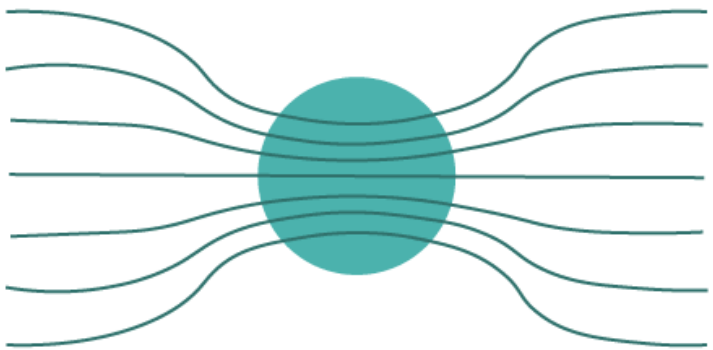


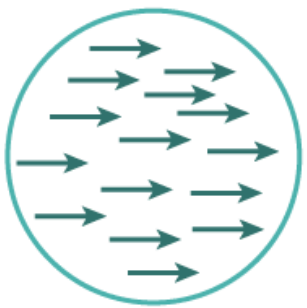
Paramagnetic Material
(eg. Chromium, Aluminium)



Random Alignment
of Magnetic Moments

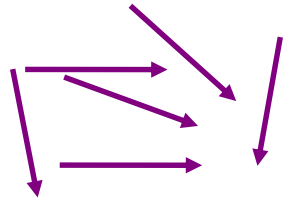
Magnetic Field Present





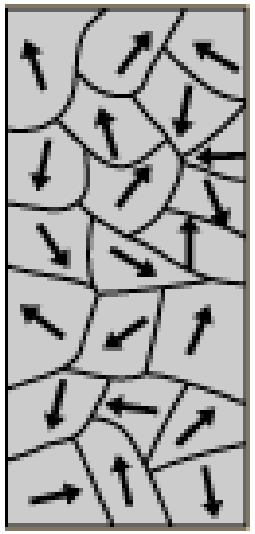
✓ Paramagnetic susceptibility is small and positive and is independent of applied field strength.

✓ Spin alignment is random.

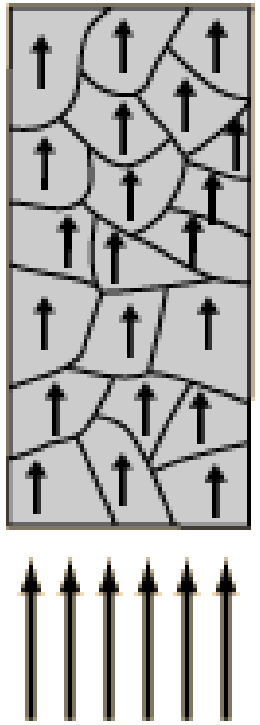


Magnitude of susceptibility	Temperature dependence	Examples
Small & positive	Dependence	Alkali metals, transition metals, rare earths
	$\chi = \frac{C}{T}$ Curie law C- Curie constant	

Ferromagnetism

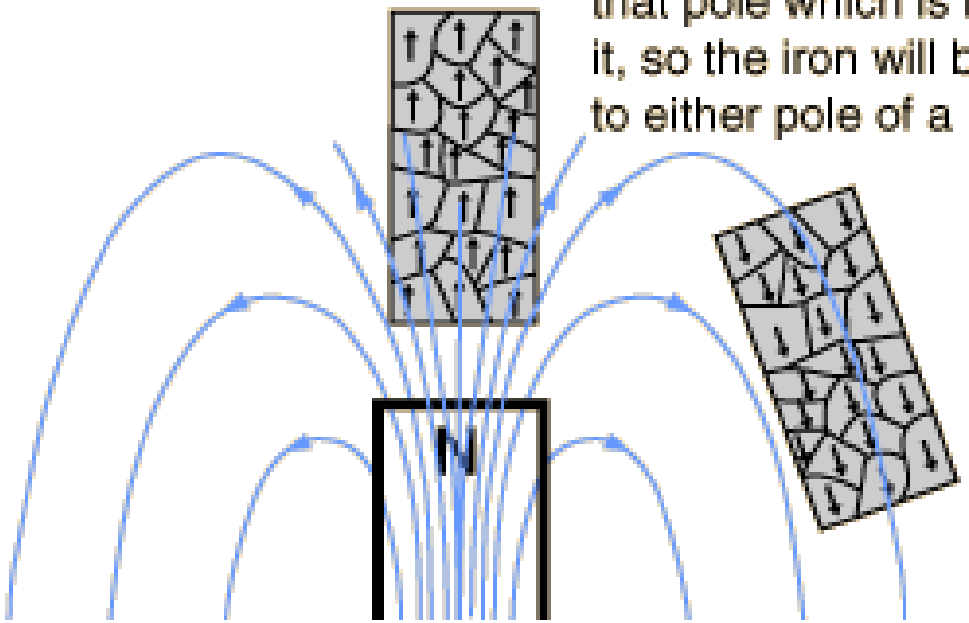


In bulk material the domains usually cancel, leaving the material unmagnetized.



Externally applied magnetic field.

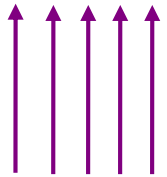
Iron will become magnetized in the direction of any applied magnetic field. This magnetization will produce a magnetic pole in the iron opposite to that pole which is nearest to it, so the iron will be attracted to either pole of a magnet.



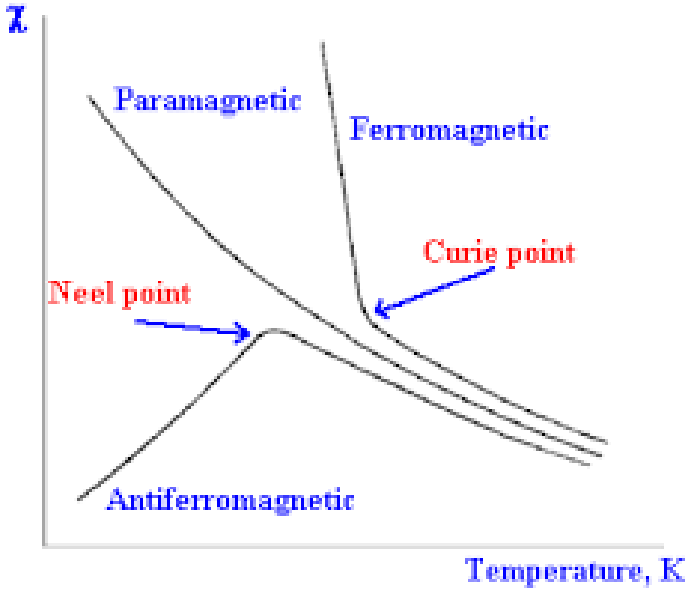
FERROMAGNETISM

- Even in the absence of external applied field, some substances exhibit strong magnetization due to unpaired electrons interacting cooperatively.
- This is due to a special form of interaction called exchange coupling between adjacent atoms that results in spontaneous magnetization of the substance.
- When placed inside a magnetic field, it attracts the magnetic lines of force very strongly.
- Each ferromagnetic material has a characteristic temperature called the ferromagnetic Curie temperature θ_f. Below this temperature the spontaneous magnetization exists.

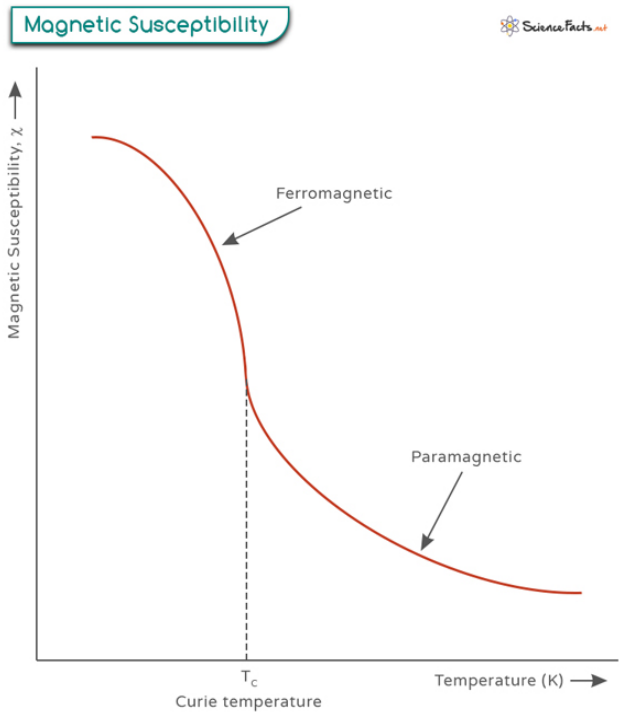
✓ Spin alignment is parallel.



✓ Ferromagnetic materials exhibit Hysteresis.
✓ They Consists of a number of small regions which are called domains.

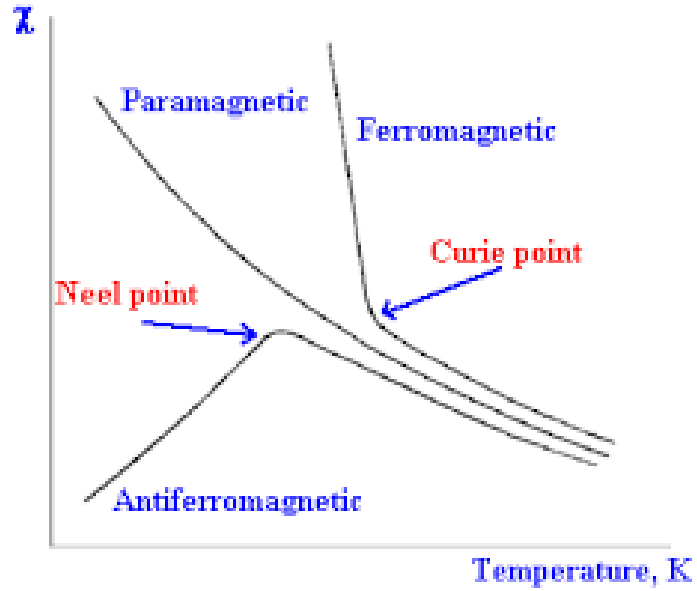


Magnitude of susceptibility	Temperature dependence	Examples
Very large & positive	Curie-Weiss Law. $\chi = \frac{C}{T-\theta}$ θ Curie temperature C- Curie Constant. For $T>\theta_f$ paramagnetic behavior For $T<\theta_f$ ferromagnetic behavior	Fe, Co, Ni, Gd

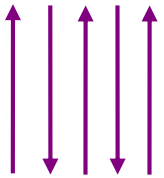


ANTI-FERROMAGNETISM

- ✓ Antiferromagnetism macroscopically similar to paramagnetism, is a weak form of magnetism.
- ✓ In certain materials when the distance between the interacting atoms is small the exchange forces produce a tendency for antiparallel alignment of electron spins of neighboring atoms.
- ✓ The magnetic susceptibility increase with the increase of temperature and reaches maximum at a certain temperature. This temperature is known as Neel temperature (T_N). Above this temperature the susceptibility again decreases.



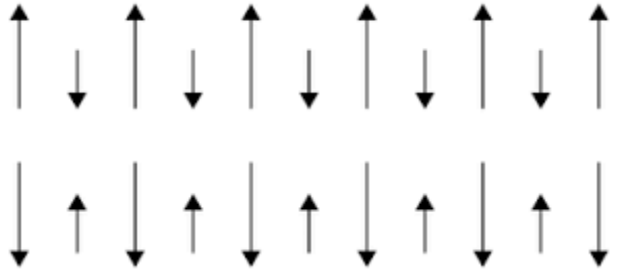
Spins are aligned antiparallel

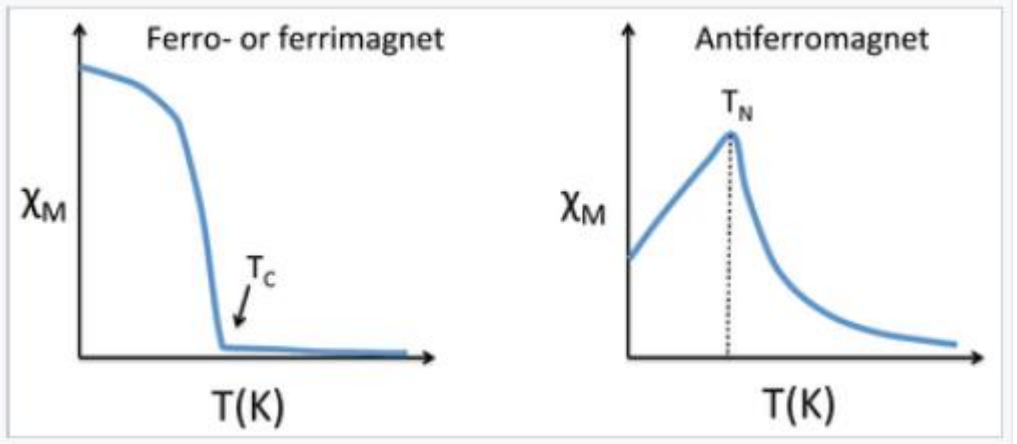


Magnitude of susceptibility	Temperature dependence	Examples
small & positive	$\chi = \frac{C}{T+\theta}$ when $T>T_N$ $\chi \propto T$ when $T<T_N$	NiO, MnO

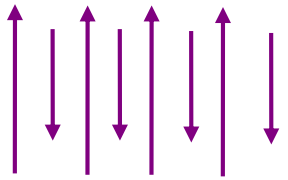
FERRIMAGNETISM

- ✓ This is a special case of antiferromagnetism.
- ✓ The net magnetization of magnetic sublattices is not zero, since antiparallel moments are of different magnitudes.
- ✓ This moment disappears above a Curie temperature analogous to the Neel temperature.
- ✓ Above T_C, thermal energy randomizes the individual magnetic moments and the material becomes paramagnetic.





- ✓ Ferrimagnetic domains become magnetic bubbles to act as memory elements.
- ✓ Spin alignment is antiparallel of different magnitudes.



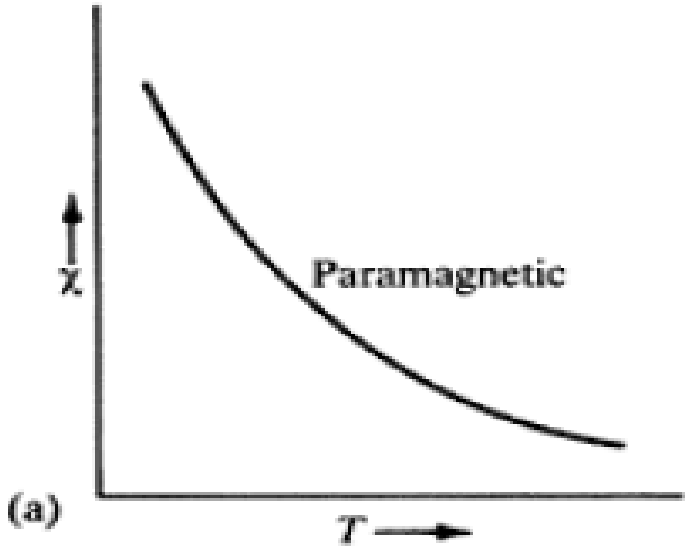
Magnitude of susceptibility	**Temperature dependence**	**Examples**
Very large & positive	$\chi = \frac{C}{T \pm \theta}$ when $T > T_N$ when $T < T_N$ behaves as paramagnetic material	Ferrites

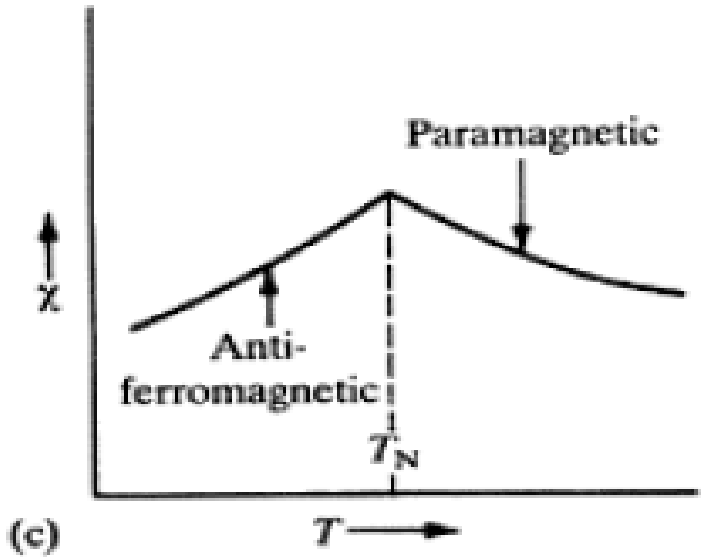
T_C and T_N values of some elements

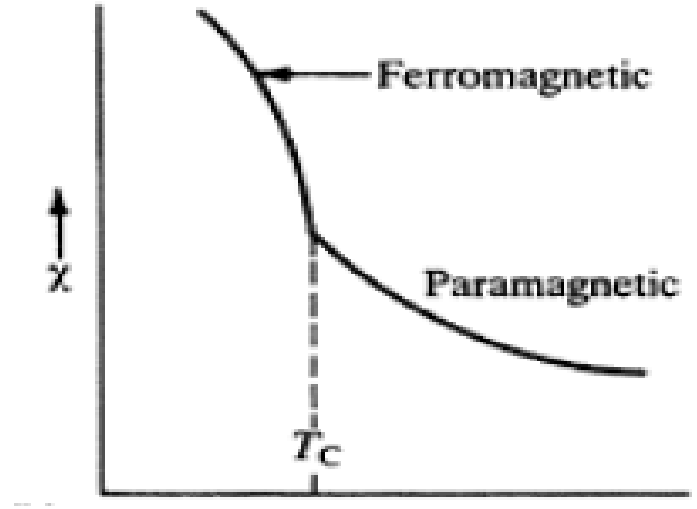
Table 9.2 ***Some Curie and Néel temperatures***

Element	T_c/K	T_N/K
Cr		308
Mn		100
Fe	1043	
Co	1404	
Ni	631	
Ce		12.5
Pr		25
Nd		19
Sm		14.8
Eu		90
Gd	293	
Tb	222	229
Dy	85	179
Ho	20	131
Er	20	84
Tm	25	56

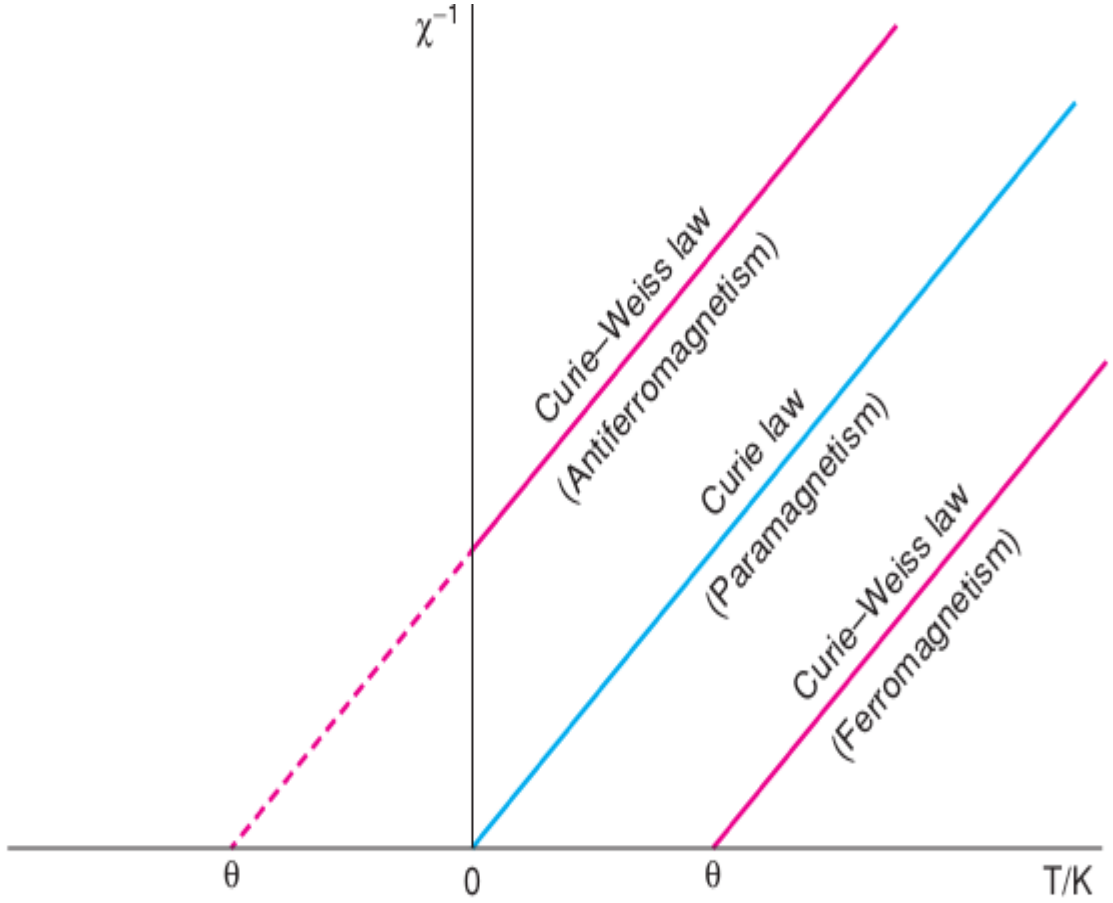
Magnetic susceptibility vs T plots





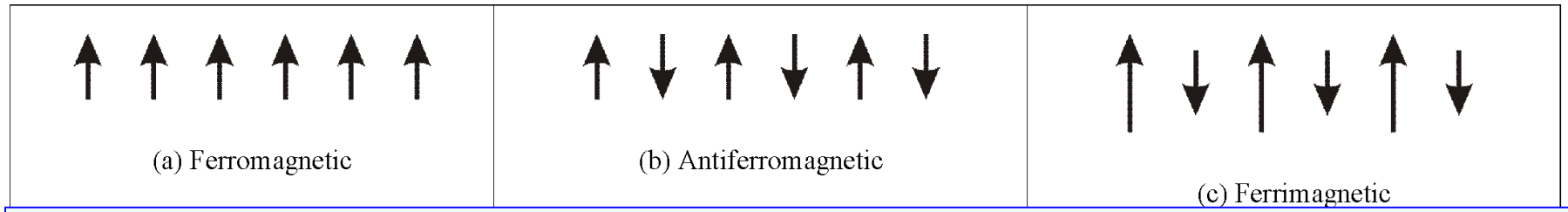


Inverse susceptibility vs T for different behaviour



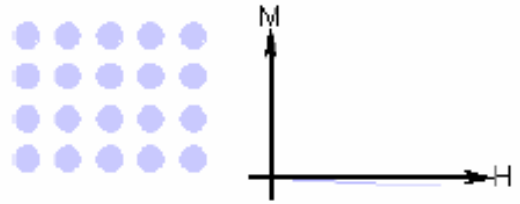
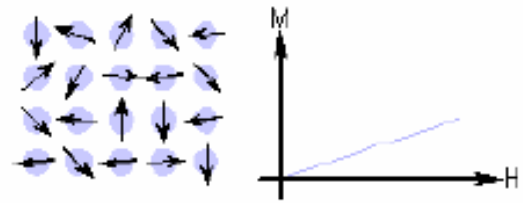
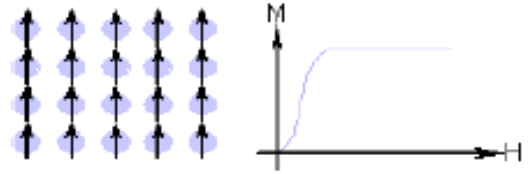
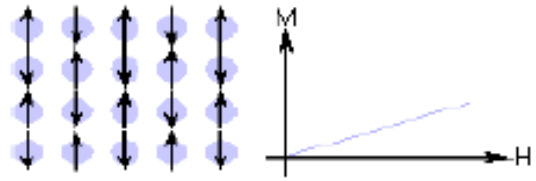
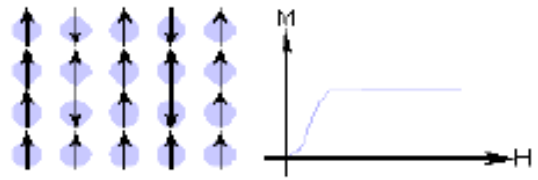
Ferromagnetism (FM)

- Ferromagnetism, Antiferromagnetism and Ferrimagnetism involve no new types of magnetic moments; but involve the way the magnetic moments are coupled (arranged).



(a) Ferromagnetic | (b) Antiferromagnetic | (c) Ferrimagnetic

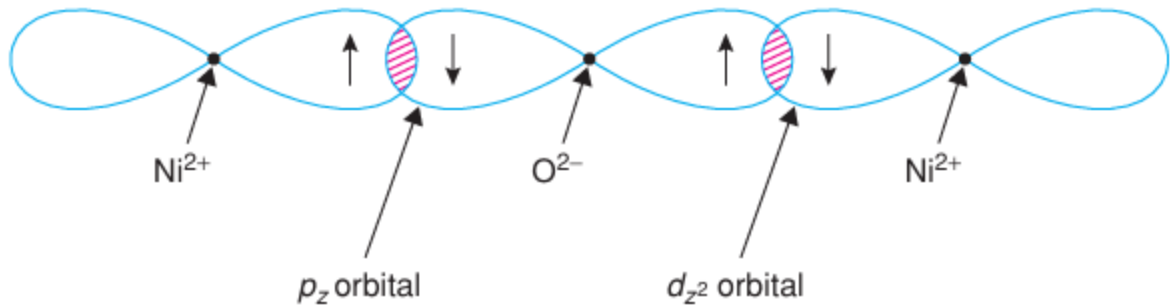
- Two important ways of understanding ferromagnetism in metals are:
  (i) assuming that moments are localized to atoms,
  (ii) using the band structure of metals (giving rise to itinerant electrons).
  The former is conceptually easier and has been assumed in the 'molecular field theory' and the Heisenberg's approach. It should be noted right at the outset that even in metals (e.g. Fe), most of the electrons behave as if they are 'localized' and the number of itinerant electrons could be a small number.
  - In Fe there are 8 valence electrons which occupy the (3d + 4s) bands. Out of these 8 electrons only ~0.95 in the 4s band are 'truly' free/itinerant and remaining ~7.05 are occupy the 'localized' 3d band.

Type of Magnetism	*Susceptibility*	*Atomic / Magnetic Behaviour*	*Example / Susceptibility*
Diamagnetism	Small & negative.	Atoms have no magnetic moment 	Au -2.74×10^{-6} Cu -0.77×10^{-6}
Paramagnetism	Small & positive.	Atoms have randomly oriented magnetic moments 	β-Sn 0.19×10^{-6} Pt 21.04×10^{-6} Mn 66.10×10^{-6}
Ferromagnetism	Large & positive, function of applied field, microstructure dependent.	Atoms have parallel aligned magnetic moments 	Fe ~100,000
Antiferromagnetism	Small & positive.	Atoms have mixed parallel and anti-parallel aligned magnetic moments 	Cr 3.6×10^{-6}
Ferrimagnetism	Large & positive, function of applied field, microstructure dependent	Atoms have anti-parallel aligned magnetic moments 	Ba ferrite ~3

Mechanism of ferro and antiferro magntic ordering – super exchange

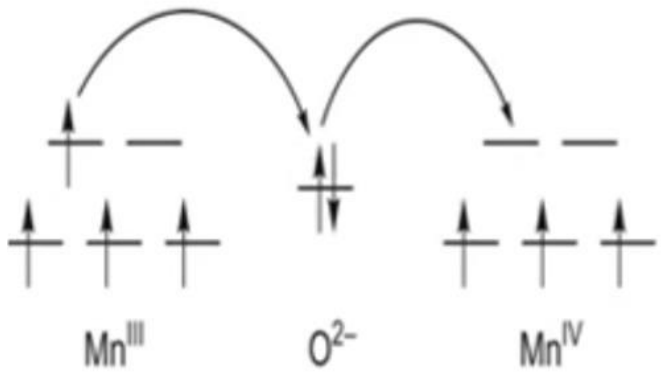
- In paramagnetic state individual moments of ions with unpaired electrons(magnetic dipoles) are oriented at random. On application of magnetic field partial alignment occurs.

- In ferro and anti ferro state magnetic diploes (unpaired electrons) align spontaneously due to the positive energy of interaction between the spins. The origin of this cooperative interaction is quantum mechanical.
- One mechanism that leads to antiferromagnetism in NiO is through super exchange (refer fig next slide)
- The singly occupied electrons in d_{z2} and d_{x2-y2} orbitals of a Ni^{2+} ion couples antiferromagnetically with that on neighbouring Ni^{2+} through the p orbital of the oxide ion. This is the superexchange



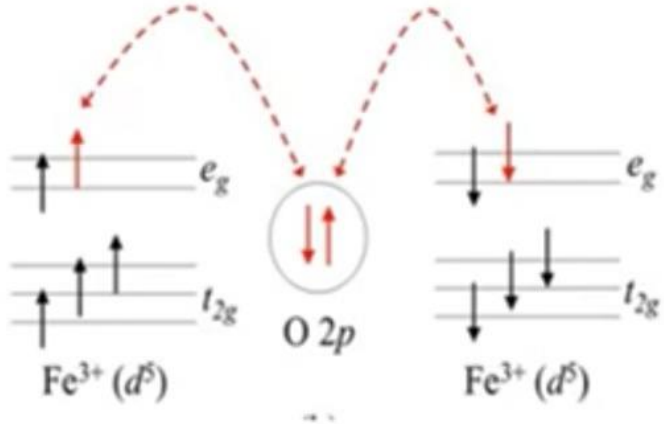
ure 9.5 ***Antiferromagnetic coupling of spins of d electrons on Ni^{2+} ions through p electrons of oxide ions.***

Double exchange



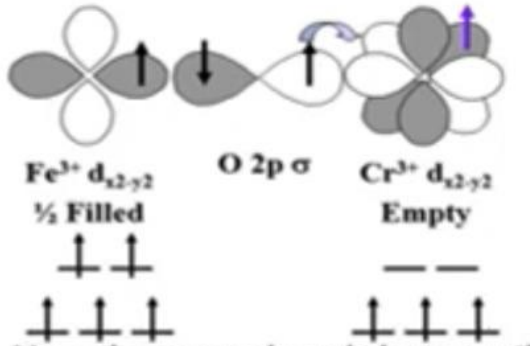
Ferromagnetic interaction

Super-exchange

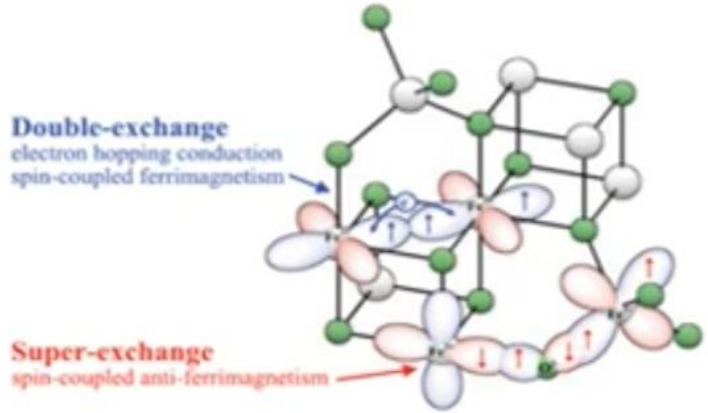


Anti-ferromagnetic coupling

Super-exchange



Here the oxygen based electron will spend some time on Cr^{3+} and due to Hund's rule polarize the t_{2g} e^- leading to **ferromagnetic coupling**.



Hysteresis

The striking property of Ferro Magnetic materials is the relation between Magnetization and the strength of Magnetic field. This property is called Hysteresis.

Ferromagnets have a domain structure and within each domain the spins are Parallel . In the saturation magnetization conditions different domains have Same orientation.

What is Hysteresis?

Hysteresis is a phenomenon where the **response of a system depends not only on its current input but also on its history**.

In other words, when you cycle a system (e.g., apply and remove a force/field), it doesn't retrace the same path — it shows a **lag** or **memory effect**.

In magnetism:

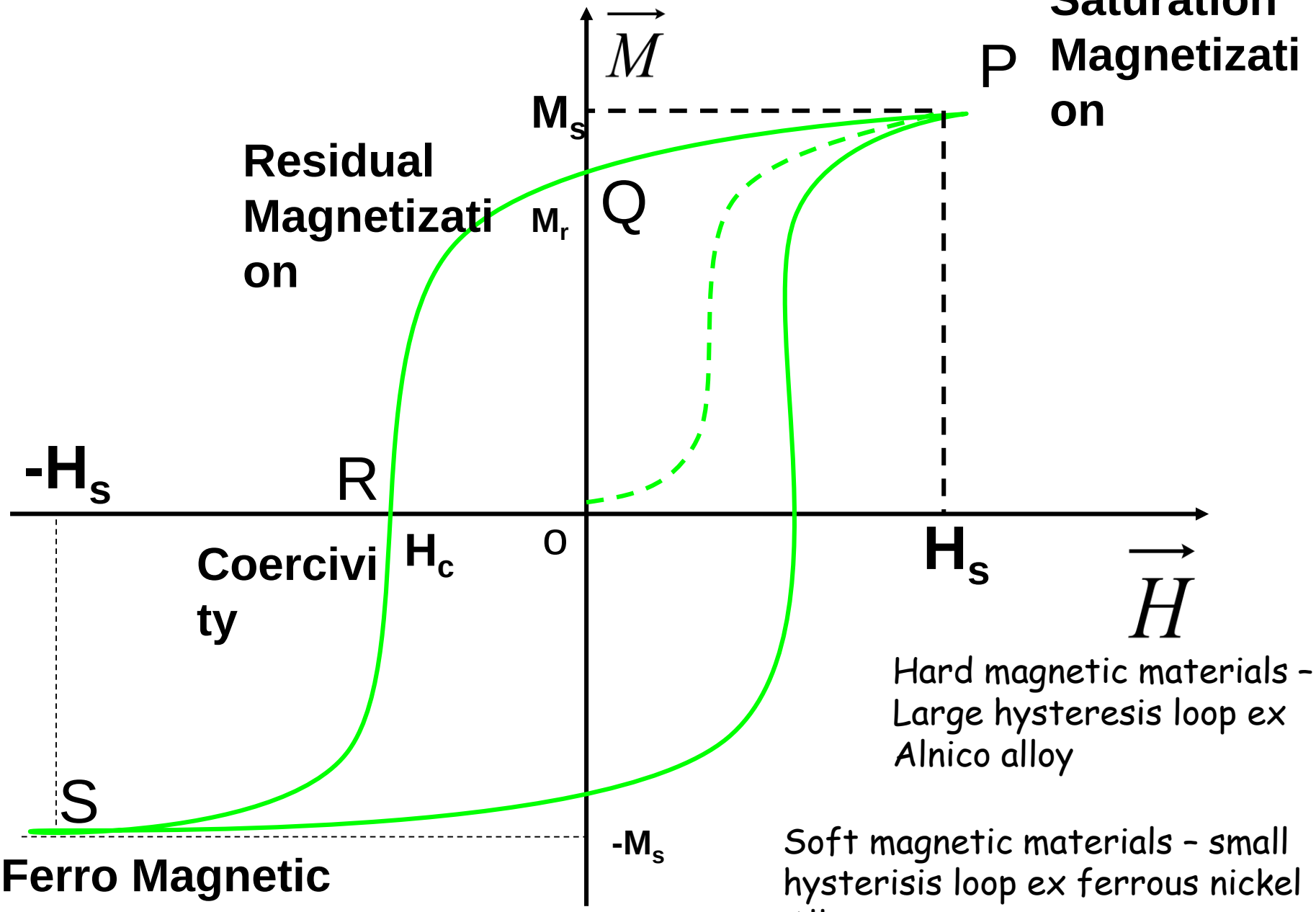
- When you apply an external magnetic field (H), the material's magnetization (M) increases.
- When you remove the field, the magnetization does **not return to zero immediately**.
- Instead, some magnetization remains → called **remanent magnetization (Mr)**.
- To demagnetize it, you must apply an opposite field → called **coercive field (Hc)**.

This creates a **hysteresis loop** when plotting M vs H.

direction changes.

Hysteresis

This effect is common in magnetic materials, where magnetization lags behind the applied magnetic field, creating a "hysteresis loop" on a graph.



Hard magnetic materials – Large hysteresis loop ex Alnico alloy

Soft magnetic materials – small hysterisis loop ex ferrous nickel Alloy

- If we start with no Magnetized specimen

(M= 0) with the increasing values of magnetizing field H.

- The Magnetization of the specimen increases from zero to higher values and attains its maximum value at a point P, at this point the Magnetization referred as Saturation Magnetization.

- When we increase Magnetic field H there is no further increment in Magnetic moment.

- When we decrease Magnetic field H to Zero, the Magnetization M attains point Q.

- At this point Magnetization referred as residual Magnetization Mr.

- Further if we increase the Magnetic field from zero to negative values, the Magnetization of material becomes zero at a point R, at that point the Magnetic field H_c is referred as Coercivity of the specimen.

- If we increase Magnetic field H in reverse direction Magnetization of material reaches its peak value at a points S.

- On reversing the polarities of Magnetic field and increasing its strength the Magnetization slowly decreases first to residual value then to zero and finally increases to saturation state and touches the original saturation curve.

- The area of loop indicates the amount of energy wasted in one cycle of operation.

Applications of Magnetic Materials

- Soft Magnetic Materials - Ferromagnetic materials are often used to enhance the magnetic flux density (B) produced when an electric current is passed through the material. Applications include cores for electromagnets, electric motors, transformers, generators, and other electrical equipment.
- Data Storage Materials - Magnetic materials are used for data storage.
- Permanent Magnets - Magnetic materials are used to make strong permanent magnets

TABLE 19-5 ■ *Typical magnetic recording materials*[16]

	Particle Length μm	Aspect Ratio	Magnetization (B_r)		Coercivity (H_c)		Surface Area m^2/g	Curie temp. (T_c) °C
			Wb/m^2	emu/cc	kA/m	Oe		
γ-Fe_2O_3	0.20	5:1	0.44	350	22–34	420	15–30	600
Co-γ-Fe_2O_3	0.20	6:1	0.48	380	30–75	940	20–35	700
CrO_2	0.20	10:1	0.50	400	30–75	950	18–55	125
Fe	0.15	10:1	1.40[a]	1100[a]	56–176	2200	20–60	770
Barium Ferrite	0.05	0.02 μm thick	0.40	320	56–240	3000	20–25	350

[a] *For overcoated, stable particles use only 50 to 80% of these values due to reduced magnetic particle volume*

(Source: From The Complete Handbook of Magnetic Recording, Fourth Edition, *by F. Jorgensen, p. 324, Table 11-1. Copyright © 1996 Reprinted by permission of The McGraw-Hill Companies.)*

Soft magnetic materials have low coercivity.

Magnetic oxides have significant applications in memory storage, particularly in non-volatile memory technologies like hard disk drives (HDDs) and magnetic random access memory (MRAM). Their ability to retain magnetization even when power is removed makes them suitable for persistent data storage.

$MgFe_2O_4$ ferrites are used in transformer core, magnetic recording and Information storage devices.

For cassettes, tapes and floppy disks, sound data are recorded as variations in the amount of magnetisation on the surface of a recording medium, which is usually either a composite consisting of a polymer film embedded with tiny particles of magnetic CrO2 or γ-Fe2O3, or a thin film of magnetic material deposited on a substrate. Each particle or domain can be magnetised in either of two directions, representing 0 or 1. The read–write recording head is a tiny electromagnet of a soft material, such as MnZn ferrite,
Schematic- next slide

FeCr multi-layers which exhibit Giant magneto resistance is used in high density Hard drives. Magneto resistance is change in resistance when there is an applied Magnetic field.

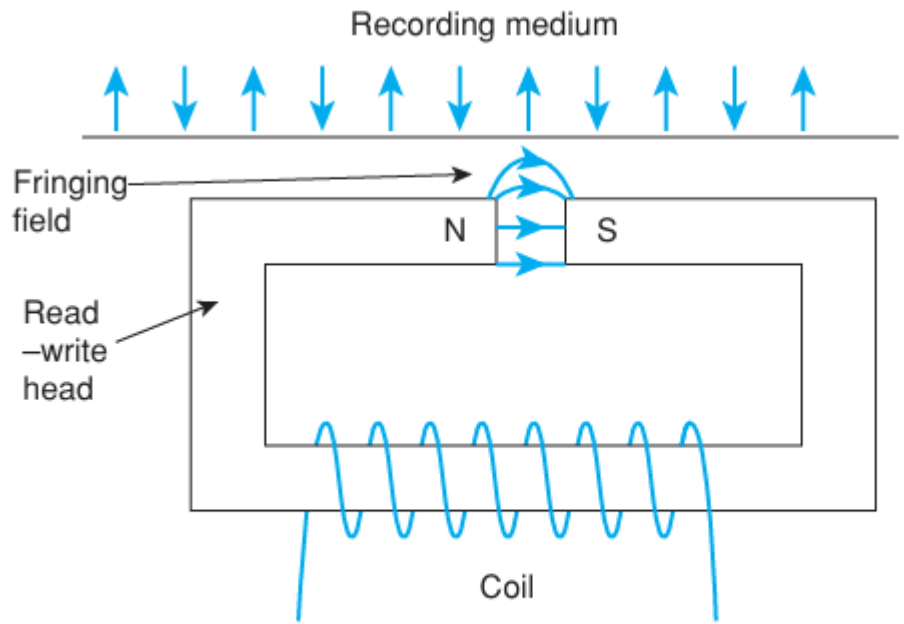


Figure 9.16 ***Read–write magnetic recording head.***

For high density hard disk, the material used is $SmCo_5$ alloy.

Functional Materials

Nanomaterials: introduction, top-down and bottom-up approaches for synthesis - bulk *vs* nano (gold), quantum dots

Nanomaterials: Introduction

- Nanomaterials are cornerstones of nanoscience and nanotechnology.
- Nanostructure science and technology is a broad and interdisciplinary area of research and development activity that has been growing explosively worldwide in the past few years.
- It has the potential for revolutionizing the ways in which materials and products are created and the range and nature of functionalities that can be accessed.
- It is already having a significant commercial impact, which will assuredly increase in the future.

What are nanomaterials?

- Nanoscale materials are defined as a set of substances where at least one dimension is less than approximately 100 nanometers.

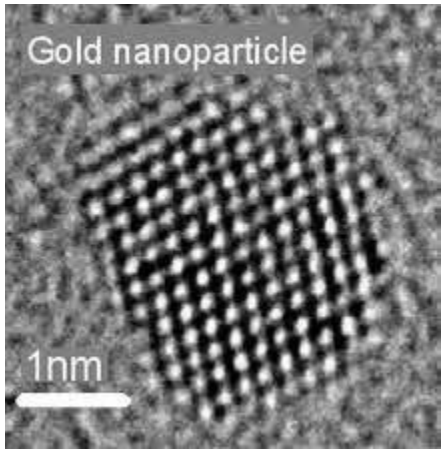
- A nanometer is one millionth of a millimeter - approximately 100,000 times smaller than the diameter of a human hair.

- Nanomaterials are of interest because at this scale unique optical, magnetic, electrical, and other properties emerge.

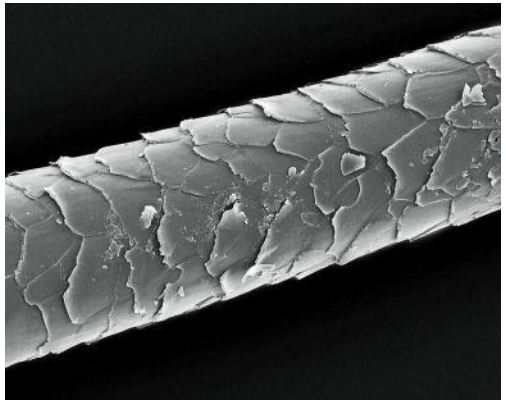
- These emergent properties have the potential for great impacts in electronics, medicine, and other fields.

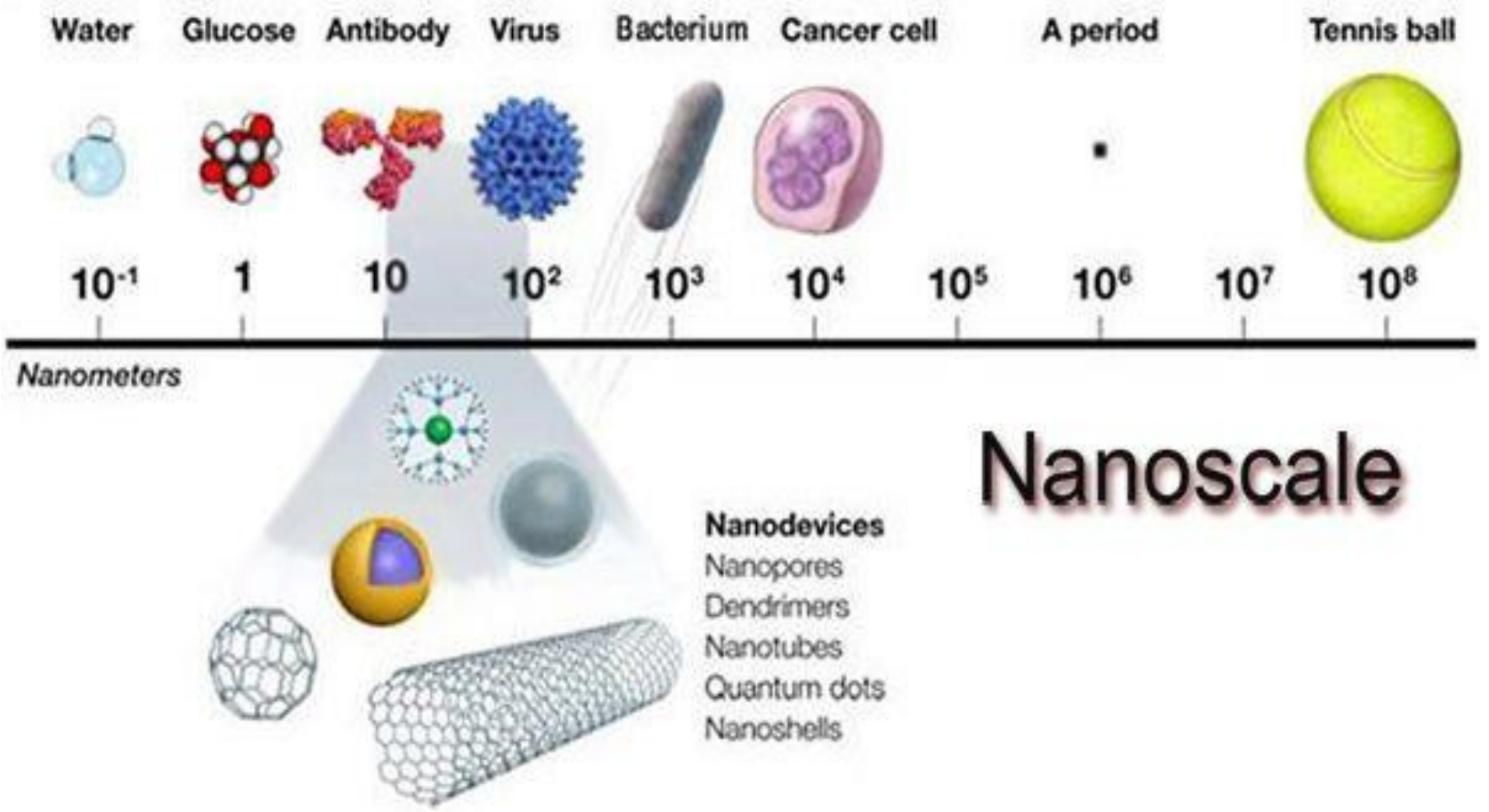
Nano is a millionth of a millimeter or a billionth of a meter, i.e. 1 nm = 10^{-9} m.

Atom is about 0.1 nanometer - 10 atoms side by side make up 1 nm.



Human hair:
50,000–100,000 nm in diameter.

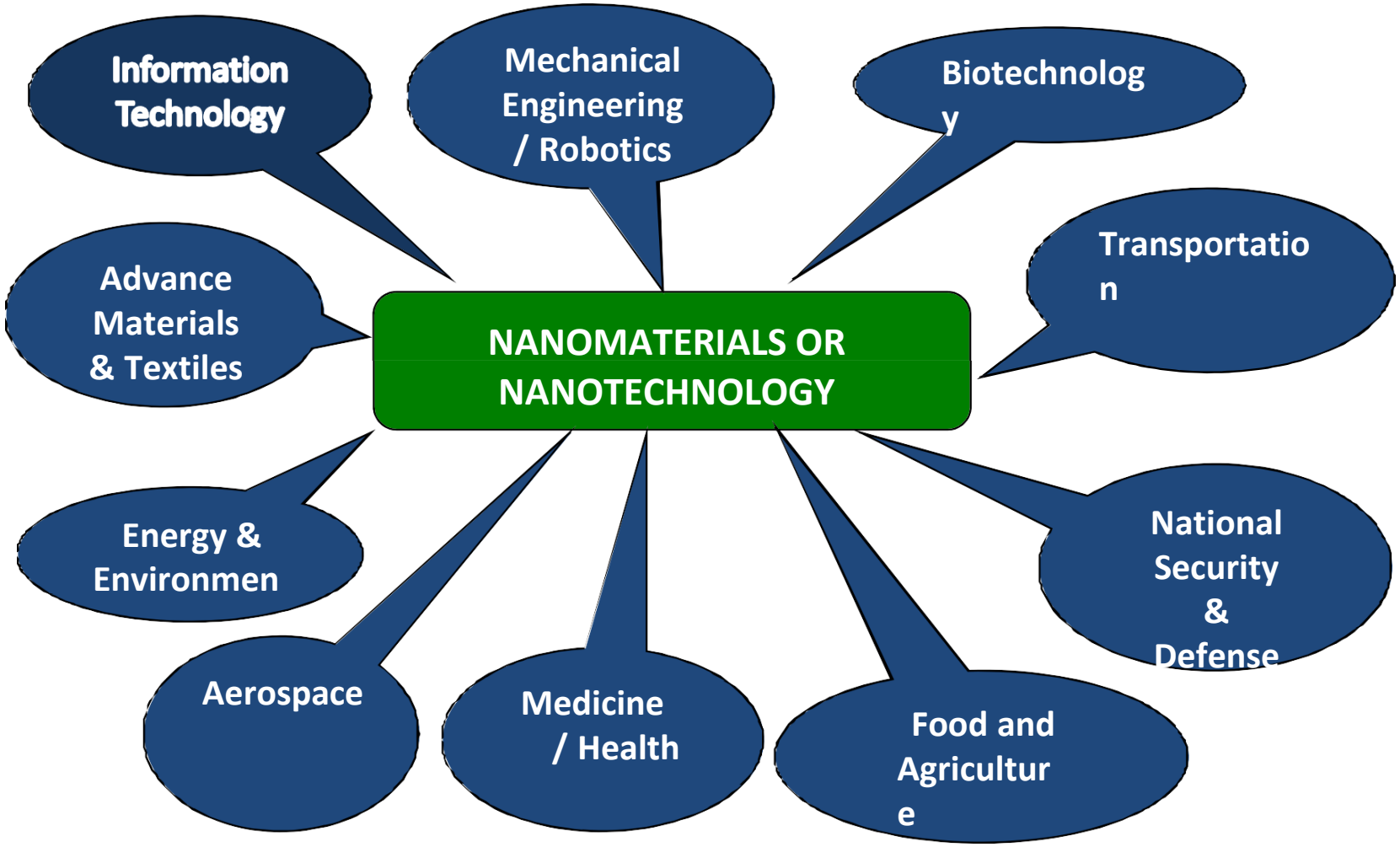




Where are nanomaterials found?

- Some nanomaterials occur naturally, but of particular interest are engineered nanomaterials (EN), which are designed for, and already being used in many commercial products and processes.
- They can be found in such things as sunscreens, cosmetics, sporting goods, stain resistant clothing, tires, electronics, as well as many other everyday items, and are used in medicine for purposes of diagnosis, imaging and drug delivery.
- Nanocoatings and nanocomposites are finding uses in diverse consumer products, such as windows, sports equipment, bicycles and automobiles
- There are novel UV-blocking coatings on glass bottles which protect beverages from damage by sunlight, and longer-lasting tennis balls using butyl-rubber/nano-clay composites.
- Nanoscale titanium dioxide, for instance, is finding applications in cosmetics, sun-block creams and self-cleaning windows, and nanoscale silica is being used as filler in a range of products, including cosmetics and dental fillings

Where do we use?



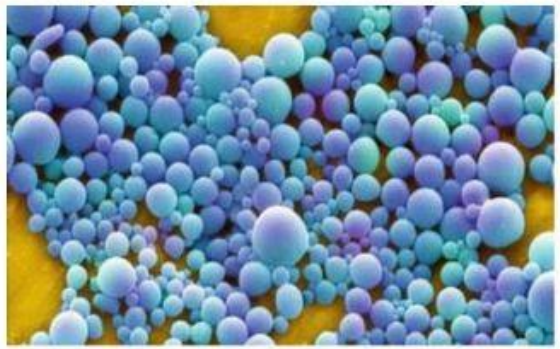
Classification: based on the number of free dimensions

0D *nanomaterial*: all three dimensions are in the nanoscale. (Nanoparticles, Colloids, Quantum dots)

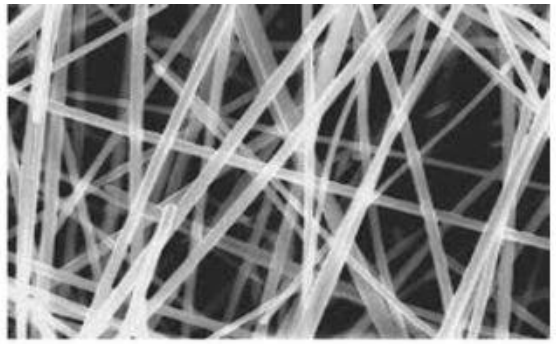
1D *nanomaterial*: one dimension beyond the nanoscale and two other dimensions in the nanoscale. (Nanowires, Nanorods, Nanotubes & Biopolymers).

2D *nanomaterial*: any two dimensions can be outside of the nanoscale and one dimension in the nanoscale. (Plate-like shapes - nanolayers, surface coatings and thin films).

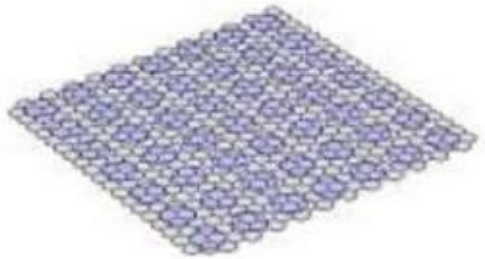
3D *nanomaterial*: all three dimensions can be outside of the nanoscale. Made of a nanomaterials. (nanoparticle dispersions, nanowire/ nanotube bundles & multiple nanolayers).



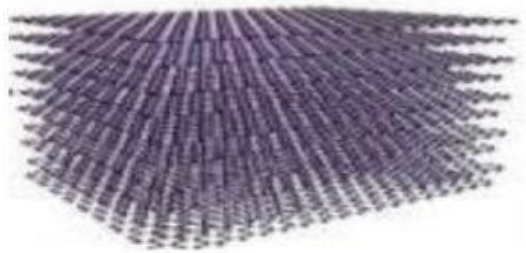
0D(Nanoparticle)



1D(Nanowire)

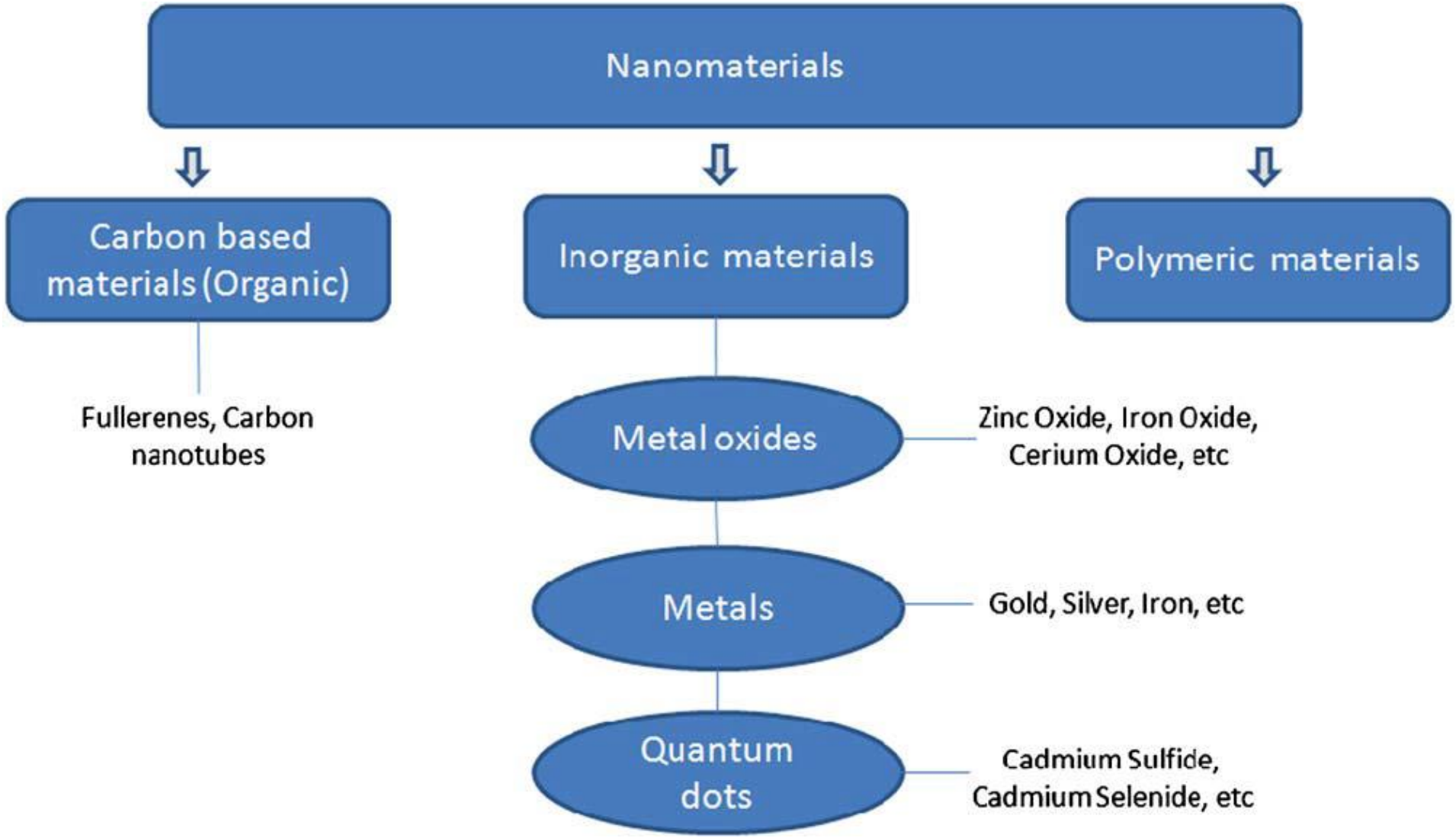


2D Nanomaterials (plate)



3D Nanomaterials

Categories of Nanomaterials



Why so much interest in nanomaterials?

These materials have created a high interest in recent years by virtue of their unusual mechanical, electrical, optical and magnetic properties. Some examples are given below:

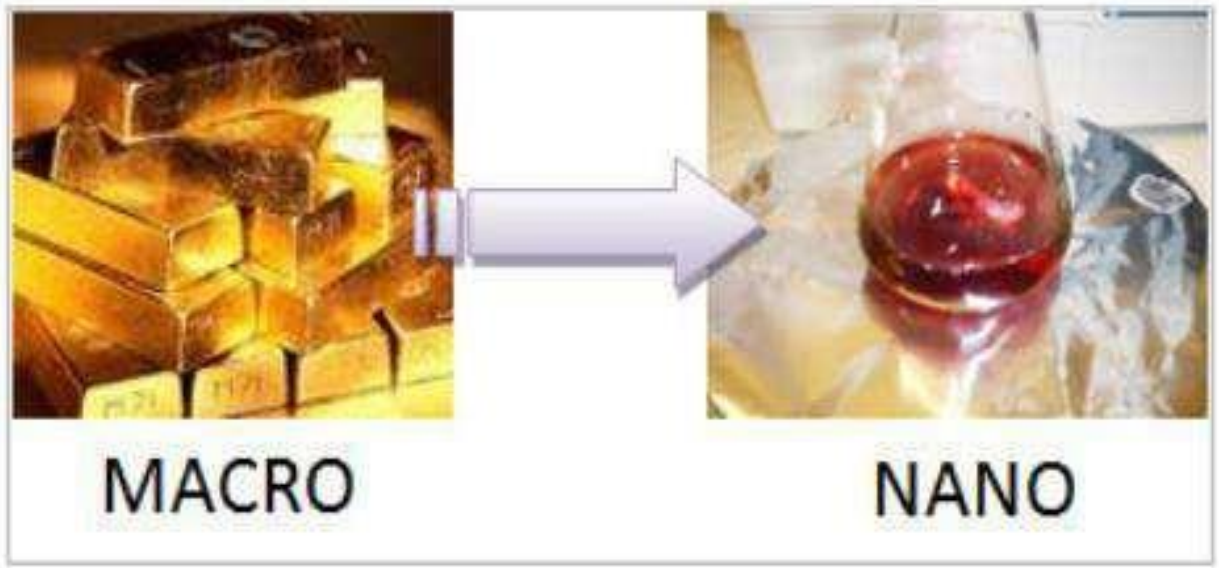
Nanophase ceramics are of particular interest because they are more ductile at elevated temperatures as compared to the coarse-grained ceramics.

Nanostructured metal-oxide (MnO_2) finds application for rechargeable batteries for cars or consumer goods

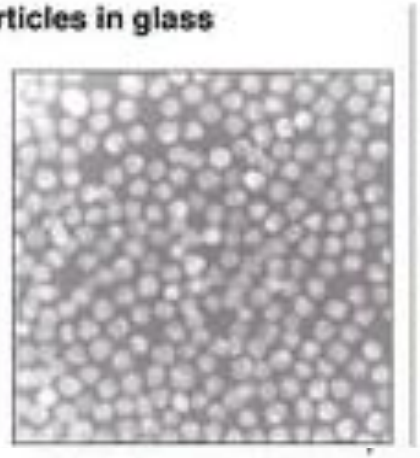
Very small particles have special atomic structures with discrete electronic states, which give rise to special properties in addition to the superparamagnetism behaviour

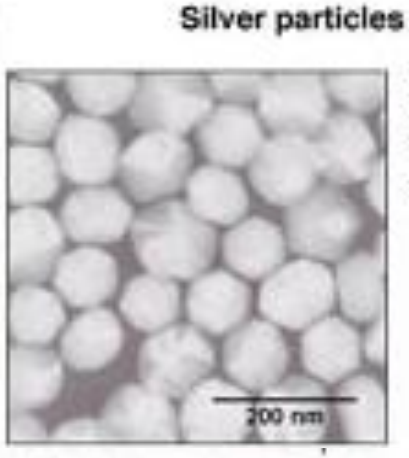
Bulk gold Vs Nano gold (In detail)

Bulk *vs* *N*ano (gold)



Size and shape dependent colors of Au and Ag nanoparticles





Gold NPs in Glass

25 nm Sphere reflected

Size*: 25 nm
Shape: sphere
Color reflected:

100 nanometers = 0.0001 millimeter

Silver NPs in Glass

Size*: 100 nm
Shape: sphere
Color reflected:

100 nm Sphere reflected

50 nm Sphere reflected

Size*: 50 nm
Shape: sphere
Color reflected:

Size*: 40 nm
Shape: sphere
Color reflected:

40 nm Sphere reflected

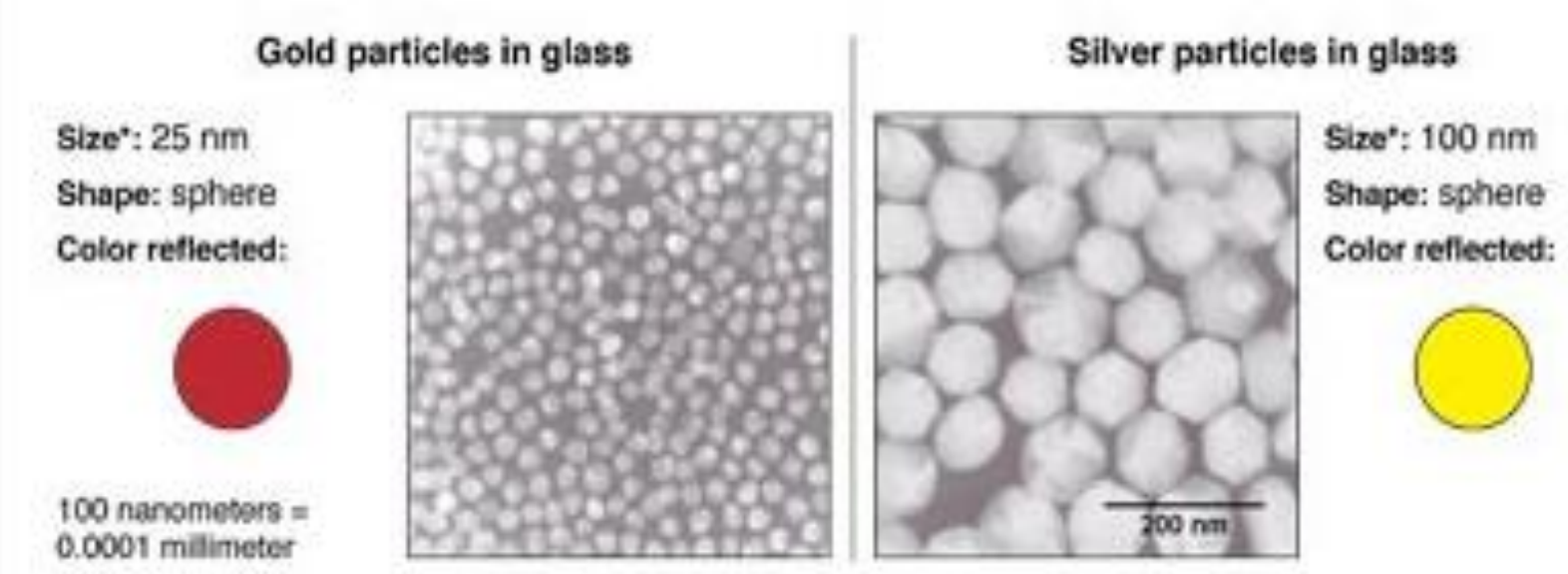
100 nm Sphere reflected

Size*: 100 nm
Shape: sphere
Color reflected:

Size*: 100 nm
Shape: prism
Color reflected:

100 nm prism reflected

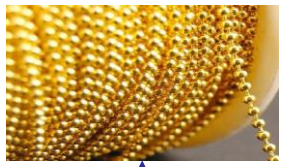
Size and shape dependent colors of Au and Ag nanoparticles



Note: **nanomaterials scatter visible light rather than absorb**

- **Distance between particles also effects colour**

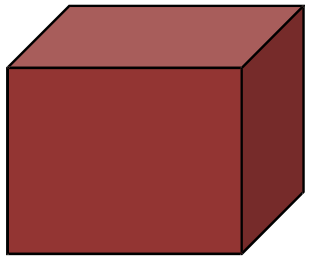
Gold : Bulk vs. Nano

	Bulk Gold	Nano Gold
1	Lustrous–they have a shiny surface when polished	Are never gold in colour but found in a range of colours
2	Malleable–they can be hammered, bent or rolled into any desired shape	Size & Shape of the nanoparticles determines the colour
3	Ductile–they can be drawn out into wires 	For example, Gold particles in glass 25 nm > red reflected 50 nm > green reflected 100 nm > orange reflected
4	Is metallic, with a yellow colour when in a mass	Are not “metals” but are semiconductors (Band gap energy = 3.4 eV)
5	Good conductors of heat and electricity	Are very good catalysts
6	Generally have high densities	
7	Have high melting point (~1080°C)	Melts at relatively low temp (~940° C)
8	Are often hard and tough with high tensile strength	
9	Having high resistance to the stresses of being stretched or drawn out	
10	Not easily breakable	
11	Inert-unaffected by air & most reagents	

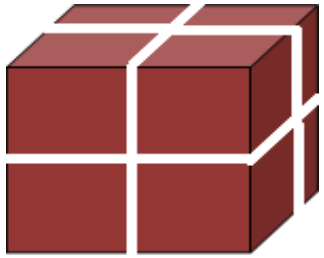
Why Nanomaterials are different

The two main reasons why materials at the nano scale can have different properties are increased relative surface area and new quantum effects.

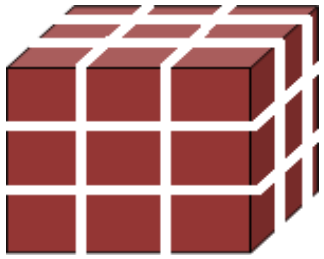
- **Increase of Surface area/Volume ratio**



Area = 6 x $1cm^2$ = 6 cm^2



Area= 6 x $(1/2cm)^2$ x 8 =12 cm^2



Area= 6 x $(1/3cm)^2$ x 27 =18 cm^2

→ Better catalytic efficiency

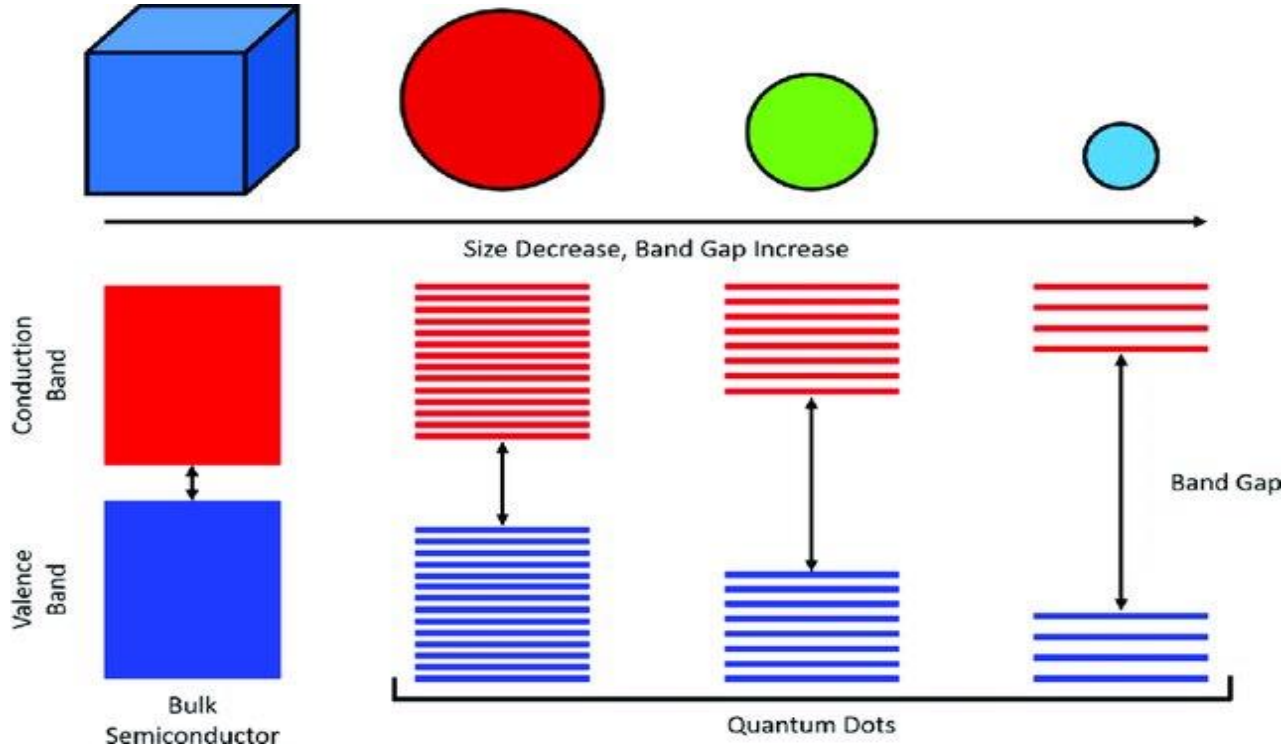
If a bulk material is subdivided into an ensemble of individual nanomaterials, the total volume rer the same, but the **collective surface area is greatly increased.** This is schematically shown in **Figure**

- **Small size effect (Quantum size effect)**

✓ The particles are so small that **electrons are not free to move** about as in bulk gold. As the movement is restricted, **the particles react differently with light**

→ **Tunable fluorescent emission**

✓ Electronic states are quite different from bulk. Discrete energy levels; Quantum confinement effect occurs, increases the energy gap between energy levels leading to metal to semiconductor to insulator transition.



Quantum confinement refers to the spatial confinement of electrons in a nanomaterial rather than the free existence within the bulk materials

Electron hole pairs become spatially confined when the diameter of the particle approaches the de Broglie wavelength of electrons in the conduction band

The quantum confinement occurs when the size of the particle is too small to be comparable to the de Broglie wavelength of the electron

- Decreased imperfections and defects: reason for changing **electrical conductivity** than bulk matter. etc.

- This more structural perfection in the nanomaterials improves their mechanical properties (increases mechanical strength e.g. hardness, toughness etc.).

- Increased paramagnetism and supermagnetism behaviour **Magnetic Property changes**

- Due to smaller size, permeability through the biological membrane Increases



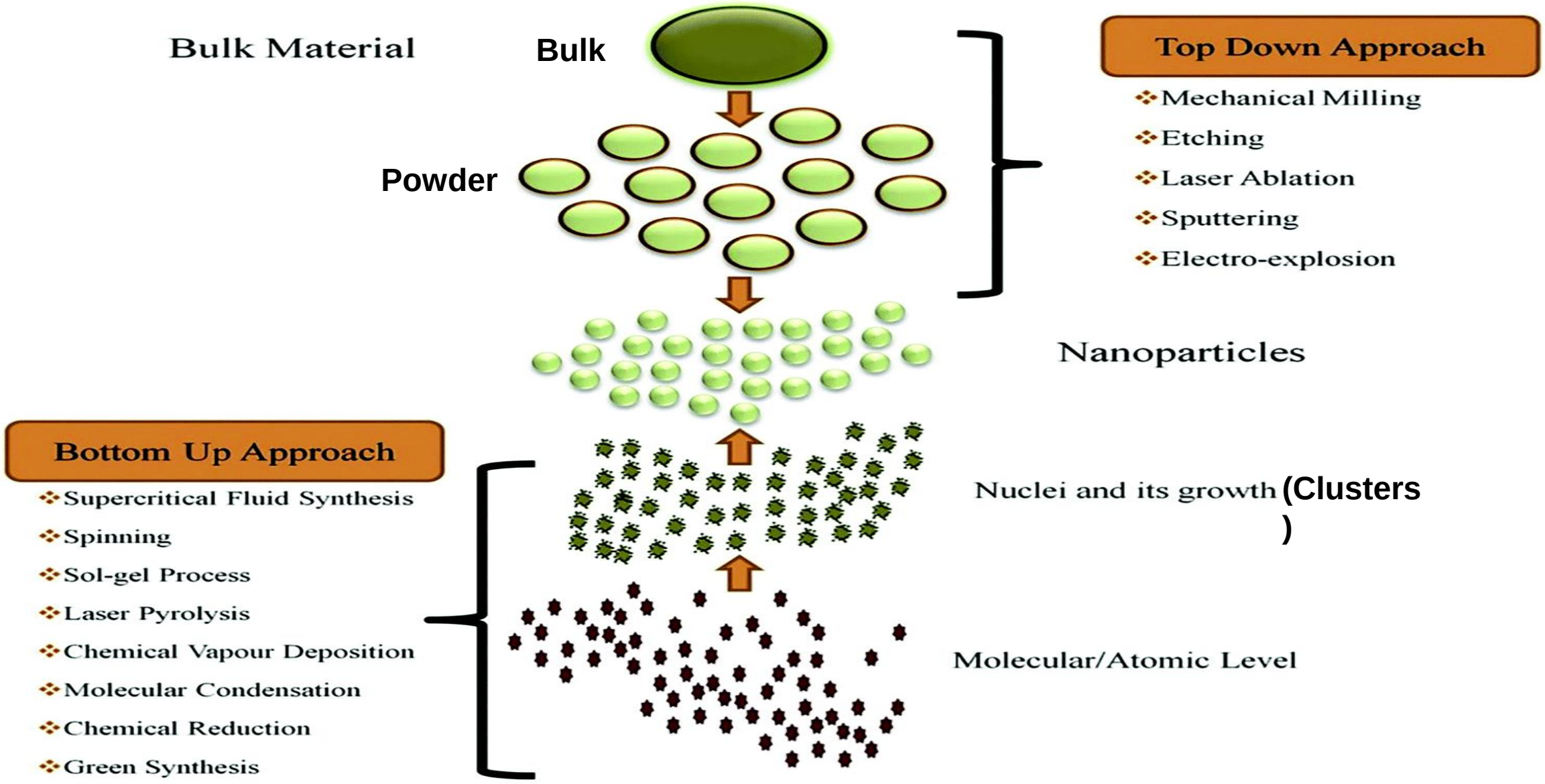
Biological Property

Top-down and bottom-up approaches for synthesis

- **Top-down**
- Involves breaking down larger pieces of material to create nanostructures. This approach is good for producing structures with long-range order and for making macroscopic connections.

- **Bottom-up**
- Involves assembling single atoms and molecules into larger nanostructures. This approach is best suited for assembly and establishing short-range order at nanoscale dimensions

- For example, in the top-down approach to produce nanosilver, a bulk piece of material is mechanically ground. In the bottom-up approach, chemical and biological methods are used to make nanostructures and nanoparticles.


Bulk Material
Bulk
Powder
Top Down Approach
Mechanical Milling
Etching
Laser Ablation
Sputtering
Electro-explosion
Nanoparticles
Bottom Up Approach
Supercritical Fluid Synthesis
Spinning
Sol-gel Process
Laser Pyrolysis
Chemical Vapour Deposition
Molecular Condensation
Chemical Reduction
Green Synthesis
Nuclei and its growth (Clusters)
Molecular/Atomic Level

Mechanical Milling method

Mechanical milling is a cost-effective method for producing materials at the nanoscale level from bulk materials.

Mechanical milling is an effective method for producing blends of different phases, and it is helpful in the production of nanocomposites.

Mechanical milling is used to produce oxide- and carbide-strengthened aluminum alloys, aluminum/nickel/magnesium/copper-based nanoalloys, and many other nanocomposite materials

Ball-milled carbon nanomaterials are considered a novel class of nanomaterial, providing the opportunity to satisfy environmental remediation, energy storage, and energy conversion demands

Mechanical milling, also known as mechanical alloying or mechanochemical synthesis, is a "top-down" method for producing nanomaterials by physically reducing the size of bulk materials through grinding and impacting.

This technique involves using high-energy mills to repeatedly impact and deform powder materials, leading to size reduction and the creation of new phases

To produce nonequilibrium structures including nanocrystalline, amorphous and quasicrystalline materials

Powders diameters of about **50 µm** with a number of hardened steel or tungsten carbide (WC) coated balls in a sealed container which is shaken or violently agitated. The most effective ratio for the ball to powder mass is 5 : 10.

Different millers are tumbler mills, attrition mills, shaker mills, vibratory mills, planetary mills etc

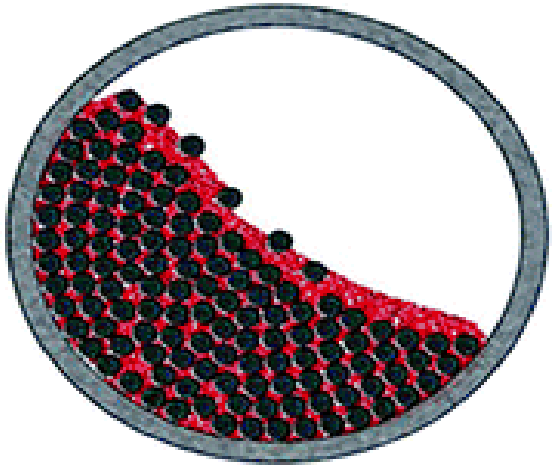
Shaker mills (e.g. SPEX model 8000) uses small batches of powder (approximately 10 cm^3 is sufficient

❖ **Advantage: High production rates**

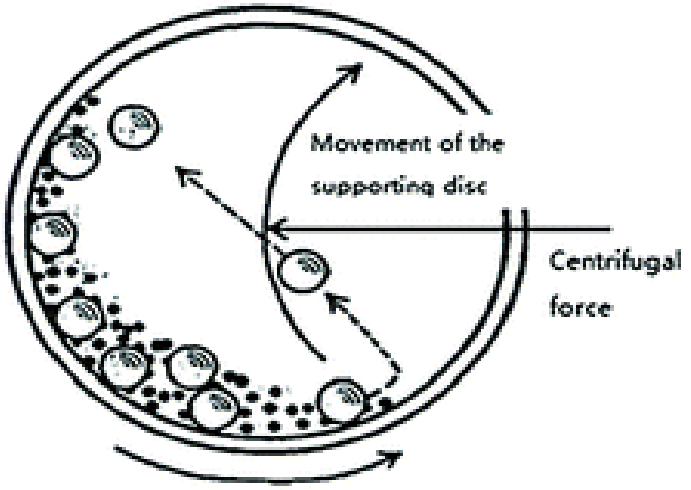
❖ **Disadvantages**

- Severe plastic deformation associated with mechanical attrition due to generation of high temp in the interphase, 100 to 200 ^{o}C.
- Difficulty in broken down to the required particle size
- Contamination by the milling tools (**Fe**) and atmosphere (trace elements of **O_2, N_2 in rare gases**) can be a problem (inert condition necessary like **Glove Box**)(Fe <1-2% and Trace elements<300 ppm)
- Protective coating to reduce milling tools contamination (MTC) increases cost of the process
- Working duration (>30 h) increases MTC (>10%)

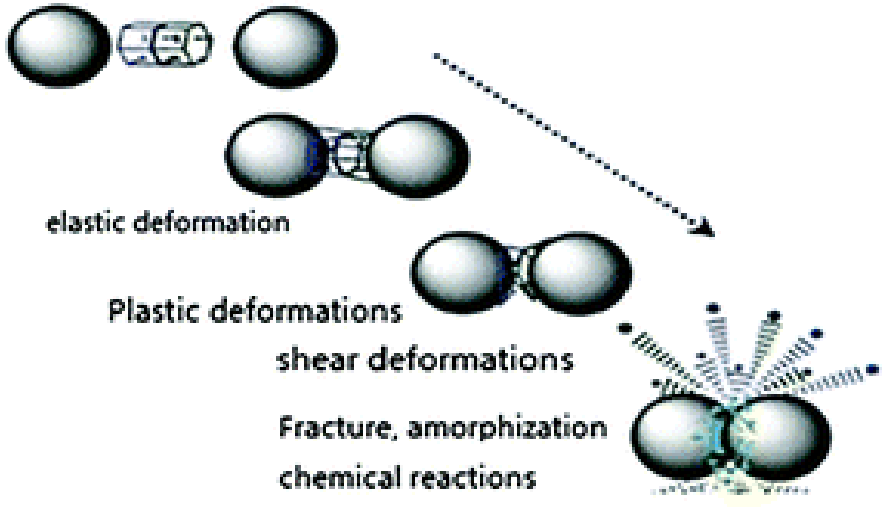




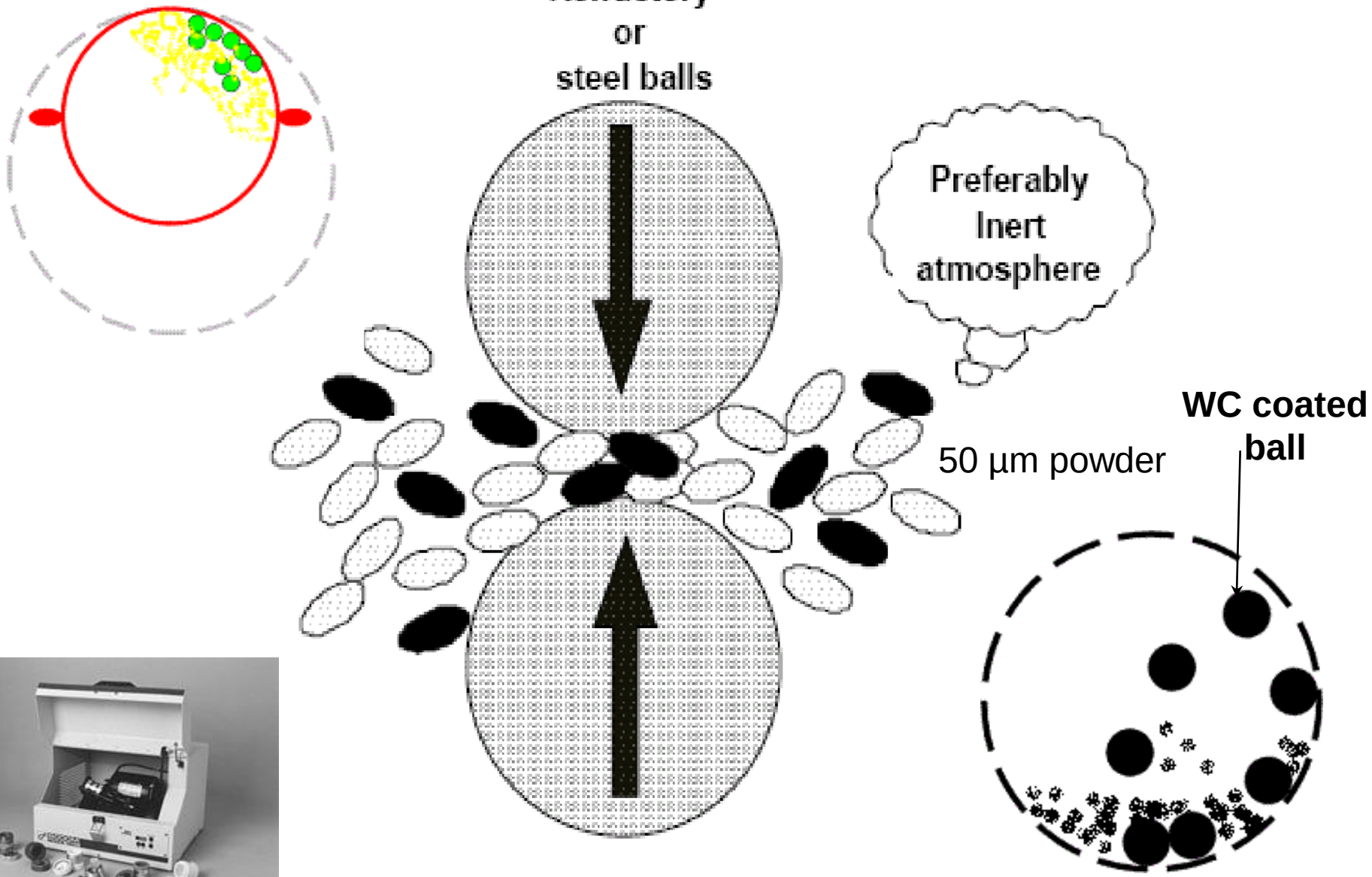
Horizontal section



Rotation of the milling bowl



Schematic representation of the principle of mechanical milling





Sol-gel method (Bottom up approach)

- This method is used for the development of various kinds of high-quality metal-oxide-based nanomaterials.
- This method is called a sol–gel method as during the synthesis of the metal-oxide nanoparticles, the liquid precursor is transformed to a sol, and the sol is ultimately converted into a network structure that is called a gel.
- Conventional precursors for the generation of nanomaterials using the sol–gel method are metal alkoxides.
- The synthesis process of nanoparticles *via* the sol–gel method can be completed in several steps.
- In the first step, the hydrolysis of the metal oxide takes place in water or with the assistance of alcohol to form a sol.

In the next step, condensation takes place, resulting in an increase in the solvent viscosity to form porous structures that are left to age.

During the condensation or polycondensation process, hydroxo- (M–OH–M) or oxo- (M–O–M) bridges form, resulting in metal–hydroxo- or metal–oxo-polymer formation in solution.

During the aging process, polycondensation continues, with changes to the structure, properties, and porosity. During aging, the porosity decreases, and the distance between the colloidal particles increases.

After the aging process, drying takes place, in which water and organic solvents are removed from the gel.

Lastly, calcination is performed to achieve nanoparticles.

Sol-Gel Processing

- Creation of Sol (solid particles in solution)
- Followed by the following two generic sol-gel processes (assuming as a precursor a metal alkoxide MOR):

hydrolysis: –MOR + H2O → –MOH + ROH

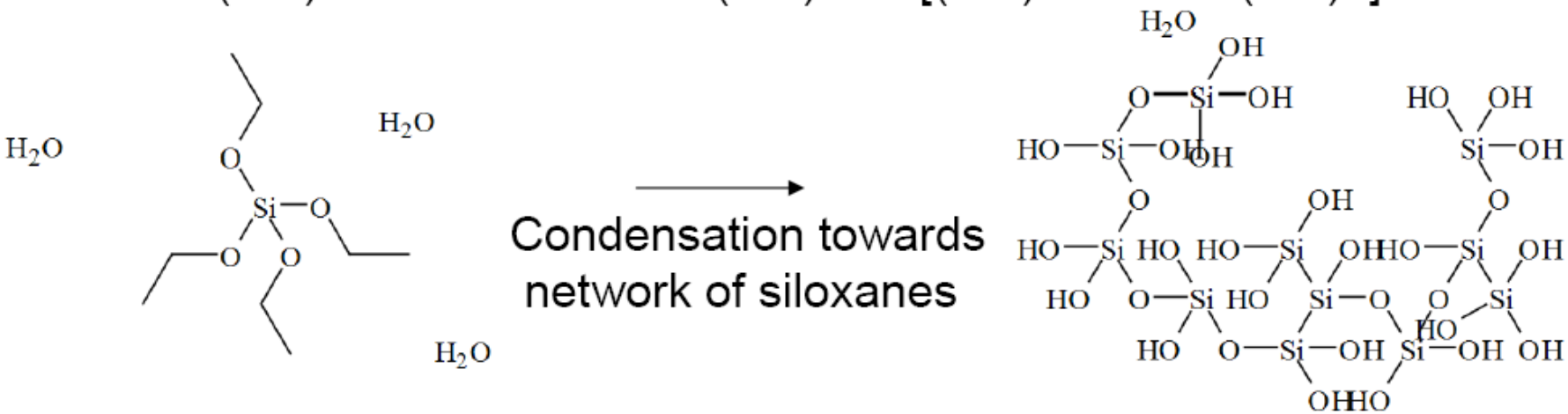
Ex.: $Si(OR)_4 + 4\ H_2O \rightarrow Si(OH)_4 + 4\ R\text{-}OH$

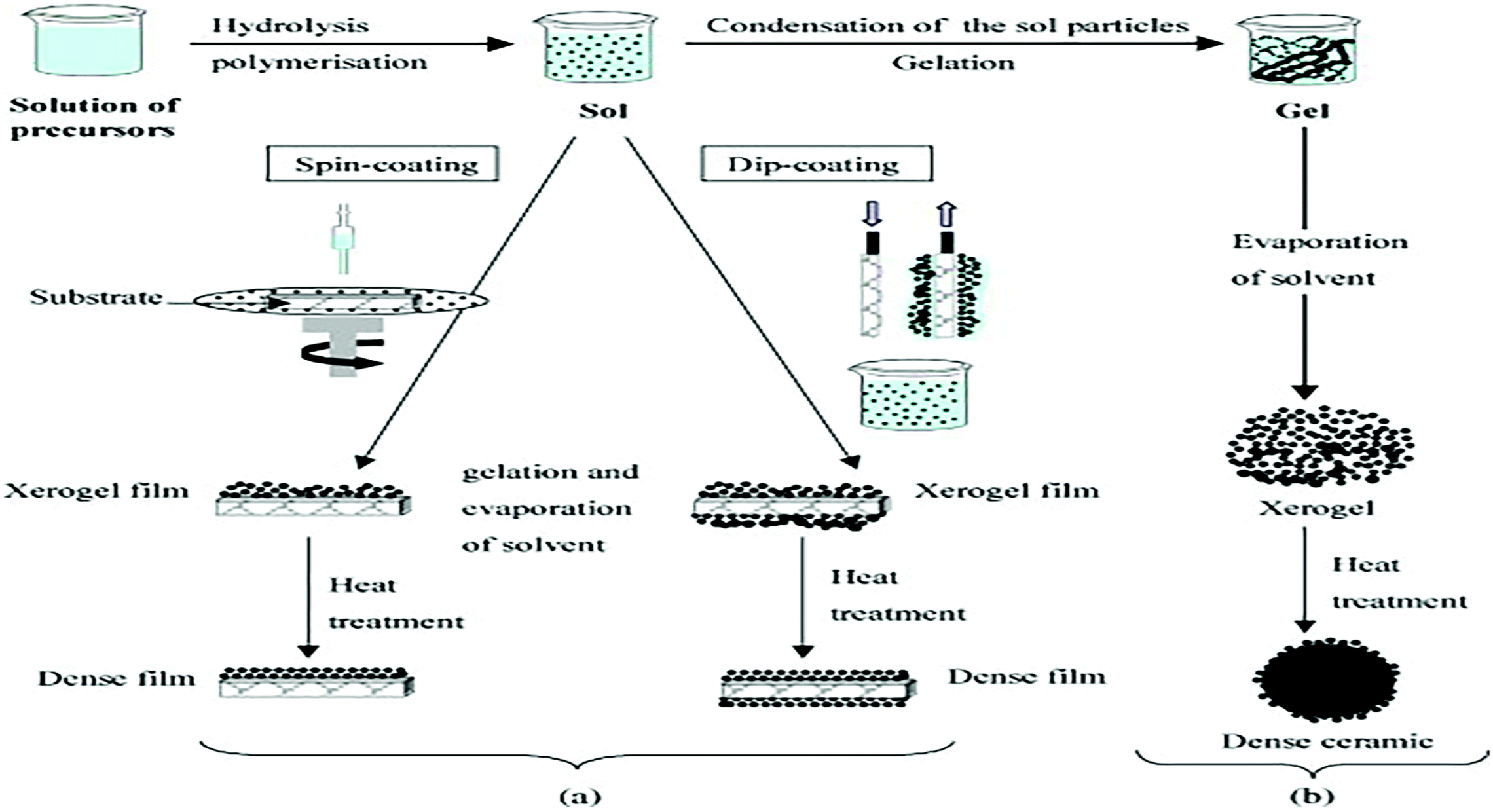
condensation: –MOH + ROM– → –MOM– + ROH

$(OR)_3–Si\text{-}OH + HO–Si\text{-}(OR)_3 \rightarrow [(OR)_3Si–O–Si(OR)_3] + H\text{-}O\text{-}H$

or –MOH + HOM– → –MOM– + H2O

(OR)3–Si-OR + HO–Si-(OR)3 → [(OR)3Si–O–Si(OR)3] + R-OH




Solution of precursors
Hydrolysis
polymerisation
Sol
Condensation of the sol particles
Gelation
Gel
Spin-coating
Dip-coating
Substrate
Evaporation of solvent
Xerogel film
gelation and evaporation of solvent
Xerogel film
Xerogel
Heat treatment
Heat treatment
Heat treatment
Dense film
Dense film
Dense ceramic
(a)
(b)

❖ Advantages

- Synthesizing nonmetallic inorganic materials like glasses, glass ceramics or ceramic materials at very low temperatures compared to melting glass or firing ceramics
- Monosized nanoparticles possible by this bottom up approach.

❖ Disadvantages

- Controlling the growth of the particles and then stopping the newly formed particles from agglomerating.
- Difficult to ensure complete reaction so that no unwanted reactant is left on the product
- Completely removal of any growth aids
- Also production rates of nanopowders are very slow by this process

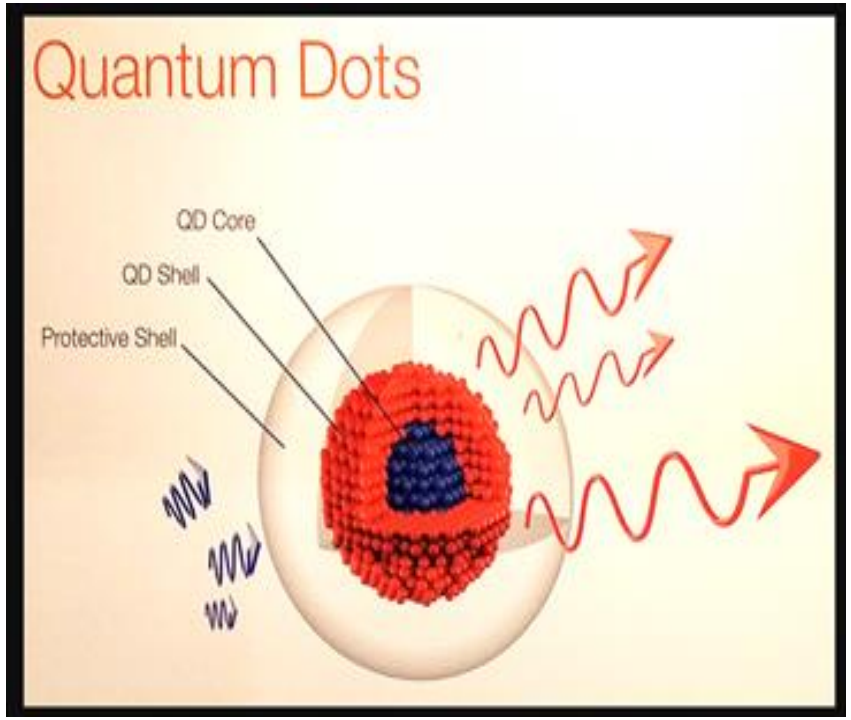
Quantum Dots

- Quantum dots are semiconductor particles with typical diameter of 2–10 nm. They are so named because, due to their nanoscale size, quantum effects play a significant part in their light emitting properties.

- Quantum dots emit light via this mechanism: under external stimulus, some of the electrons of the dot material absorb sufficient energy to escape their atomic orbit. This creates a conductance region in which the electrons can move through the material, effectively conducting electricity.

- As these electrons drop back into atomic orbit, energy is released in the form of light, the color of which depends on the amount of energy released.

- **Examples: cadmium selenide (CdSe), cadmium sulfide (CdS), and indium phosphide (InP)**

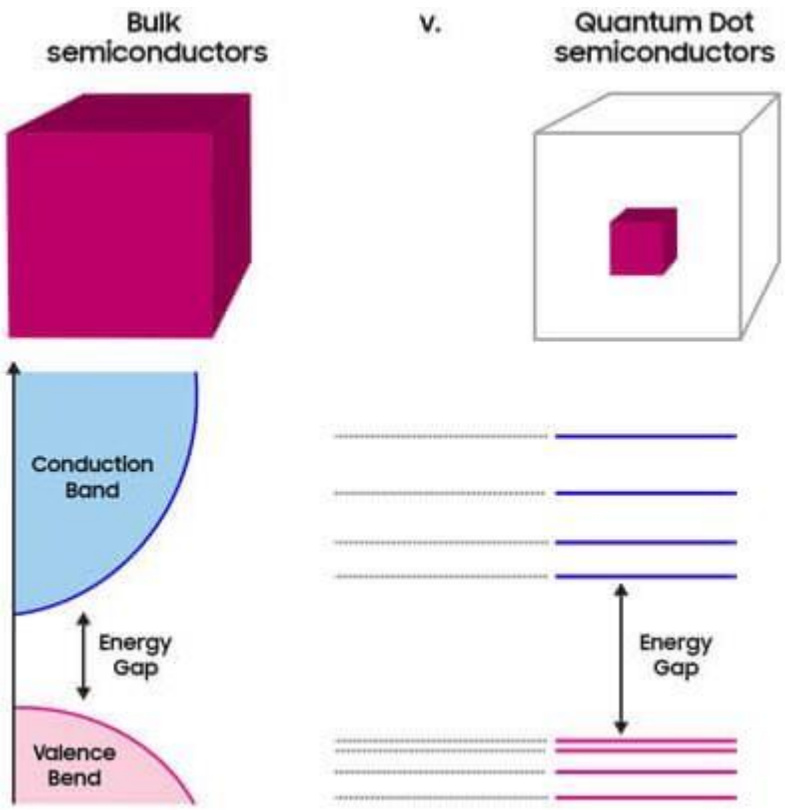


QDs contain 100 to 1000 electrons and are 2 to 10 nanometers in diameter, or 10 to 50 atoms.

Changing the band gap of semiconductors is the most attractive due to fundamental and technological importance.

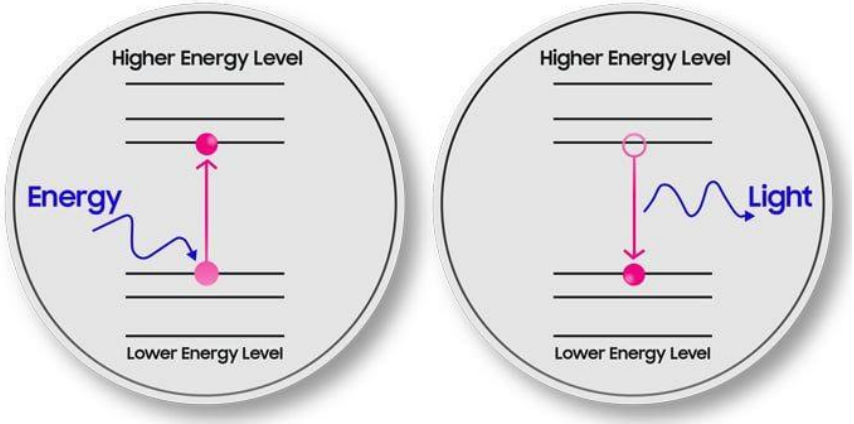
Widely tunable bandgap semiconductor are considered materials for new generation flat panel display photovoltaic, optoelectronic devices, lasers, sensors, photon bandgap devices, etc.

- High extinction coefficient
- High electron mobility
- Bandwidth and position adjustment
- Solution Process capabilities



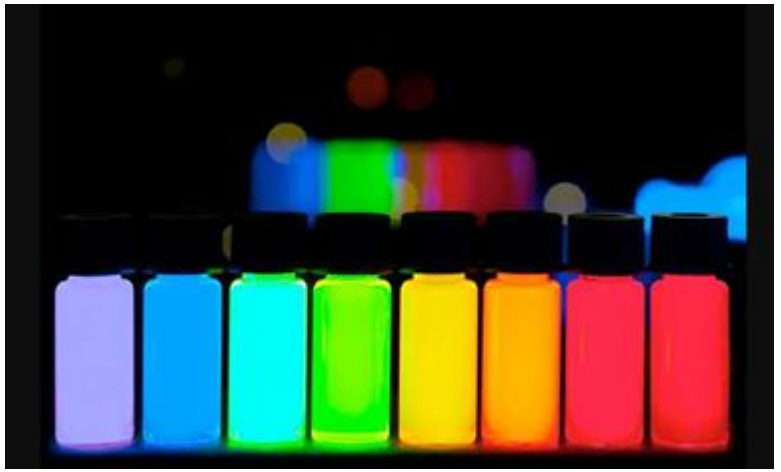
Schrödinger:
electrons in confined systems occupy quantized energy levels

The release of confinement energy is a key property of a quantum dot, which explains the positive relationship between the QD size and the frequency of the light it emits.



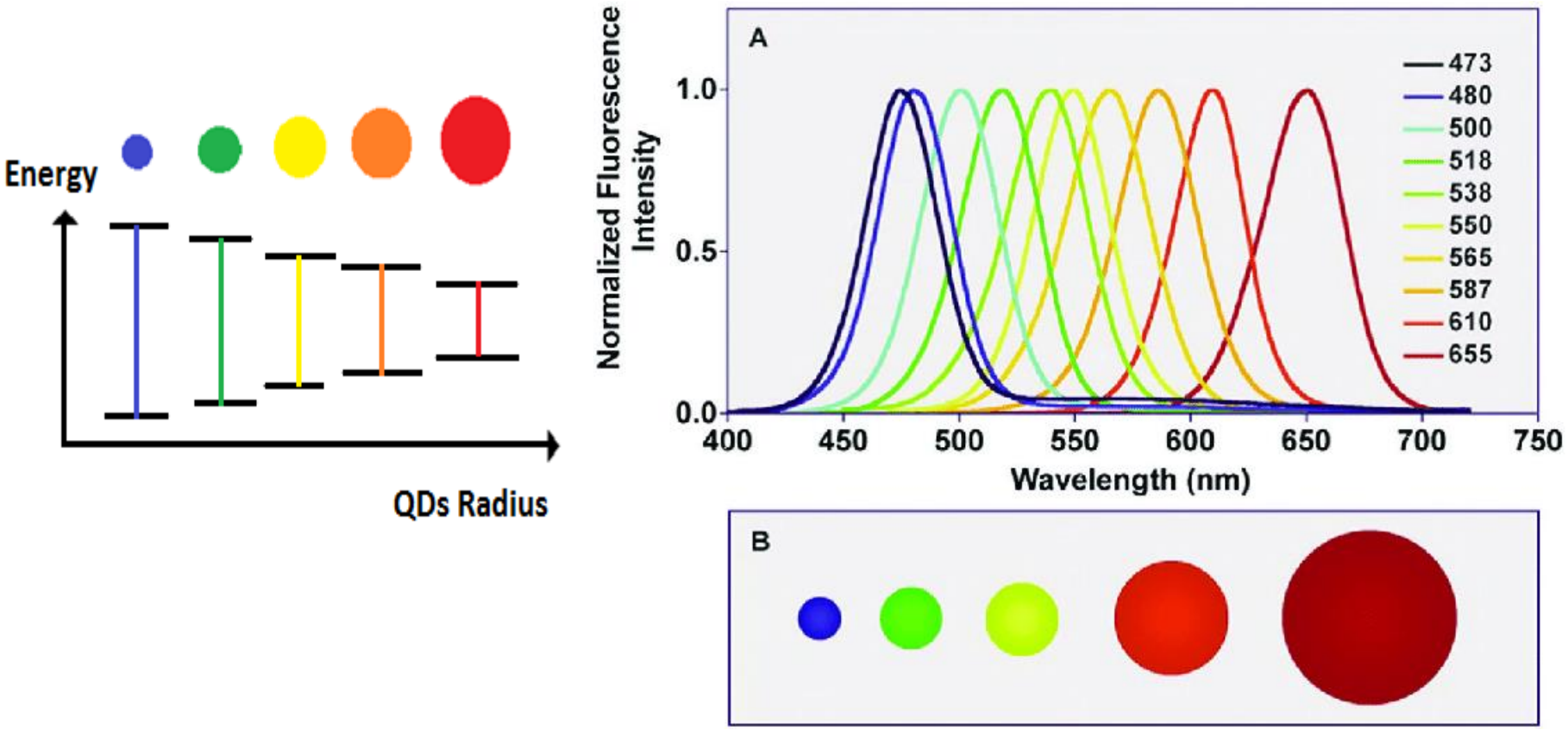
Unlike bulk semiconductors, which have continuous energy levels within the bands, the quantum confinement effect creates a large band gap with observed discrete energy levels. And because of this quantized band gap, a quantum dot can emit light at a very constant wavelength, which can be fine-tuned by changing the size of the quantum dots (or in other words, changing the energy levels).

- Because of the dots' small size, the amount of energy released is relatively consistent from electron to electron, yielding emissions of a single color.
- The color is entirely dependent upon the size of the dot, with larger dots (e.g., 5-6 nm) providing lower energy emissions (i.e., reds and oranges) and smaller dots (e.g., 2-3 nm) providing higher energy emissions (i.e., blues and violets).
- This property is sometimes referred to as "quantum confinement," indicating that constraints at the atomic level are predominant.



Quantum dot emissions from violet to deep red

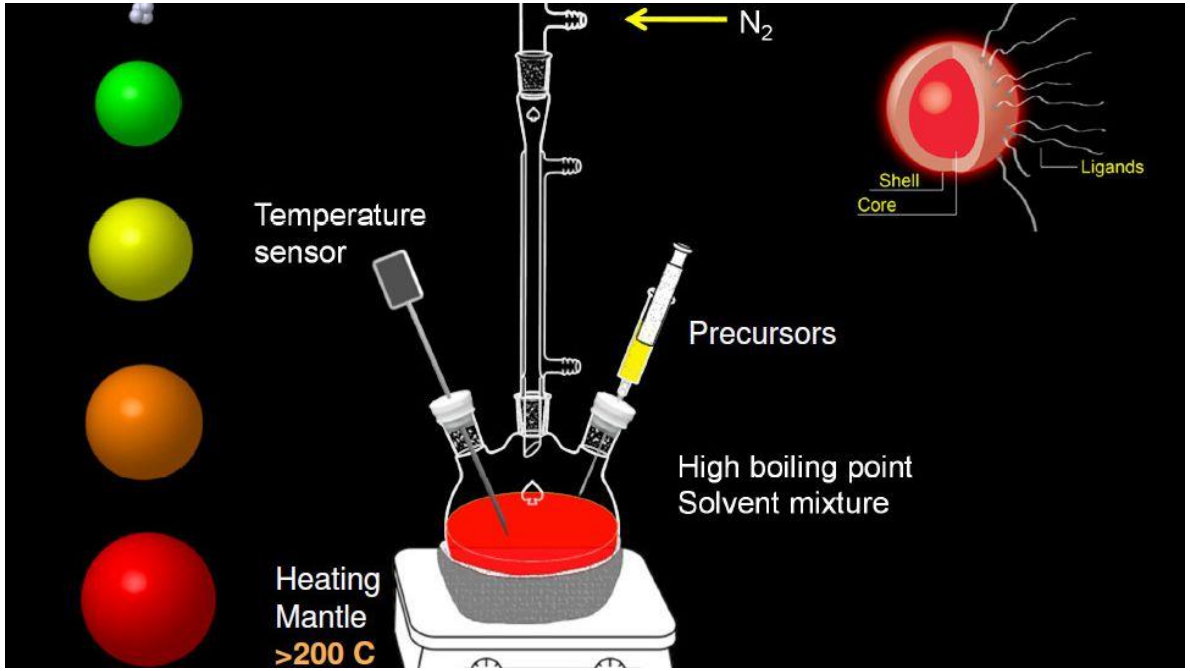
Emission properties of Quantum Dots



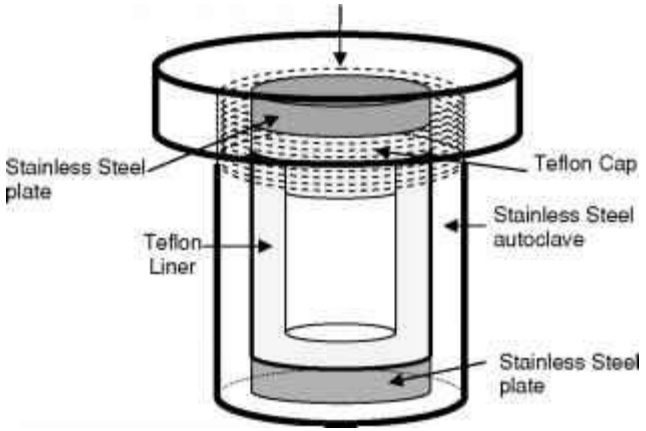
1. Band gap decreases as the QDs size increases
2. Emission wavelength becomes red shifted with increase in size of QDs

Synthetic Procedures of Quantum Dots

1. High temperature synthesis



2. Hydrothermal synthesis



3. Microwave synthesis



4. Ultrasonication synthesis

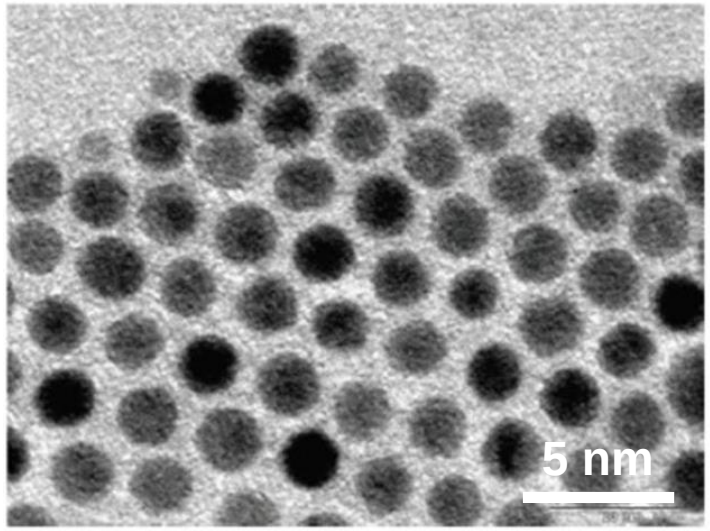


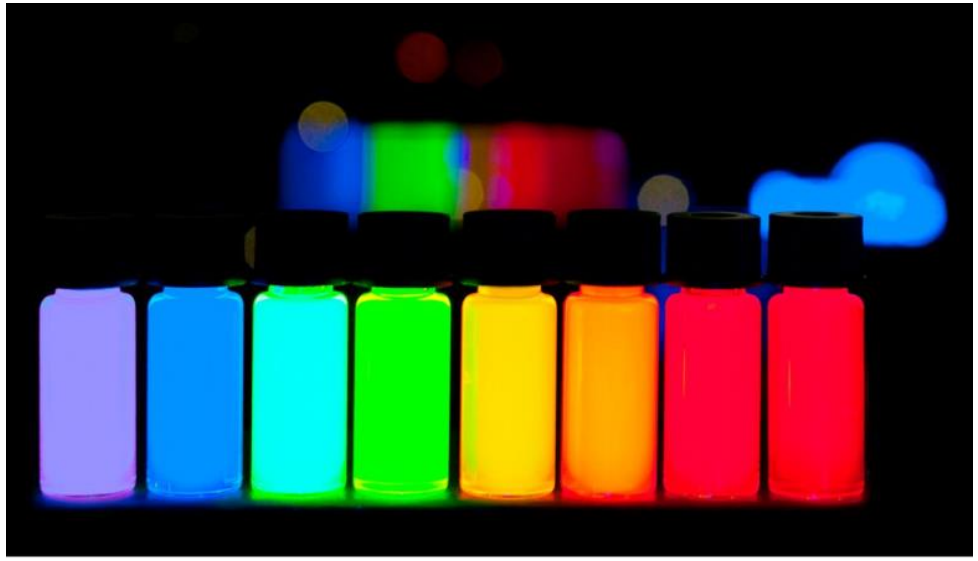
Applications:

Displays:
Quantum dots are used in displays to create brighter, more vibrant colors, particularly in QD-OLED TVs
Solar Cells:
Quantum dots can enhance the efficiency of solar cells by absorbing a wider range of light wavelengths
Biomedical Imaging:
Their tunable emission makes them useful for imaging tissues and cells, aiding in diagnostics and drug delivery.
Solid-State Lighting:
Quantum dots can be used to create more energy-efficient and colorful lighting solutions
Other applications
Quantum dots are also being explored for use in sensors, lasers, quantum computing, and other areas of research.

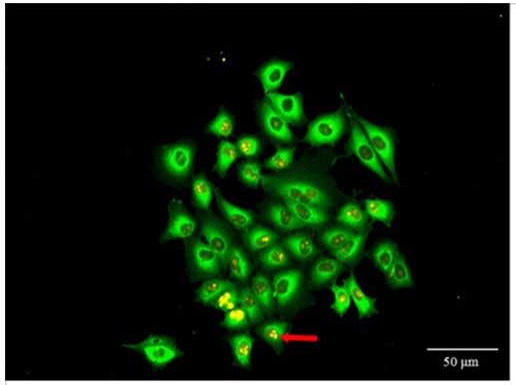
Quantum Dots and applications

Quantum dots (QDs) are semiconductor particles of few nanometers (<10 nm) in size, having optical and electronic properties that differ from larger particles.

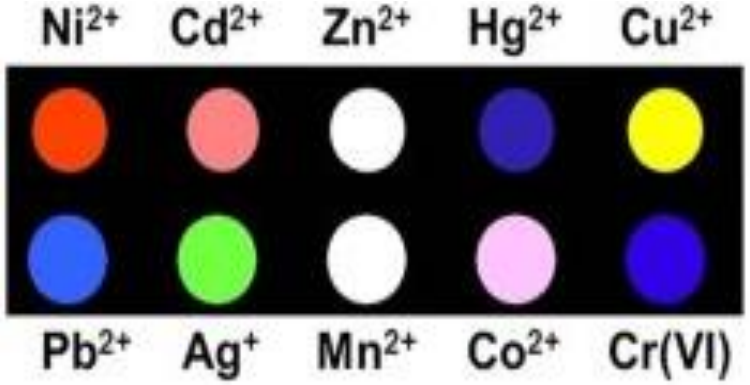




Cancer cell imaging



Metal ions sensing



Light emitting diode



Liquid Coolants

Liquid Coolants

- Types of coolants
- Thermal Conductivity
- Applications

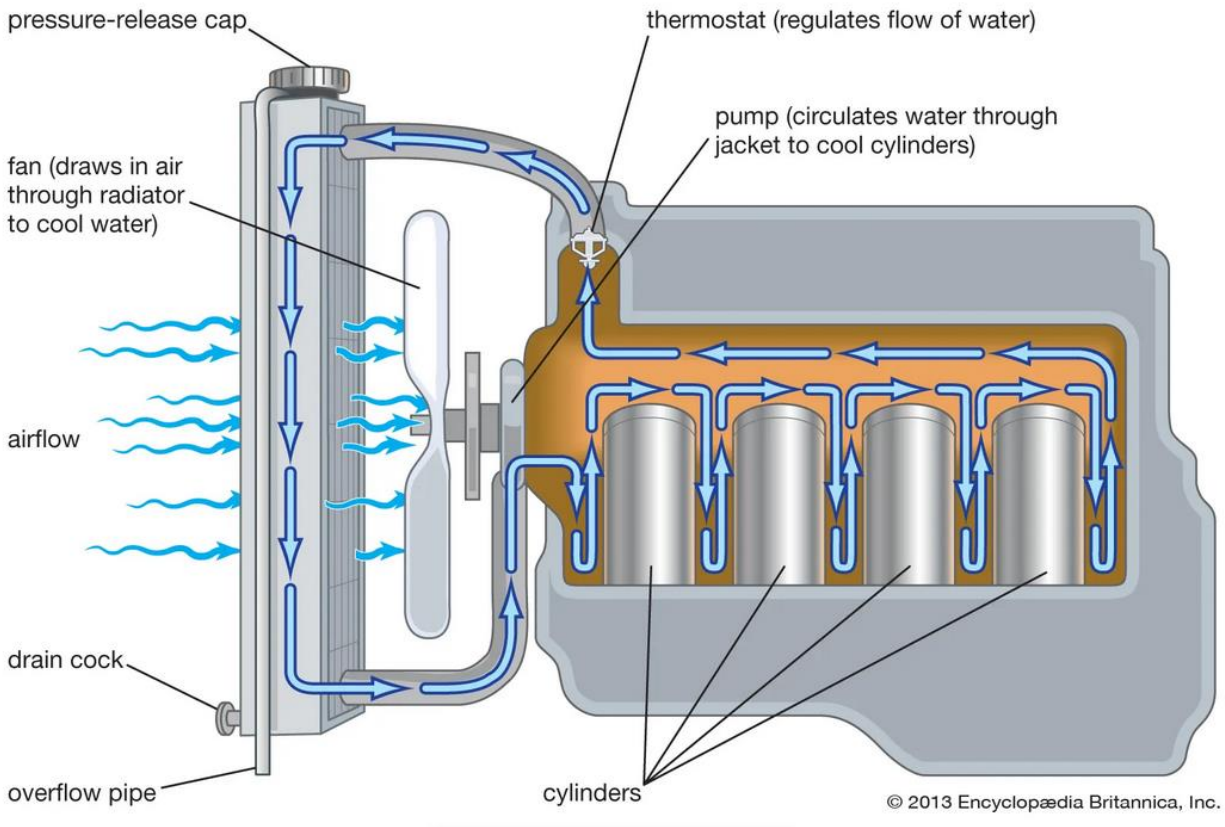
What are Coolants?

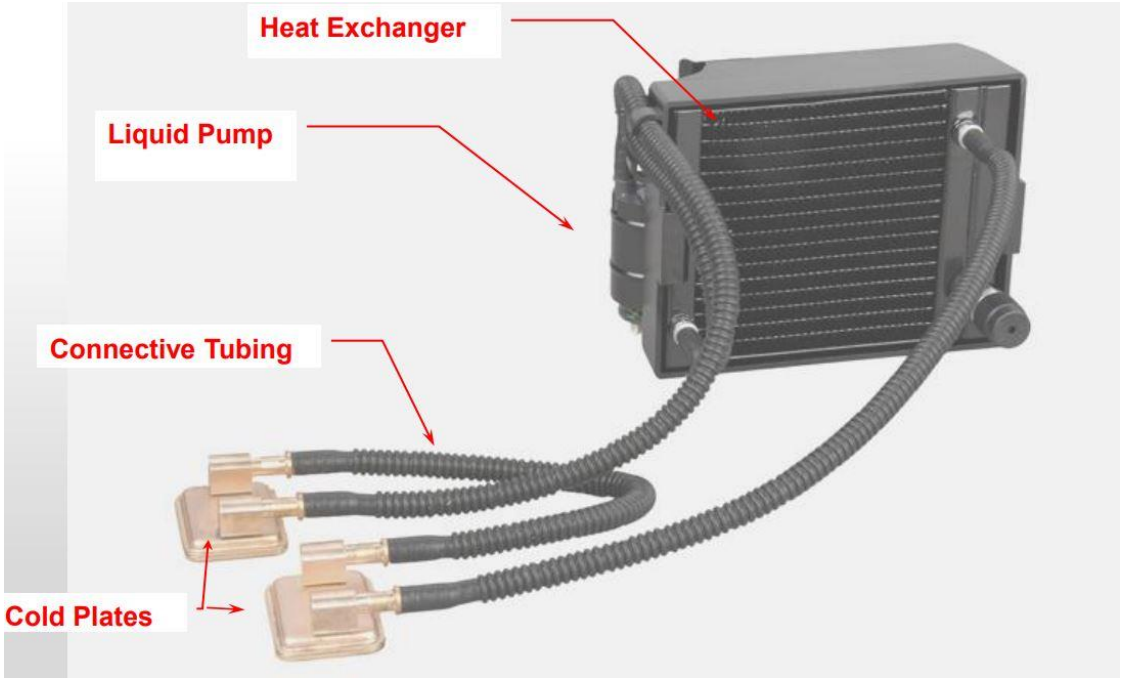
- Coolants are fluids that run across an engine to absorb the heat and carry it to the radiator where it can be cooled and recycled. Coolants, cool machinery that produces heat during operation.
- Electronics, especially miniature electronics, require cooling because the chips are highly power-dense, and they can generate heat flux of up to 1000 W/cm^2.
- High power density leads to an increase in heat generation, which can bring down the performance of the engine/device.

Coolants are Everywhere

- Automobiles
- HVAC systems
- Home air-conditioners
- Computers
- Supercomputers
- Data centers
- Miniature electronics
- Military devices
- Mobile devices

Examples of Cooling Systems





https://www.qats.com/cms/2017/11/13/fluids-can-used-liquid-cold-plates-electronics-cooling/

Thermal Conductivity

Thermal conductivity is the intensive property of a material that indicates its ability to conduct heat. For one-dimensional heat flow in the x-direction the steady state heat transfer can be described by Fourier's law of heat conduction:

$$Q = -\kappa \frac{dT}{dx}$$

where Q heat flux in the x-direction per unit area perpendicular to the heat flow (J (m2 s)$^{-1}$); κ = thermal conductivity (J (m s K)$^{-1}$); $\frac{dT}{dx}$ = temperature gradient in the x-direction(K m^{-1})

https://doi.org/10.1016/B978-0-08-054819-7.00017-0

Thermal Conductivity

As proposed by P. W. Bridgman in 1923 the heat transfer can be predicted and controlled by the means of changing its thermal conductivity by the equation

$$\kappa_L = 3k_B v_s n^{2/3}$$

Where κ_L=thermal conductivity of liquid. k_B=Boltzmann constant, v_s=velocity of sound and n = number density of molecule

DOI 10.1088/0256-307X/37/10/104401

Requirements for Coolants

- Thermophysical properties
  - Thermal conductivity, specific heat, latent heat of vaporization, surface tension, and viscosity
- Electrical conductivity
  - Water and glycol-water are highly conductive and spills may damage the electronics
  - Non-aqueous coolants are preferred when electronics are involved
- Flammability
  - Coolants for electronics must be non-flammable
  - Any liquid that has a flash point of <93 °C is considered flammable
  - Ethanol, methanol, and benzene are all flammable
  - PFC and HFE are inflammable and suitable for cooling applications

Requirements for Coolants

- Toxicity
  - Coolant should be non-toxic as leakages may happen and harm biological organisms
  - Fluorocarbons and hydrofluoroether-based coolants have low toxicity compared to ethylene glycol-based coolants
- Material Compatibility
  - Wetted surface and the coolant must be compatible to minimize corrosion
  - Copper and aluminum are preferred materials to build devices as they have high thermal conductivity and are compatible with most coolants

Requirements for Coolants

- Environmental Impact
  - Coolant should have low global warming potential (GWP) and ozone layer depletion potential (ODP)
  - Perflouoro carbon based coolants have high GWP but zero ODP
  - Hydrofluoroether based coolants have low GWP and zero ODP
- Cost
  - Coolants based on water are usually low-cost, but addition of antifreee, glycols and anti-corrosion materials may increase the cost

Classification of Coolants

- Aqueous coolants
  - Water has a good thermal conductivity, high specific heat, high latent heat of vaporization, and low viscosity
  - It is a low-risk coolant
  - Water is compatible with most materials
  - Freezing temperature of water makes it unsuitable in certain conditions
  - In such cases, additives such as glycol help lower the freezing point
  - Water – NaCl or water-$CaCl_2$ have low FP
  - Salt solutions are highly corrosive

Classification of Coolants

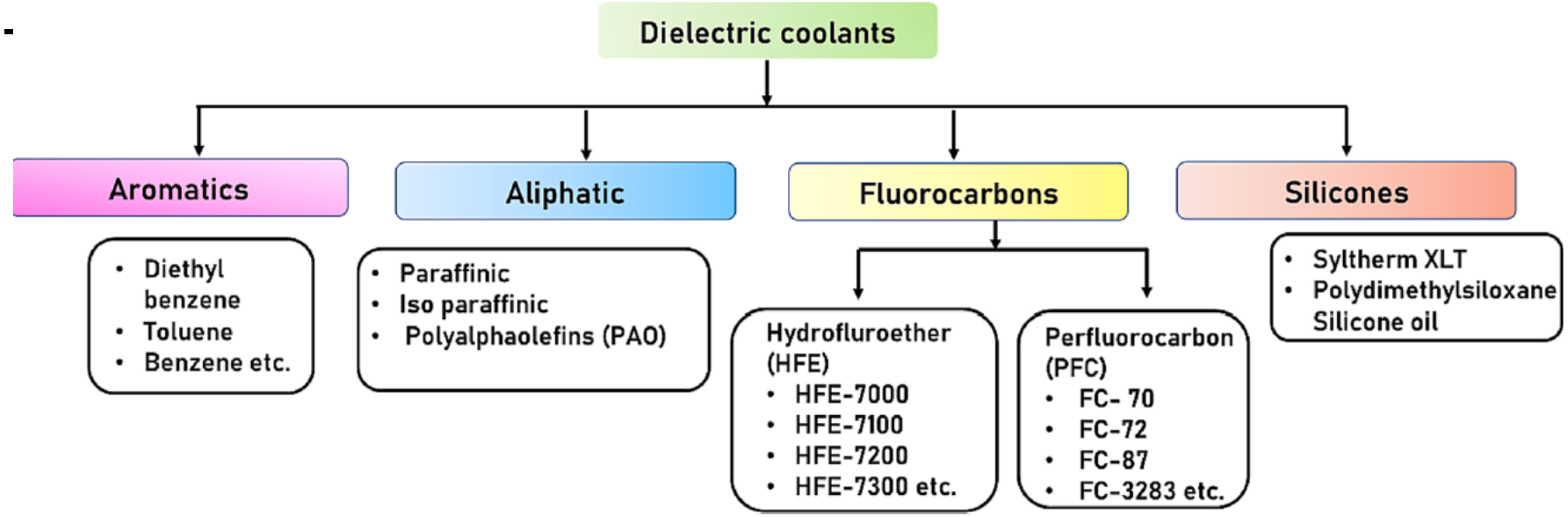
- Deionized water
  - Tap water contains minerals and pathogens
  - Deionized water is less corrosive than tap water
  - One should add corrosion inhibitors even to deionized water because exposure to air can make it corrosive
  - Common corrosion inhibitors are: phosphates, tolytriazole, and 2-ethyl hexanoic acid.
- Water-glycol solutions
  - Water-ethylene glycol and water-propylene glycol have low freezing point compared to water
  - High thermal conductivity, boiling point, specific heat, and boiling point

Classification of Coolants

- Water potassium formate/acetate solutions
  - Less corrosive compared to water-NaCl or water-$CaCl_2$
  - Their low freezing point (-70 C) is ideal for many applications
  - Food, beverage, chemical and pharmaceutical industries use these solutions.

Classification of Coolants

- Dielectric Coolants
  - High electrical conductivity of water restricts direct usage in contact with chips and circuits.
  - Data centers use these coolants in immersion cooling or direct-to-



Thermal Conductivity Measurements

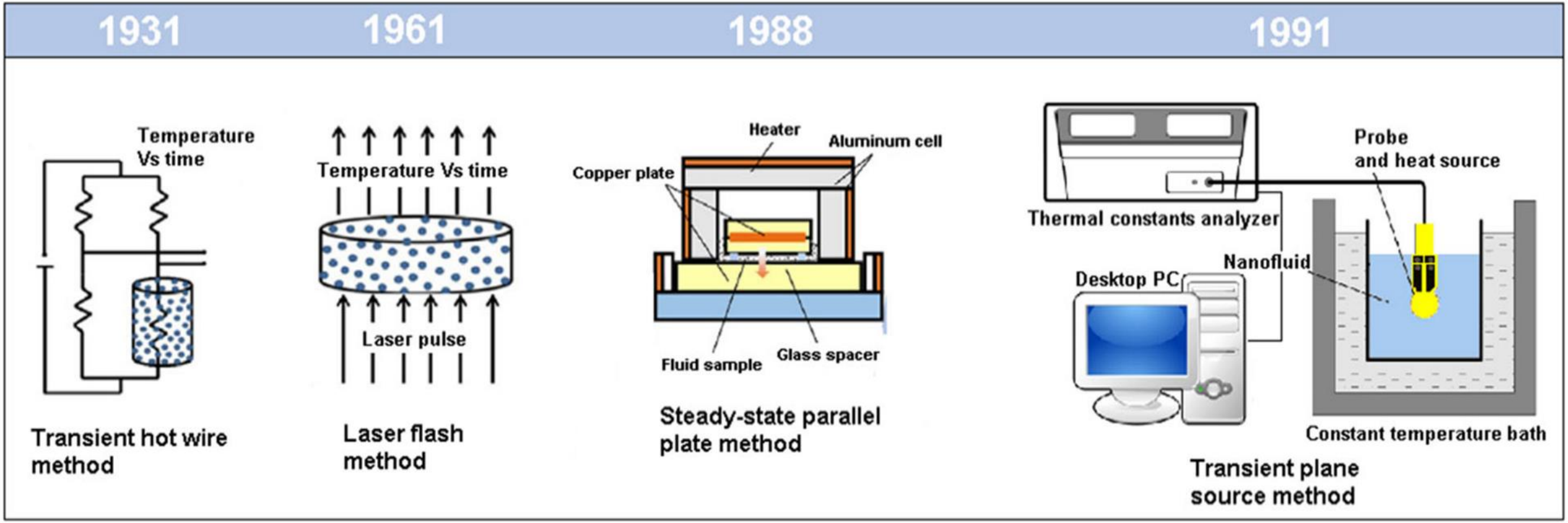
The current methods developed for thermal conductivity measurements are static methods and can be divided in Steady State Methods and Transient State Methods to avoid convective heat transfer in liquid.

Steady State Methods:

the parallel plane and coaxial cylinders methods

Transient State Methods: the transient hot wire, the transient plane source theory, the temperature oscillation method, the laser flash method and the 3ω method

Thermal Conductivity Measurements



https://doi.org/10.3390/app11062525

- https://termopasty.com/en/thermal-conductivity-in-cooling-systems/
- https://www.sciencedirect.com/science/article/pii/S1359431124004757